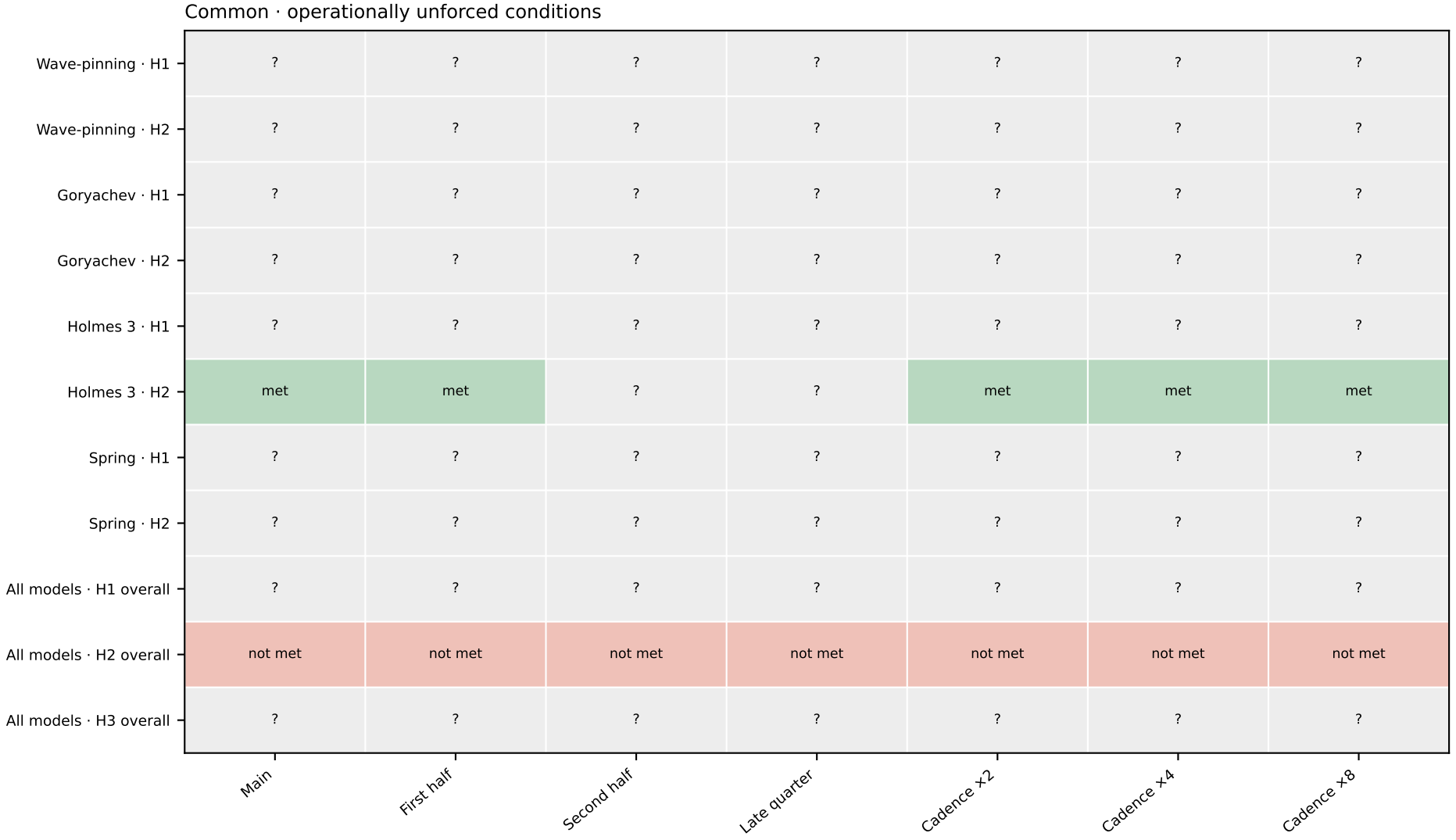
S0 · Completed audited Figure 4 criteria and qualifications · Declared hypothesis criteria across every view



Green: criterion met. Red: declared criterion not met. ?: inconclusive.
Raw criterion truth is shown here. A7 window, horizon and seed-cohort qualifications are separate S12 evidence.
An unmet criterion does not establish its biological opposite; no main framing is selected by this renderer.

Completed audited Figure 4 criteria and qualifications. Both forcing views are retained. All seven declared windows/cadences are displayed; missing work is refused. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_001_S0_common_primary.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · common · operationally unforced conditions

Wave-pinning · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 11; gain inputs: 5; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.03; large ci: 0.0001, 0.1

Wave-pinning · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 11

Goryachev · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 5; gain inputs: 0; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, unavailable; large ci: 0.0001, unavailable

Goryachev · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 5

Holmes 3 · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 11; gain inputs: 4; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.03

Holmes 3 · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: earlier f: 0.03; later f: 0.3; fold drop: 2.228; observation bound certified: yes, earlier f: 0.05; later f: 0.3; fold drop: 2.202; observation bound certified: yes
evaluable inputs: 11

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_002_S0_common_primary_details_01.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · common · operationally unforced conditions

Spring · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 10; gain inputs: 7; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.003; large ci: 0.0001, 0.01

Spring · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 10

all · H1 overall · inconclusive
reason: Predeclared model count
passes: 0
inconclusive: 4
required: 3
total: 4

all · H2 overall · criterion not met
reason: All required criteria are reported separately
components: criterion: H2 model count; status: inconclusive; reason: Predeclared model count; passes: 1; inconclusive: 3; required: 2; total: 4, criterion: H2 A6 predictor gate; status: criterion not met; reason: Missing folds are not successes; incomplete coverage is distinguished from evaluated disagreement; features: criterion: multiple front fraction; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: 0.09091; successful folds: 0; unavailable folds: 1; planned folds: 4, criterion: region count mean; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.389; successful folds: 3; unavailable folds: 1; planned folds: 4

all · H3 overall · inconclusive
reason: At least two different models must show distinct corroborated, non-forced failure modes with both seed fingerprints separated
All 65 preset distinct-model/mode pairs are displayed separately on H3 pair pages, including every inconclusive or unmet component.

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_003_S0_common_primary_details_02.csv

S0 · Completed audited Figure 4 criteria and qualifications · Declared hypothesis criteria across every view



	Main	First half	Second half	Late quarter	Cadence x2	Cadence x4	Cadence x8
Wave-pinning · H1	?	?	?	?	?	?	?
Wave-pinning · H2	?	?	?	?	?	?	?
Goryachev · H1	?	?	?	?	?	?	?
Goryachev · H2	?	?	?	?	?	?	?
Holmes 3 · H1	?	?	?	?	?	?	?
Holmes 3 · H2	met	met	met	?	met	met	met
Spring · H1	not met	?	not met	?	not met	not met	?
Spring · H2	met	?	met	?	met	met	?
All models · H1 overall	?	?	?	?	?	?	?
All models · H2 overall	not met	not met	not met	not met	not met	not met	not met
All models · H3 overall	?	?	?	?	?	?	?

Green: criterion met. Red: declared criterion not met. ?: inconclusive.
Raw criterion truth is shown here. A7 window, horizon and seed-cohort qualifications are separate S12 evidence.
An unmet criterion does not establish its biological opposite; no main framing is selected by this renderer.

Completed audited Figure 4 criteria and qualifications. Both forcing views are retained. All seven declared windows/cadences are displayed; missing work is refused. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_004_S0_common_inclusive.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · common · all conditions including operationally forced

Wave-pinning · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 6; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.03; large ci: 0.0001, 0.1

Wave-pinning · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 12

Goryachev · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 7; gain inputs: 2; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.01; large ci: 0.0001, unavailable

Goryachev · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 7

Holmes 3 · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 5; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.03

Holmes 3 · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: earlier f: 0.03; later f: 0.3; fold drop: 2.228; observation bound certified: yes, earlier f: 0.05; later f: 0.3; fold drop: 2.202; observation bound certified: yes
evaluable inputs: 12

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_005_S0_common_inclusive_details_01.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · common · all conditions including operationally forced

Spring · H1 model · criterion not met
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: criterion not met; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 7; reversal inputs: 1; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion not met; reason: All designed comparisons remain in the coverage table; reversal inputs: 1, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.003; large ci: 0.0001, 0.01

Spring · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: 1
peak drop ranges: none observed
evaluable inputs: 12

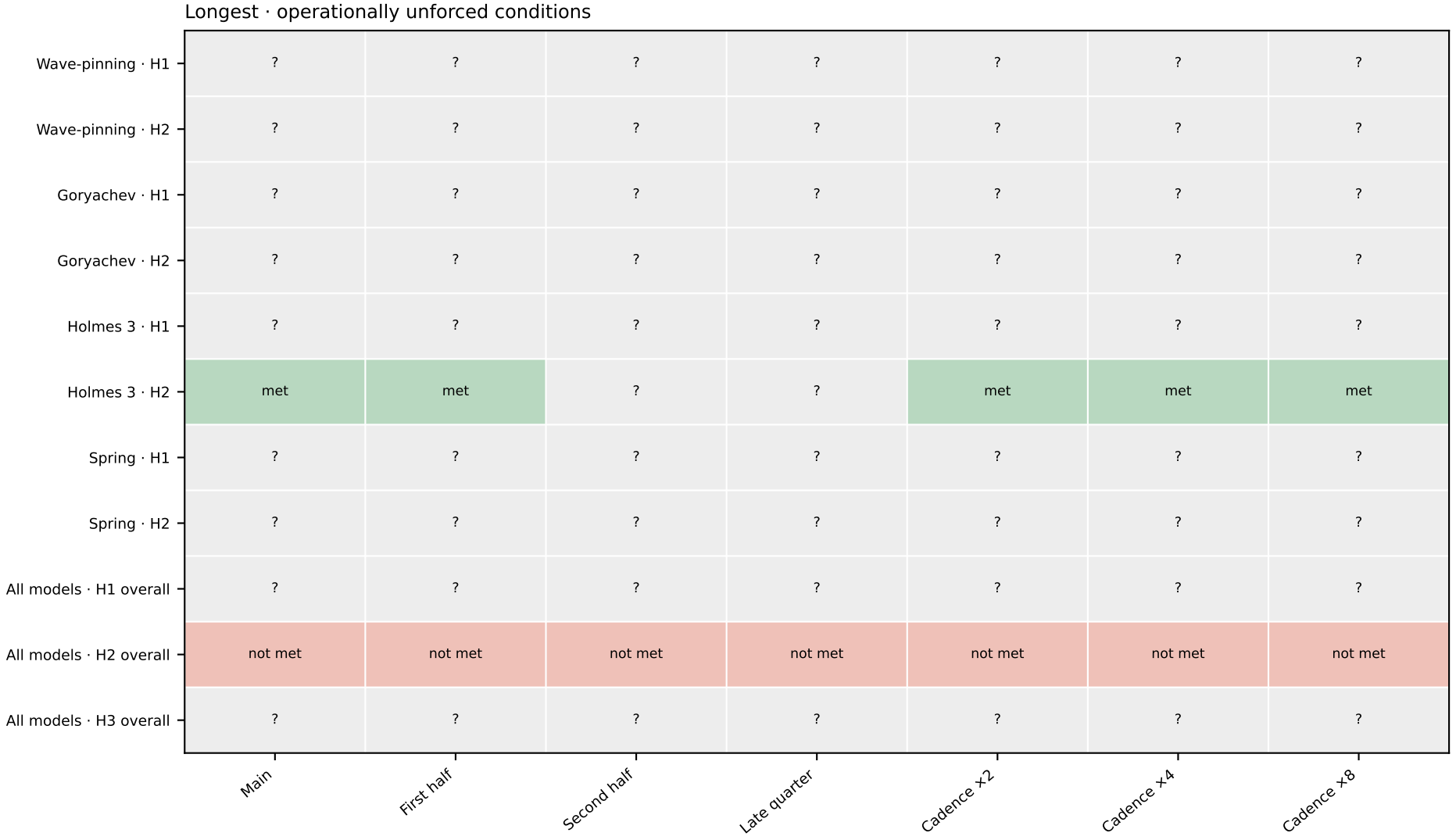
all · H1 overall · inconclusive
reason: Predeclared model count
passes: 0
inconclusive: 3
required: 3
total: 4

all · H2 overall · criterion not met
reason: All required criteria are reported separately
components: criterion: H2 model count; status: criterion met; reason: Predeclared model count; passes: 2; inconclusive: 2; required: 2; total: 4, criterion: H2 A6 predictor gate; status: criterion not met; reason: Missing folds are not successes; incomplete coverage is distinguished from evaluated disagreement; features: criterion: multiple front fraction; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.02143; successful folds: 0; unavailable folds: 1; planned folds: 4, criterion: region count mean; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.3416; successful folds: 3; unavailable folds: 1; planned folds: 4

all · H3 overall · inconclusive
reason: At least two different models must show distinct corroborated, non-forced failure modes with both seed fingerprints separated
All 65 preset distinct-model/mode pairs are displayed separately on H3 pair pages, including every inconclusive or unmet component.

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_006_S0_common_inclusive_details_02.csv

S0 · Completed audited Figure 4 criteria and qualifications · Declared hypothesis criteria across every view



Green: criterion met. Red: declared criterion not met. ?: inconclusive.
Raw criterion truth is shown here. A7 window, horizon and seed-cohort qualifications are separate S12 evidence.
An unmet criterion does not establish its biological opposite; no main framing is selected by this renderer.

Completed audited Figure 4 criteria and qualifications. Both forcing views are retained. All seven declared windows/cadences are displayed; missing work is refused. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_007_S0_longest_primary.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · longest · operationally unforced conditions

Wave-pinning · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 11; gain inputs: 5; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.05

Wave-pinning · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 11

Goryachev · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 5; gain inputs: 0; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, unavailable; large ci: 0.0001, unavailable

Goryachev · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 5

Holmes 3 · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 11; gain inputs: 4; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.02

Holmes 3 · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: earlier f: 0.03; later f: 0.3; fold drop: 2.228; observation bound certified: yes, earlier f: 0.05; later f: 0.3; fold drop: 2.202; observation bound certified: yes
evaluable inputs: 11

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_008_S0_longest_primary_details_01.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · longest · operationally unforced conditions

Spring · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 10; gain inputs: 7; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.003; large ci: 0.0001, 0.01

Spring · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 10

all · H1 overall · inconclusive
reason: Predeclared model count
passes: 0
inconclusive: 4
required: 3
total: 4

all · H2 overall · criterion not met
reason: All required criteria are reported separately
components: criterion: H2 model count; status: inconclusive; reason: Predeclared model count; passes: 1; inconclusive: 3; required: 2; total: 4, criterion: H2 A6 predictor gate; status: criterion not met; reason: Missing folds are not successes; incomplete coverage is distinguished from evaluated disagreement; features: criterion: multiple front fraction; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: 0.09091; successful folds: 0; unavailable folds: 1; planned folds: 4, criterion: region count mean; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.389; successful folds: 3; unavailable folds: 1; planned folds: 4

all · H3 overall · inconclusive
reason: At least two different models must show distinct corroborated, non-forced failure modes with both seed fingerprints separated
All 65 preset distinct-model/mode pairs are displayed separately on H3 pair pages, including every inconclusive or unmet component.

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_009_S0_longest_primary_details_02.csv

S0 · Completed audited Figure 4 criteria and qualifications · Declared hypothesis criteria across every view



Green: criterion met. Red: declared criterion not met. ?: inconclusive.
Raw criterion truth is shown here. A7 window, horizon and seed-cohort qualifications are separate S12 evidence.
An unmet criterion does not establish its biological opposite; no main framing is selected by this renderer.

Completed audited Figure 4 criteria and qualifications. Both forcing views are retained. All seven declared windows/cadences are displayed; missing work is refused. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_010_S0_longest_inclusive.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · longest · all conditions including operationally forced

Wave-pinning · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 6; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.05

Wave-pinning · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 12

Goryachev · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 7; gain inputs: 2; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.01; large ci: 0.0001, unavailable

Goryachev · H2 model · inconclusive
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: none observed
evaluable inputs: 7

Holmes 3 · H1 model · inconclusive
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: inconclusive; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 5; reversal inputs: none observed; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion met; reason: All designed comparisons remain in the coverage table; reversal inputs: none observed, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.02; large ci: 0.0001, 0.02

Holmes 3 · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: none observed
peak drop ranges: earlier f: 0.03; later f: 0.3; fold drop: 2.228; observation bound certified: yes, earlier f: 0.05; later f: 0.3; fold drop: 2.202; observation bound certified: yes
evaluable inputs: 12

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_011_S0_longest_inclusive_details_01.csv

S0 · Completed audited Figure 4 criteria and qualifications · Main-view criterion components · longest · all conditions including operationally forced

Spring · H1 model · criterion not met
reason: All required criteria are reported separately
components: criterion: H1 all evaluable gains; status: criterion not met; reason: Every evaluable ratio CI must exclude1 above; undefined or missing qualification cannot pass; evaluable inputs: 12; gain inputs: 7; reversal inputs: 1; unknown forced qualification: no, criterion: H1 no supported reversal; status: criterion not met; reason: All designed comparisons remain in the coverage table; reversal inputs: 1, criterion: H1 noise tolerance; status: inconclusive; reason: Largest/smallest weakest-input tolerance CIs must be non-overlapping in the increasing direction; small ci: 0.0001, 0.003; large ci: 0.0001, 0.01

Spring · H2 model · criterion met
reason: Explicit existence rule over all designed input pairs; no forced or unavailable gap is silently bridged
reversal inputs: 1
peak drop ranges: none observed
evaluable inputs: 12

all · H1 overall · inconclusive
reason: Predeclared model count
passes: 0
inconclusive: 3
required: 3
total: 4

all · H2 overall · criterion not met
reason: All required criteria are reported separately
components: criterion: H2 model count; status: criterion met; reason: Predeclared model count; passes: 2; inconclusive: 2; required: 2; total: 4, criterion: H2 A6 predictor gate; status: criterion not met; reason: Missing folds are not successes; incomplete coverage is distinguished from evaluated disagreement; features: criterion: multiple front fraction; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.02143; successful folds: 0; unavailable folds: 1; planned folds: 4, criterion: region count mean; status: criterion not met; reason: Only the two frozen direct-tau negative-association predictors can support H2; pooled spearman: -0.3416; successful folds: 3; unavailable folds: 1; planned folds: 4

all · H3 overall · inconclusive
reason: At least two different models must show distinct corroborated, non-forced failure modes with both seed fingerprints separated
All 65 preset distinct-model/mode pairs are displayed separately on H3 pair pages, including every inconclusive or unmet component.

Completed audited Figure 4 criteria and qualifications. Criterion components are copied from the audited final tables, without recomputing truth. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_012_S0_longest_inclusive_details_02.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009372 [0, 0.002187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0171 [0.007002, 0.03033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9893 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2347 [0.124, 0.3757] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9795 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6389 [0.5518, 0.7217] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.5, 505.5] resolved	[8000, ∞] right censored	S 0.9425 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9545 [0.9444, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[127.9, 127.9] resolved	[8000, ∞] right censored	S 0.9218 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.9853, 0.9872] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[83.26, 83.26] resolved	[8000, ∞] right censored	S 0.9328 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9613 [0.9603, 0.9622] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_013_S0_H3_candidates_common_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[474.5, 474.5] resolved	[8000, ∞] right censored	S 0.2376 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07688 [0.06599, 0.08612] S low 0.856 [0.8388, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[564.4, 564.4] resolved	[8000, ∞] right censored	S 0.1314 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0624 [0.05545, 0.06938] S low 0.9004 [0.8883, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[734.3, 734.3] resolved	[8000, ∞] right censored	S 0.05307 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02684 [0.02106, 0.03265] S low 0.9579 [0.9502, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1257, 1257] resolved	[8000, ∞] right censored	S 0.01543 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004842 [0.001874, 0.00781] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_014_S0_H3_candidates_common_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0003905 [0, 0.001119] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9789 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1968 [0.0342, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3901, 3901] resolved	[8000, ∞] right censored	S 0.9798 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4833 [0.3906, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[8000, ∞] right censored	S 0.985 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7162 [0.6475, 0.7784] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.5, 986.5] resolved	[8000, ∞] right censored	S 0.9754 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8569 [0.8322, 0.8799] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.1, 388.1] resolved	[8000, ∞] right censored	S 0.9626 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9196 [0.901, 0.9378] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[103.8, 103.8] resolved	[8000, ∞] right censored	S 0.964 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9311 [0.9288, 0.9335] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_015_S0_H3_candidates_common_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[51.37, 51.37] resolved	[8000, ∞] right censored	S 0.9693 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8552 [0.8485, 0.8619] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.000781 [7.81e-05, 0.001744] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.4, 533.4] resolved	[800, ∞] right censored	S 0.9981 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02353 [0.006378, 0.04493] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146, 146] resolved	[800, ∞] right censored	S 0.9921 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1367 [0.1076, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.92, 37.92] resolved	[800, ∞] right censored	S 0.9845 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3529 [0.3182, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[800, ∞] right censored	S 0.9828 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4992 [0.4707, 0.531] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.65, 10.65] resolved	[800, ∞] right censored	S 0.9792 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6586 [0.6433, 0.6732] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.794, 6.794] resolved	[800, ∞] right censored	S 0.9745 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7892 [0.7759, 0.8008] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.964, 4.964] resolved	[800, ∞] right censored	S 0.9746 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.8399 [0.8372, 0.8425] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_016_S0_H3_candidates_common_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.121, 4.121] resolved	[800, ∞] right censored	S 0.9903 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.768 [0.7476, 0.7885] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_017_S0_H3_candidates_common_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8679, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_018_S0_H3_fingerprints_common_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3857, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9133, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_019_S0_H3_fingerprints_common_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01544; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008175, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_020_S0_H3_fingerprints_common_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01148, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8567, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7005, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8086, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_021_S0_H3_fingerprints_common_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_022_S0_H3_fingerprints_common_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5114, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5599, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.744, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6043, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_023_S0_H3_fingerprints_common_main_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_024_S0_H3_pairs_common_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_025_S0_H3_pairs_common_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_026_S0_H3_pairs_common_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_027_S0_H3_pairs_common_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_028_S0_H3_pairs_common_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_029_S0_H3_pairs_common_main_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_030_S0_H3_pairs_common_main_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_031_S0_H3_pairs_common_main_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_032_S0_H3_pairs_common_main_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009372 [0, 0.002187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0171 [0.007002, 0.03033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9893 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2347 [0.124, 0.3757] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9795 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6389 [0.5518, 0.7217] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.5, 505.5] resolved	[8000, ∞] right censored	S 0.9425 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9545 [0.9444, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[127.9, 127.9] resolved	[8000, ∞] right censored	S 0.9218 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.9853, 0.9872] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[83.26, 83.26] resolved	[8000, ∞] right censored	S 0.9328 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9613 [0.9603, 0.9622] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_033_S0_H3_candidates_common_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[474.5, 474.5] resolved	[8000, ∞] right censored	S 0.2376 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07688 [0.06599, 0.08612] S low 0.856 [0.8388, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[564.4, 564.4] resolved	[8000, ∞] right censored	S 0.1314 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0624 [0.05545, 0.06938] S low 0.9004 [0.8883, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[734.3, 734.3] resolved	[8000, ∞] right censored	S 0.05307 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02684 [0.02106, 0.03265] S low 0.9579 [0.9502, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1257, 1257] resolved	[8000, ∞] right censored	S 0.01543 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004842 [0.001874, 0.00781] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_034_S0_H3_candidates_common_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0003905 [0, 0.001119] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9789 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1968 [0.0342, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3901, 3901] resolved	[8000, ∞] right censored	S 0.9798 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4833 [0.3906, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[8000, ∞] right censored	S 0.985 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7162 [0.6475, 0.7784] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.5, 986.5] resolved	[8000, ∞] right censored	S 0.9754 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8569 [0.8322, 0.8799] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.1, 388.1] resolved	[8000, ∞] right censored	S 0.9626 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9196 [0.901, 0.9378] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[103.8, 103.8] resolved	[8000, ∞] right censored	S 0.964 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9311 [0.9288, 0.9335] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_035_S0_H3_candidates_common_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[51.37, 51.37] resolved	[8000, ∞] right censored	S 0.9693 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8552 [0.8485, 0.8619] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.000781 [7.81e-05, 0.001744] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.4, 533.4] resolved	[800, ∞] right censored	S 0.9981 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02353 [0.006378, 0.04493] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146, 146] resolved	[800, ∞] right censored	S 0.9921 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1367 [0.1076, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.92, 37.92] resolved	[800, ∞] right censored	S 0.9845 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3529 [0.3182, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[800, ∞] right censored	S 0.9828 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4992 [0.4707, 0.531] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.65, 10.65] resolved	[800, ∞] right censored	S 0.9792 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6586 [0.6433, 0.6732] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.794, 6.794] resolved	[800, ∞] right censored	S 0.9745 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7892 [0.7759, 0.8008] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.964, 4.964] resolved	[800, ∞] right censored	S 0.9746 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.8399 [0.8372, 0.8425] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_036_S0_H3_candidates_common_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.121, 4.121] resolved	[800, ∞] right censored	S 0.9903 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.768 [0.7476, 0.7885] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_037_S0_H3_candidates_common_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8679, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_038_S0_H3_fingerprints_common_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3857, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9133, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_039_S0_H3_fingerprints_common_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01544; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008175, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_040_S0_H3_fingerprints_common_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01148, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8567, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7005, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8086, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_041_S0_H3_fingerprints_common_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_042_S0_H3_fingerprints_common_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5114, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5599, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.744, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6043, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_043_S0_H3_fingerprints_common_main_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_044_S0_H3_pairs_common_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_045_S0_H3_pairs_common_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_046_S0_H3_pairs_common_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_047_S0_H3_pairs_common_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_048_S0_H3_pairs_common_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_049_S0_H3_pairs_common_main_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_050_S0_H3_pairs_common_main_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_051_S0_H3_pairs_common_main_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_052_S0_H3_pairs_common_main_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0002082 [0, 0.0006246] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.02441 [0.006141, 0.0519] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9964 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2057 [0.0673, 0.3891] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2704, 2704] resolved	[4000, ∞] right censored	S 0.9964 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6192 [0.5258, 0.7165] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[487, 487] resolved	[4000, ∞] right censored	S 0.9483 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9579 [0.946, 0.9681] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[132.3, 132.3] resolved	[4000, ∞] right censored	S 0.9308 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.985 [0.985, 0.985] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[77.8, 77.8] resolved	[4000, ∞] right censored	S 0.9445 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9622 [0.9594, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_053_S0_H3_candidates_common_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[540.8, 540.8] resolved	[4000, ∞] right censored	S 0.2132 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06949 [0.05278, 0.08584] S low 0.8661 [0.844, 0.8911] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[606, 606] resolved	[4000, ∞] right censored	S 0.1347 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.05569 [0.04679, 0.06673] S low 0.9029 [0.8896, 0.9143] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[748.5, 748.5] resolved	[4000, ∞] right censored	S 0.05051 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0266 [0.01764, 0.03618] S low 0.9619 [0.9517, 0.9717] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable lag gap	[4000, ∞] right censored	S 0.0003448 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.00812 [0.003748, 0.01249] S low 0.9866 [0.9781, 0.995] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_054_S0_H3_candidates_common_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[4000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9932 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1835 [0.03965, 0.375] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	inconclusive	inconclusive	[3999, 3999] resolved	[4000, ∞] right censored	S 0.998 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.367 [0.2444, 0.4962] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[1827, 1827] resolved	[4000, ∞] right censored	S 0.9906 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.7034 [0.6057, 0.7941] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[975.5, 975.5] resolved	[4000, ∞] right censored	S 0.9642 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8384 [0.7979, 0.8804] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[397.5, 397.5] resolved	[4000, ∞] right censored	S 0.9622 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9261 [0.9032, 0.9492] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[98.02, 98.02] resolved	[4000, ∞] right censored	S 0.9469 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9341 [0.9288, 0.9394] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_055_S0_H3_candidates_common_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[49.28, 49.28] resolved	[4000, ∞] right censored	S 0.9661 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8695 [0.8682, 0.8707] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0009369 [0, 0.002655] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[347.3, 347.3] resolved	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02056 [0.002342, 0.05127] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[134.2, 134.2] resolved	[400, ∞] right censored	S 0.9921 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.1433 [0.09447, 0.193] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[34.54, 34.54] resolved	[400, ∞] right censored	S 0.9857 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3674 [0.3309, 0.4097] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.22, 18.22] resolved	[400, ∞] right censored	S 0.9847 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4844 [0.4473, 0.5205] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.31, 10.31] resolved	[400, ∞] right censored	S 0.9774 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6649 [0.6391, 0.6876] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.701, 6.701] resolved	[400, ∞] right censored	S 0.975 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7977 [0.7819, 0.814] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.789, 4.789] resolved	[400, ∞] right censored	S 0.9742 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.8267 [0.8226, 0.8307] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_056_S0_H3_candidates_common_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.337, 4.337] resolved	[400, ∞] right censored	S 0.9893 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7561 [0.727, 0.7851] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_057_S0_H3_candidates_common_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.018 [1.002, 1.042] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8815, low 0; forced mean no	1.178 [1.031, 1.372] N 12/12; valid 1	1.007 [1.003, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.651 [1.505, 1.778] N 12/12; valid 1	1.012 [1.007, 1.017] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_058_S0_H3_fingerprints_common_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3779, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9141, low 0; forced mean no	2.309 [2.253, 2.364] N 12/12; valid 1	1.023 [1.014, 1.033] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.506 [2.388, 2.633] N 2/2; valid 1	1.112 [1.078, 1.147] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.459 [2.368, 2.554] N 2/2; valid 1	1.286 [1.259, 1.314] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.009 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.032] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.112 [1.103, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9412 [0.9057, 0.9772] N 12/12; valid 1	1.593 [1.418, 1.838] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7942 [0.7712, 0.8197] N 12/12; valid 1	3.393 [3.313, 3.462] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_059_S0_H3_fingerprints_common_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0.08333; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7682 [0.7542, 0.7843] N 12/12; valid 1	2.601 [2.511, 2.698] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.01228, low 0; forced mean yes	1.429 [1.342, 1.51] N 12/12; valid 1	0.4705 [0.4524, 0.4893] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.381 [1.271, 1.485] N 12/12; valid 1	0.4573 [0.4425, 0.4726] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.219 [1.139, 1.304] N 12/12; valid 1	0.4295 [0.4176, 0.4423] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.6434 [0.6143, 0.6742] N 2/2; valid 1	0.4216 [0.4195, 0.4237] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.04676 [0, 0.09559] N 2/2; valid 1	0.4164 [0.3826, 0.4562] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_060_S0_H3_fingerprints_common_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.179 [1.032, 1.363] N 12/12; valid 1	1.012 [1.011, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.351 [1.211, 1.499] N 12/12; valid 1	1.029 [1.025, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8559, low 0; forced mean no	1.771 [1.622, 1.937] N 12/12; valid 1	1.049 [1.044, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.019 [1.911, 2.12] N 12/12; valid 1	1.073 [1.066, 1.08] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7088, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8021, low 0; forced mean no	2.151 [2.065, 2.239] N 12/12; valid 1	1.135 [1.119, 1.15] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_061_S0_H3_fingerprints_common_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.496 [2.397, 2.6] N 2/2; valid 1	1.265 [1.254, 1.276] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.185 [2.056, 2.315] N 2/2; valid 1	1.6 [1.54, 1.663] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1.001] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1.001 [1, 1.003] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.019 [1.002, 1.049] N 12/12; valid 1	1.005 [1.002, 1.007] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.128 [1.082, 1.177] N 12/12; valid 1	1.001 [0.995, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.312 [1.274, 1.352] N 12/12; valid 1	1 [0.9924, 1.009] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_062_S0_H3_fingerprints_common_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1632, low 0; forced mean no	no eligible check single front; H 3200 multi 0.4919, low 0; forced mean no	1.418 [1.371, 1.465] N 12/12; valid 1	1.003 [0.9936, 1.013] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.569 [1.521, 1.612] N 12/12; valid 1	1.006 [0.989, 1.022] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5564, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7449, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5985, low 0; forced mean no	1.7 [1.675, 1.726] N 12/12; valid 1	1.039 [1.017, 1.061] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.738 [1.693, 1.782] N 2/2; valid 1	1.17 [1.097, 1.253] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.723 [1.693, 1.752] N 2/2; valid 1	1.271 [1.236, 1.306] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_063_S0_H3_fingerprints_common_analysis_first_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_064_S0_H3_pairs_common_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_065_S0_H3_pairs_common_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_066_S0_H3_pairs_common_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_067_S0_H3_pairs_common_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_068_S0_H3_pairs_common_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_069_S0_H3_pairs_common_analysis_first_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_070_S0_H3_pairs_common_analysis_first_half_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_071_S0_H3_pairs_common_analysis_first_half_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_072_S0_H3_pairs_common_analysis_first_half_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0002082 [0, 0.0006246] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.02441 [0.006141, 0.0519] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9964 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2057 [0.0673, 0.3891] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2704, 2704] resolved	[4000, ∞] right censored	S 0.9964 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6192 [0.5258, 0.7165] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[487, 487] resolved	[4000, ∞] right censored	S 0.9483 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9579 [0.946, 0.9681] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[132.3, 132.3] resolved	[4000, ∞] right censored	S 0.9308 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.985 [0.985, 0.985] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[77.8, 77.8] resolved	[4000, ∞] right censored	S 0.9445 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9622 [0.9594, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_073_S0_H3_candidates_common_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[540.8, 540.8] resolved	[4000, ∞] right censored	S 0.2132 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06949 [0.05278, 0.08584] S low 0.8661 [0.844, 0.8911] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[606, 606] resolved	[4000, ∞] right censored	S 0.1347 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.05569 [0.04679, 0.06673] S low 0.9029 [0.8896, 0.9143] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[748.5, 748.5] resolved	[4000, ∞] right censored	S 0.05051 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0266 [0.01764, 0.03618] S low 0.9619 [0.9517, 0.9717] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable lag gap	[4000, ∞] right censored	S 0.0003448 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.00812 [0.003748, 0.01249] S low 0.9866 [0.9781, 0.995] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_074_S0_H3_candidates_common_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[4000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9932 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1835 [0.03965, 0.375] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	inconclusive	inconclusive	[3999, 3999] resolved	[4000, ∞] right censored	S 0.998 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.367 [0.2444, 0.4962] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[1827, 1827] resolved	[4000, ∞] right censored	S 0.9906 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.7034 [0.6057, 0.7941] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[975.5, 975.5] resolved	[4000, ∞] right censored	S 0.9642 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8384 [0.7979, 0.8804] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[397.5, 397.5] resolved	[4000, ∞] right censored	S 0.9622 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9261 [0.9032, 0.9492] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[98.02, 98.02] resolved	[4000, ∞] right censored	S 0.9469 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9341 [0.9288, 0.9394] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_075_S0_H3_candidates_common_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[49.28, 49.28] resolved	[4000, ∞] right censored	S 0.9661 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8695 [0.8682, 0.8707] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0009369 [0, 0.002655] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[347.3, 347.3] resolved	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02056 [0.002342, 0.05127] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[134.2, 134.2] resolved	[400, ∞] right censored	S 0.9921 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.1433 [0.09447, 0.193] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[34.54, 34.54] resolved	[400, ∞] right censored	S 0.9857 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3674 [0.3309, 0.4097] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.22, 18.22] resolved	[400, ∞] right censored	S 0.9847 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4844 [0.4473, 0.5205] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.31, 10.31] resolved	[400, ∞] right censored	S 0.9774 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6649 [0.6391, 0.6876] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.701, 6.701] resolved	[400, ∞] right censored	S 0.975 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7977 [0.7819, 0.814] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.789, 4.789] resolved	[400, ∞] right censored	S 0.9742 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.8267 [0.8226, 0.8307] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_076_S0_H3_candidates_common_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.337, 4.337] resolved	[400, ∞] right censored	S 0.9893 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7561 [0.727, 0.7851] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_077_S0_H3_candidates_common_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.018 [1.002, 1.042] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8815, low 0; forced mean no	1.178 [1.031, 1.372] N 12/12; valid 1	1.007 [1.003, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.651 [1.505, 1.778] N 12/12; valid 1	1.012 [1.007, 1.017] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_078_S0_H3_fingerprints_common_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3779, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9141, low 0; forced mean no	2.309 [2.253, 2.364] N 12/12; valid 1	1.023 [1.014, 1.033] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.506 [2.388, 2.633] N 2/2; valid 1	1.112 [1.078, 1.147] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.459 [2.368, 2.554] N 2/2; valid 1	1.286 [1.259, 1.314] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.009 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.032] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.112 [1.103, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9412 [0.9057, 0.9772] N 12/12; valid 1	1.593 [1.418, 1.838] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7942 [0.7712, 0.8197] N 12/12; valid 1	3.393 [3.313, 3.462] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_079_S0_H3_fingerprints_common_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0.08333; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7682 [0.7542, 0.7843] N 12/12; valid 1	2.601 [2.511, 2.698] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.01228, low 0; forced mean yes	1.429 [1.342, 1.51] N 12/12; valid 1	0.4705 [0.4524, 0.4893] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.381 [1.271, 1.485] N 12/12; valid 1	0.4573 [0.4425, 0.4726] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.219 [1.139, 1.304] N 12/12; valid 1	0.4295 [0.4176, 0.4423] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.6434 [0.6143, 0.6742] N 2/2; valid 1	0.4216 [0.4195, 0.4237] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.04676 [0, 0.09559] N 2/2; valid 1	0.4164 [0.3826, 0.4562] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_080_S0_H3_fingerprints_common_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.179 [1.032, 1.363] N 12/12; valid 1	1.012 [1.011, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.351 [1.211, 1.499] N 12/12; valid 1	1.029 [1.025, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8559, low 0; forced mean no	1.771 [1.622, 1.937] N 12/12; valid 1	1.049 [1.044, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.019 [1.911, 2.12] N 12/12; valid 1	1.073 [1.066, 1.08] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7088, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8021, low 0; forced mean no	2.151 [2.065, 2.239] N 12/12; valid 1	1.135 [1.119, 1.15] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_081_S0_H3_fingerprints_common_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.496 [2.397, 2.6] N 2/2; valid 1	1.265 [1.254, 1.276] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.185 [2.056, 2.315] N 2/2; valid 1	1.6 [1.54, 1.663] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1.001] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1.001 [1, 1.003] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.019 [1.002, 1.049] N 12/12; valid 1	1.005 [1.002, 1.007] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.128 [1.082, 1.177] N 12/12; valid 1	1.001 [0.995, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.312 [1.274, 1.352] N 12/12; valid 1	1 [0.9924, 1.009] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_082_S0_H3_fingerprints_common_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1632, low 0; forced mean no	no eligible check single front; H 3200 multi 0.4919, low 0; forced mean no	1.418 [1.371, 1.465] N 12/12; valid 1	1.003 [0.9936, 1.013] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.569 [1.521, 1.612] N 12/12; valid 1	1.006 [0.989, 1.022] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5564, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7449, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5985, low 0; forced mean no	1.7 [1.675, 1.726] N 12/12; valid 1	1.039 [1.017, 1.061] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.738 [1.693, 1.782] N 2/2; valid 1	1.17 [1.097, 1.253] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.723 [1.693, 1.752] N 2/2; valid 1	1.271 [1.236, 1.306] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_083_S0_H3_fingerprints_common_analysis_first_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_084_S0_H3_pairs_common_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_085_S0_H3_pairs_common_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.04676 [0, 0.09559] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4164 [0.3826, 0.4562] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_086_S0_H3_pairs_common_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_087_S0_H3_pairs_common_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_088_S0_H3_pairs_common_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_089_S0_H3_pairs_common_analysis_first_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_090_S0_H3_pairs_common_analysis_first_half_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_091_S0_H3_pairs_common_analysis_first_half_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04676 [0, 0.09559] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4164 [0.3826, 0.4562] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_092_S0_H3_pairs_common_analysis_first_half_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.001666 [0, 0.003956] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.009786 [0.003748, 0.01713] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9823 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2636 [0.1409, 0.4316] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2591, 2591] resolved	[4000, ∞] right censored	S 0.9641 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6587 [0.5212, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[519.6, 519.6] resolved	[4000, ∞] right censored	S 0.9356 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9512 [0.9327, 0.9693] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[120.8, 120.8] resolved	[4000, ∞] right censored	S 0.9141 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9875 [0.9856, 0.9894] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[89.93, 89.93] resolved	[4000, ∞] right censored	S 0.921 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9603 [0.9557, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_093_S0_H3_candidates_common_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[445.8, 445.8] resolved	[4000, ∞] right censored	S 0.261 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.08432 [0.07131, 0.09593] S low 0.8457 [0.8243, 0.8687] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[513, 513] resolved	[4000, ∞] right censored	S 0.1159 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06907 [0.05767, 0.08078] S low 0.898 [0.8794, 0.9162] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[778.2, 778.2] resolved	[4000, ∞] right censored	S 0.05398 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02712 [0.01973, 0.03409] S low 0.9539 [0.9411, 0.9668] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1169, 1169] resolved	[4000, ∞] right censored	S 0.02487 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.001562 [0, 0.003123] S low 0.9984 [0.9969, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_094_S0_H3_candidates_common_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[4000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0007808 [0, 0.002238] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9646 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2101 [0.01353, 0.4176] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[3164, 3164] resolved	[4000, ∞] right censored	S 0.9536 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.5997 [0.4581, 0.7432] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2436, 2436] resolved	[4000, ∞] right censored	S 0.9774 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.729 [0.6299, 0.8188] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[932.3, 932.3] resolved	[4000, ∞] right censored	S 0.9844 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8754 [0.8284, 0.9147] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[382.3, 382.3] resolved	[4000, ∞] right censored	S 0.9625 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9132 [0.8915, 0.9359] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[110.7, 110.7] resolved	[4000, ∞] right censored	S 0.9823 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9279 [0.9269, 0.9288] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_095_S0_H3_candidates_common_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[53.4, 53.4] resolved	[4000, ∞] right censored	S 0.972 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8407 [0.8289, 0.8526] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0006246 [0, 0.00177] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 0.995 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02649 [0.00406, 0.05945] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[199, 199] resolved	[400, ∞] right censored	S 0.9903 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.13 [0.08468, 0.182] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[42.25, 42.25] resolved	[400, ∞] right censored	S 0.9823 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3384 [0.2924, 0.388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.06, 18.06] resolved	[400, ∞] right censored	S 0.9805 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.5139 [0.4664, 0.562] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.07, 11.07] resolved	[400, ∞] right censored	S 0.9809 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6522 [0.6354, 0.669] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.896, 6.896] resolved	[400, ∞] right censored	S 0.9739 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7807 [0.759, 0.8005] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.19, 5.19] resolved	[400, ∞] right censored	S 0.9749 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.8532 [0.8438, 0.8626] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_096_S0_H3_candidates_common_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[3.881, 3.881] resolved	[400, ∞] right censored	S 0.9912 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7801 [0.7683, 0.792] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_097_S0_H3_candidates_common_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.003 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.004 [1, 1.009] N 12/12; valid 1	1.006 [1.005, 1.008] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8543, low 0; forced mean no	1.246 [1.112, 1.416] N 12/12; valid 1	1.01 [1.007, 1.014] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.685 [1.497, 1.846] N 12/12; valid 1	1.007 [1.001, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_098_S0_H3_fingerprints_common_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3933, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9126, low 0; forced mean no	2.31 [2.228, 2.389] N 12/12; valid 1	1.025 [1.017, 1.035] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.459 [2.364, 2.555] N 2/2; valid 1	1.137 [1.132, 1.143] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.556 [2.481, 2.633] N 2/2; valid 1	1.216 [1.208, 1.224] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.009] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.032] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.119 [1.114, 1.124] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9592, 0.9901] N 12/12; valid 1	1.39 [1.33, 1.457] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7884 [0.7767, 0.8] N 12/12; valid 1	3.299 [3.213, 3.39] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_099_S0_H3_fingerprints_common_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.03087; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1978; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7687 [0.7458, 0.7905] N 12/12; valid 1	2.631 [2.551, 2.704] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.00406, low 0; forced mean yes	1.478 [1.369, 1.585] N 12/12; valid 1	0.4817 [0.4616, 0.5027] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.316 [1.238, 1.397] N 12/12; valid 1	0.4644 [0.4506, 0.4779] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.193 [1.074, 1.325] N 12/12; valid 1	0.4449 [0.4296, 0.4626] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.6643 [0.5959, 0.7388] N 2/2; valid 1	0.4279 [0.4122, 0.4438] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.09375 [0.07432, 0.1143] N 2/2; valid 1	0.4292 [0.4084, 0.4522] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_100_S0_H3_fingerprints_common_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.208 [1.002, 1.431] N 12/12; valid 1	1.014 [1.013, 1.016] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.669 [1.476, 1.851] N 12/12; valid 1	1.028 [1.024, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.02295, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.77 [1.621, 1.919] N 12/12; valid 1	1.049 [1.042, 1.057] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.004 [1.889, 2.119] N 12/12; valid 1	1.079 [1.071, 1.088] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.6923, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8151, low 0; forced mean no	2.07 [1.996, 2.152] N 12/12; valid 1	1.134 [1.12, 1.148] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_101_S0_H3_fingerprints_common_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.218 [2.101, 2.339] N 2/2; valid 1	1.347 [1.313, 1.381] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.245 [2.098, 2.405] N 2/2; valid 1	1.787 [1.703, 1.878] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [0.9999, 1.001] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.023 [1.002, 1.056] N 12/12; valid 1	1.009 [1.005, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.113 [1.068, 1.163] N 12/12; valid 1	1.001 [0.9955, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.284 [1.242, 1.334] N 12/12; valid 1	0.9997 [0.9923, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_102_S0_H3_fingerprints_common_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1823, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5307, low 0; forced mean no	1.449 [1.399, 1.503] N 12/12; valid 1	0.9915 [0.9815, 1] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.564 [1.538, 1.588] N 12/12; valid 1	1.019 [0.9997, 1.038] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5634, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7431, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.61, low 0; forced mean no	1.702 [1.667, 1.739] N 12/12; valid 1	1.027 [1, 1.053] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.787 [1.716, 1.86] N 2/2; valid 1	1.173 [1.09, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.796 [1.762, 1.83] N 2/2; valid 1	1.23 [1.172, 1.291] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_103_S0_H3_fingerprints_common_analysis_second_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_104_S0_H3_pairs_common_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_105_S0_H3_pairs_common_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_106_S0_H3_pairs_common_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_107_S0_H3_pairs_common_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_108_S0_H3_pairs_common_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_109_S0_H3_pairs_common_analysis_second_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_110_S0_H3_pairs_common_analysis_second_half_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_111_S0_H3_pairs_common_analysis_second_half_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_112_S0_H3_pairs_common_analysis_second_half_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_113_S0_H3_pairs_common_analysis_second_half_primary_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.001666 [0, 0.003956] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.009786 [0.003748, 0.01713] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9823 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2636 [0.1409, 0.4316] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2591, 2591] resolved	[4000, ∞] right censored	S 0.9641 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6587 [0.5212, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[519.6, 519.6] resolved	[4000, ∞] right censored	S 0.9356 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9512 [0.9327, 0.9693] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[120.8, 120.8] resolved	[4000, ∞] right censored	S 0.9141 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9875 [0.9856, 0.9894] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[89.93, 89.93] resolved	[4000, ∞] right censored	S 0.921 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9603 [0.9557, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_114_S0_H3_candidates_common_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[445.8, 445.8] resolved	[4000, ∞] right censored	S 0.261 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.08432 [0.07131, 0.09593] S low 0.8457 [0.8243, 0.8687] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[513, 513] resolved	[4000, ∞] right censored	S 0.1159 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06907 [0.05767, 0.08078] S low 0.898 [0.8794, 0.9162] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[778.2, 778.2] resolved	[4000, ∞] right censored	S 0.05398 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02712 [0.01973, 0.03409] S low 0.9539 [0.9411, 0.9668] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1169, 1169] resolved	[4000, ∞] right censored	S 0.02487 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.001562 [0, 0.003123] S low 0.9984 [0.9969, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_115_S0_H3_candidates_common_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[4000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0007808 [0, 0.002238] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[4000, ∞] right censored	S 0.9646 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2101 [0.01353, 0.4176] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[3164, 3164] resolved	[4000, ∞] right censored	S 0.9536 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.5997 [0.4581, 0.7432] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2436, 2436] resolved	[4000, ∞] right censored	S 0.9774 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.729 [0.6299, 0.8188] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[932.3, 932.3] resolved	[4000, ∞] right censored	S 0.9844 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8754 [0.8284, 0.9147] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[382.3, 382.3] resolved	[4000, ∞] right censored	S 0.9625 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9132 [0.8915, 0.9359] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[110.7, 110.7] resolved	[4000, ∞] right censored	S 0.9823 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9279 [0.9269, 0.9288] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_116_S0_H3_candidates_common_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[53.4, 53.4] resolved	[4000, ∞] right censored	S 0.972 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8407 [0.8289, 0.8526] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0006246 [0, 0.00177] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[400, ∞] right censored	[400, ∞] right censored	S 0.995 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02649 [0.00406, 0.05945] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[199, 199] resolved	[400, ∞] right censored	S 0.9903 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.13 [0.08468, 0.182] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[42.25, 42.25] resolved	[400, ∞] right censored	S 0.9823 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3384 [0.2924, 0.388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.06, 18.06] resolved	[400, ∞] right censored	S 0.9805 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.5139 [0.4664, 0.562] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.07, 11.07] resolved	[400, ∞] right censored	S 0.9809 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6522 [0.6354, 0.669] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.896, 6.896] resolved	[400, ∞] right censored	S 0.9739 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7807 [0.759, 0.8005] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.19, 5.19] resolved	[400, ∞] right censored	S 0.9749 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.8532 [0.8438, 0.8626] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_117_S0_H3_candidates_common_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[3.881, 3.881] resolved	[400, ∞] right censored	S 0.9912 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7801 [0.7683, 0.792] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_118_S0_H3_candidates_common_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.003 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.004 [1, 1.009] N 12/12; valid 1	1.006 [1.005, 1.008] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8543, low 0; forced mean no	1.246 [1.112, 1.416] N 12/12; valid 1	1.01 [1.007, 1.014] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.685 [1.497, 1.846] N 12/12; valid 1	1.007 [1.001, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_119_S0_H3_fingerprints_common_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3933, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9126, low 0; forced mean no	2.31 [2.228, 2.389] N 12/12; valid 1	1.025 [1.017, 1.035] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.459 [2.364, 2.555] N 2/2; valid 1	1.137 [1.132, 1.143] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.556 [2.481, 2.633] N 2/2; valid 1	1.216 [1.208, 1.224] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.009] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.032] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.119 [1.114, 1.124] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9592, 0.9901] N 12/12; valid 1	1.39 [1.33, 1.457] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7884 [0.7767, 0.8] N 12/12; valid 1	3.299 [3.213, 3.39] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_120_S0_H3_fingerprints_common_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.03087; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1978; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7687 [0.7458, 0.7905] N 12/12; valid 1	2.631 [2.551, 2.704] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.00406, low 0; forced mean yes	1.478 [1.369, 1.585] N 12/12; valid 1	0.4817 [0.4616, 0.5027] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.316 [1.238, 1.397] N 12/12; valid 1	0.4644 [0.4506, 0.4779] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.193 [1.074, 1.325] N 12/12; valid 1	0.4449 [0.4296, 0.4626] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.6643 [0.5959, 0.7388] N 2/2; valid 1	0.4279 [0.4122, 0.4438] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.09375 [0.07432, 0.1143] N 2/2; valid 1	0.4292 [0.4084, 0.4522] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_121_S0_H3_fingerprints_common_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.208 [1.002, 1.431] N 12/12; valid 1	1.014 [1.013, 1.016] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.669 [1.476, 1.851] N 12/12; valid 1	1.028 [1.024, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.02295, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.77 [1.621, 1.919] N 12/12; valid 1	1.049 [1.042, 1.057] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.004 [1.889, 2.119] N 12/12; valid 1	1.079 [1.071, 1.088] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.6923, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8151, low 0; forced mean no	2.07 [1.996, 2.152] N 12/12; valid 1	1.134 [1.12, 1.148] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_122_S0_H3_fingerprints_common_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.218 [2.101, 2.339] N 2/2; valid 1	1.347 [1.313, 1.381] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.245 [2.098, 2.405] N 2/2; valid 1	1.787 [1.703, 1.878] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [0.9999, 1.001] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.023 [1.002, 1.056] N 12/12; valid 1	1.009 [1.005, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.113 [1.068, 1.163] N 12/12; valid 1	1.001 [0.9955, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.284 [1.242, 1.334] N 12/12; valid 1	0.9997 [0.9923, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_123_S0_H3_fingerprints_common_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1823, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5307, low 0; forced mean no	1.449 [1.399, 1.503] N 12/12; valid 1	0.9915 [0.9815, 1] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.564 [1.538, 1.588] N 12/12; valid 1	1.019 [0.9997, 1.038] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5634, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7431, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.61, low 0; forced mean no	1.702 [1.667, 1.739] N 12/12; valid 1	1.027 [1, 1.053] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.787 [1.716, 1.86] N 2/2; valid 1	1.173 [1.09, 1.268] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.796 [1.762, 1.83] N 2/2; valid 1	1.23 [1.172, 1.291] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_124_S0_H3_fingerprints_common_analysis_second_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_125_S0_H3_pairs_common_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_126_S0_H3_pairs_common_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.09375 [0.07432, 0.1143] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4292 [0.4084, 0.4522] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_127_S0_H3_pairs_common_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_128_S0_H3_pairs_common_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_129_S0_H3_pairs_common_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_130_S0_H3_pairs_common_analysis_second_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_131_S0_H3_pairs_common_analysis_second_half_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_132_S0_H3_pairs_common_analysis_second_half_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_133_S0_H3_pairs_common_analysis_second_half_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.09375 [0.07432, 0.1143] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4292 [0.4084, 0.4522] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_134_S0_H3_pairs_common_analysis_second_half_inclusive_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.002081 [0, 0.006242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.009155 [0.001144, 0.01987] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9757 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2917 [0.1244, 0.5021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9902 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6283 [0.4561, 0.7684] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[523.6, 523.6] resolved	[2000, ∞] right censored	S 0.9389 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9597 [0.9421, 0.9774] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[113.6, 113.6] resolved	[2000, ∞] right censored	S 0.9101 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.985 [0.9825, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[90.81, 90.81] resolved	[2000, ∞] right censored	S 0.9227 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9638 [0.9563, 0.9713] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_135_S0_H3_candidates_common_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[450, 450] resolved	[2000, ∞] right censored	S 0.2955 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0748 [0.0594, 0.09124] S low 0.8537 [0.8408, 0.8707] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[428, 428] resolved	[2000, ∞] right censored	S 0.128 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0671 [0.05452, 0.07886] S low 0.9087 [0.892, 0.926] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[978, 978] resolved	[2000, ∞] right censored	S 0.05124 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.03069 [0.01987, 0.04203] S low 0.9511 [0.9319, 0.9681] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[2000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_136_S0_H3_candidates_common_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[2000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9919 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.162 [0.01007, 0.3558] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9537 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.5145 [0.3285, 0.706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9877 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6821 [0.5204, 0.8242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	inconclusive	inconclusive	[1347, 1347] resolved	[2000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.8719 [0.8317, 0.9134] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[367.8, 367.8] resolved	[2000, ∞] right censored	S 0.9576 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9181 [0.8925, 0.944] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[97.62, 97.62] resolved	[2000, ∞] right censored	S 0.9923 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9295 [0.9201, 0.9388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_137_S0_H3_candidates_common_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[58.16, 58.16] resolved	[2000, ∞] right censored	S 0.9759 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8102 [0.7928, 0.8277] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.00104 [0, 0.003121] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 0.99 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.04359 [0.00437, 0.1033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	inconclusive	inconclusive	[179.1, 179.1] resolved	[200, ∞] right censored	S 0.9928 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.1256 [0.0723, 0.1927] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[48.74, 48.74] resolved	[200, ∞] right censored	S 0.9833 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3343 [0.27, 0.4022] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.09, 18.09] resolved	[200, ∞] right censored	S 0.9776 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.533 [0.4707, 0.6013] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.29, 11.29] resolved	[200, ∞] right censored	S 0.9828 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6471 [0.6185, 0.6756] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.988, 6.988] resolved	[200, ∞] right censored	S 0.9724 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7762 [0.7507, 0.7998] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.513, 5.513] resolved	[200, ∞] right censored	S 0.9747 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.829 [0.7903, 0.8677] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_138_S0_H3_candidates_common_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[3.991, 3.991] resolved	[200, ∞] right censored	S 0.9924 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7709 [0.7191, 0.8227] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_139_S0_H3_candidates_common_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [0.9999, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [0.9999, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.002 [1, 1.005] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.005 [1, 1.012] N 12/12; valid 1	1.008 [1.005, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8688, low 0; forced mean no	1.29 [1.103, 1.521] N 12/12; valid 1	1.011 [1.006, 1.015] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.641 [1.428, 1.853] N 12/12; valid 1	1.008 [0.9981, 1.018] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_140_S0_H3_fingerprints_common_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3682, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9189, low 0; forced mean no	2.312 [2.243, 2.38] N 12/12; valid 1	1.027 [1.017, 1.04] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.442 [2.348, 2.54] N 2/2; valid 1	1.143 [1.112, 1.175] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.617 [2.484, 2.75] N 2/2; valid 1	1.187 [1.123, 1.253] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.028, 1.033] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.107, 1.125] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9419, 1] N 12/12; valid 1	1.39 [1.313, 1.511] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7926 [0.7663, 0.8219] N 12/12; valid 1	3.225 [3.093, 3.362] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_141_S0_H3_fingerprints_common_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.06169; forced mean no	no eligible check single front; H 32000 multi 0, low 0.2398; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7782 [0.7472, 0.8097] N 12/12; valid 1	2.615 [2.542, 2.694] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.007074, low 0; forced mean yes	1.452 [1.346, 1.561] N 12/12; valid 1	0.472 [0.4473, 0.4987] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.317 [1.203, 1.428] N 12/12; valid 1	0.4675 [0.45, 0.4849] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.191 [1.04, 1.369] N 12/12; valid 1	0.4325 [0.4136, 0.4551] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.5441 [0.4932, 0.6032] N 2/2; valid 1	0.4432 [0.4413, 0.4451] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.03546 [0.02778, 0.04348] N 2/2; valid 1	0.4206 [0.3952, 0.4478] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_142_S0_H3_fingerprints_common_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.009 [1.008, 1.01] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.161 [0.9983, 1.358] N 12/12; valid 1	1.015 [1.012, 1.018] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.614 [1.355, 1.89] N 12/12; valid 1	1.028 [1.022, 1.035] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8859, low 0; forced mean no	1.678 [1.477, 1.875] N 12/12; valid 1	1.05 [1.041, 1.059] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	1.975 [1.841, 2.108] N 12/12; valid 1	1.081 [1.069, 1.093] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7209, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8092, low 0; forced mean no	2.112 [2.008, 2.218] N 12/12; valid 1	1.132 [1.114, 1.153] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_143_S0_H3_fingerprints_common_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.363 [2.228, 2.5] N 2/2; valid 1	1.379 [1.31, 1.45] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.164 [2.06, 2.279] N 2/2; valid 1	1.827 [1.787, 1.867] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.038 [1, 1.107] N 12/12; valid 1	1.009 [1.003, 1.014] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.107 [1.053, 1.173] N 12/12; valid 1	1.004 [0.9956, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.27 [1.203, 1.352] N 12/12; valid 1	0.9951 [0.9836, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_144_S0_H3_fingerprints_common_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1828, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5309, low 0; forced mean no	1.473 [1.403, 1.553] N 12/12; valid 1	0.9857 [0.9721, 0.9981] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.548 [1.522, 1.573] N 12/12; valid 1	1.025 [1.001, 1.048] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5624, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6064, low 0; forced mean no	1.677 [1.64, 1.716] N 12/12; valid 1	1.023 [0.9881, 1.059] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.743 [1.686, 1.8] N 2/2; valid 1	1.153 [1.086, 1.221] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.752 [1.686, 1.82] N 2/2; valid 1	1.277 [1.102, 1.491] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_145_S0_H3_fingerprints_common_late_quarter_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_146_S0_H3_pairs_common_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_147_S0_H3_pairs_common_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_148_S0_H3_pairs_common_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_149_S0_H3_pairs_common_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_150_S0_H3_pairs_common_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_151_S0_H3_pairs_common_late_quarter_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_152_S0_H3_pairs_common_late_quarter_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_153_S0_H3_pairs_common_late_quarter_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_154_S0_H3_pairs_common_late_quarter_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_155_S0_H3_pairs_common_late_quarter_primary_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.002081 [0, 0.006242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.009155 [0.001144, 0.01987] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9757 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2917 [0.1244, 0.5021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9902 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6283 [0.4561, 0.7684] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[523.6, 523.6] resolved	[2000, ∞] right censored	S 0.9389 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9597 [0.9421, 0.9774] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[113.6, 113.6] resolved	[2000, ∞] right censored	S 0.9101 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.985 [0.9825, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[90.81, 90.81] resolved	[2000, ∞] right censored	S 0.9227 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9638 [0.9563, 0.9713] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_156_S0_H3_candidates_common_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[450, 450] resolved	[2000, ∞] right censored	S 0.2955 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0748 [0.0594, 0.09124] S low 0.8537 [0.8408, 0.8707] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[428, 428] resolved	[2000, ∞] right censored	S 0.128 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0671 [0.05452, 0.07886] S low 0.9087 [0.892, 0.926] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[978, 978] resolved	[2000, ∞] right censored	S 0.05124 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.03069 [0.01987, 0.04203] S low 0.9511 [0.9319, 0.9681] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[2000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_157_S0_H3_candidates_common_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[2000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9919 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.162 [0.01007, 0.3558] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9537 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.5145 [0.3285, 0.706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2000, ∞] right censored	[2000, ∞] right censored	S 0.9877 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6821 [0.5204, 0.8242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	inconclusive	inconclusive	[1347, 1347] resolved	[2000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.8719 [0.8317, 0.9134] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[367.8, 367.8] resolved	[2000, ∞] right censored	S 0.9576 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9181 [0.8925, 0.944] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[97.62, 97.62] resolved	[2000, ∞] right censored	S 0.9923 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9295 [0.9201, 0.9388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_158_S0_H3_candidates_common_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[58.16, 58.16] resolved	[2000, ∞] right censored	S 0.9759 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8102 [0.7928, 0.8277] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.00104 [0, 0.003121] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[200, ∞] right censored	[200, ∞] right censored	S 0.99 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.04359 [0.00437, 0.1033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	inconclusive	inconclusive	[179.1, 179.1] resolved	[200, ∞] right censored	S 0.9928 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.1256 [0.0723, 0.1927] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[48.74, 48.74] resolved	[200, ∞] right censored	S 0.9833 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3343 [0.27, 0.4022] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.09, 18.09] resolved	[200, ∞] right censored	S 0.9776 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.533 [0.4707, 0.6013] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.29, 11.29] resolved	[200, ∞] right censored	S 0.9828 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6471 [0.6185, 0.6756] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.988, 6.988] resolved	[200, ∞] right censored	S 0.9724 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7762 [0.7507, 0.7998] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.513, 5.513] resolved	[200, ∞] right censored	S 0.9747 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.829 [0.7903, 0.8677] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_159_S0_H3_candidates_common_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[3.991, 3.991] resolved	[200, ∞] right censored	S 0.9924 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7709 [0.7191, 0.8227] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_160_S0_H3_candidates_common_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [0.9999, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [0.9999, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.002 [1, 1.005] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.005 [1, 1.012] N 12/12; valid 1	1.008 [1.005, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8688, low 0; forced mean no	1.29 [1.103, 1.521] N 12/12; valid 1	1.011 [1.006, 1.015] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.641 [1.428, 1.853] N 12/12; valid 1	1.008 [0.9981, 1.018] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_161_S0_H3_fingerprints_common_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3682, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9189, low 0; forced mean no	2.312 [2.243, 2.38] N 12/12; valid 1	1.027 [1.017, 1.04] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.442 [2.348, 2.54] N 2/2; valid 1	1.143 [1.112, 1.175] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.617 [2.484, 2.75] N 2/2; valid 1	1.187 [1.123, 1.253] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.028, 1.033] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.107, 1.125] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9419, 1] N 12/12; valid 1	1.39 [1.313, 1.511] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7926 [0.7663, 0.8219] N 12/12; valid 1	3.225 [3.093, 3.362] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_162_S0_H3_fingerprints_common_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.06169; forced mean no	no eligible check single front; H 32000 multi 0, low 0.2398; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7782 [0.7472, 0.8097] N 12/12; valid 1	2.615 [2.542, 2.694] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.007074, low 0; forced mean yes	1.452 [1.346, 1.561] N 12/12; valid 1	0.472 [0.4473, 0.4987] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.317 [1.203, 1.428] N 12/12; valid 1	0.4675 [0.45, 0.4849] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.191 [1.04, 1.369] N 12/12; valid 1	0.4325 [0.4136, 0.4551] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.5441 [0.4932, 0.6032] N 2/2; valid 1	0.4432 [0.4413, 0.4451] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.03546 [0.02778, 0.04348] N 2/2; valid 1	0.4206 [0.3952, 0.4478] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_163_S0_H3_fingerprints_common_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.009 [1.008, 1.01] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.161 [0.9983, 1.358] N 12/12; valid 1	1.015 [1.012, 1.018] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.614 [1.355, 1.89] N 12/12; valid 1	1.028 [1.022, 1.035] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8859, low 0; forced mean no	1.678 [1.477, 1.875] N 12/12; valid 1	1.05 [1.041, 1.059] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	1.975 [1.841, 2.108] N 12/12; valid 1	1.081 [1.069, 1.093] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7209, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8092, low 0; forced mean no	2.112 [2.008, 2.218] N 12/12; valid 1	1.132 [1.114, 1.153] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_164_S0_H3_fingerprints_common_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.363 [2.228, 2.5] N 2/2; valid 1	1.379 [1.31, 1.45] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.164 [2.06, 2.279] N 2/2; valid 1	1.827 [1.787, 1.867] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.038 [1, 1.107] N 12/12; valid 1	1.009 [1.003, 1.014] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.107 [1.053, 1.173] N 12/12; valid 1	1.004 [0.9956, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.27 [1.203, 1.352] N 12/12; valid 1	0.9951 [0.9836, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_165_S0_H3_fingerprints_common_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1828, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5309, low 0; forced mean no	1.473 [1.403, 1.553] N 12/12; valid 1	0.9857 [0.9721, 0.9981] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.548 [1.522, 1.573] N 12/12; valid 1	1.025 [1.001, 1.048] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5624, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6064, low 0; forced mean no	1.677 [1.64, 1.716] N 12/12; valid 1	1.023 [0.9881, 1.059] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.743 [1.686, 1.8] N 2/2; valid 1	1.153 [1.086, 1.221] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.752 [1.686, 1.82] N 2/2; valid 1	1.277 [1.102, 1.491] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_166_S0_H3_fingerprints_common_late_quarter_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_167_S0_H3_pairs_common_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_168_S0_H3_pairs_common_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.03546 [0.02778, 0.04348] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4206 [0.3952, 0.4478] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_169_S0_H3_pairs_common_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_170_S0_H3_pairs_common_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_171_S0_H3_pairs_common_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_172_S0_H3_pairs_common_late_quarter_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_173_S0_H3_pairs_common_late_quarter_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_174_S0_H3_pairs_common_late_quarter_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_175_S0_H3_pairs_common_late_quarter_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03546 [0.02778, 0.04348] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4206 [0.3952, 0.4478] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_176_S0_H3_pairs_common_late_quarter_inclusive_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009369 [0, 0.002186] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01712 [0.007027, 0.0304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9892 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.235 [0.1245, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9798 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6387 [0.5516, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.9, 505.9] resolved	[8000, ∞] right censored	S 0.9422 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9548 [0.9448, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.3, 128.3] resolved	[8000, ∞] right censored	S 0.9207 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.9844, 0.9881] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[83.27, 83.27] resolved	[8000, ∞] right censored	S 0.9319 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9606 [0.9582, 0.9631] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_177_S0_H3_candidates_common_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[475.4, 475.4] resolved	[8000, ∞] right censored	S 0.2373 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07594 [0.06543, 0.08484] S low 0.8565 [0.8397, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.6, 563.6] resolved	[8000, ∞] right censored	S 0.131 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06283 [0.05564, 0.07001] S low 0.9004 [0.8879, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[733.2, 733.2] resolved	[8000, ∞] right censored	S 0.05288 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0266 [0.02082, 0.03206] S low 0.9579 [0.9503, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1251, 1251] resolved	[8000, ∞] right censored	S 0.01525 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004997 [0.001874, 0.00812] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_178_S0_H3_candidates_common_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004164 [0, 0.001145] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.979 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1969 [0.03435, 0.3848] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3903, 3903] resolved	[8000, ∞] right censored	S 0.9796 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4837 [0.3914, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[8000, ∞] right censored	S 0.985 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7161 [0.6474, 0.7783] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.9, 986.9] resolved	[8000, ∞] right censored	S 0.9749 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.857 [0.832, 0.8798] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.5, 388.5] resolved	[8000, ∞] right censored	S 0.9627 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.919 [0.9007, 0.9373] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104, 104] resolved	[8000, ∞] right censored	S 0.9616 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9335 [0.9332, 0.9338] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_179_S0_H3_candidates_common_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[51.19, 51.19] resolved	[8000, ∞] right censored	S 0.9708 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.852 [0.8463, 0.8576] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0007287 [5.205e-05, 0.001666] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.5, 533.5] resolved	[800, ∞] right censored	S 0.9981 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02363 [0.006558, 0.04497] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[800, ∞] right censored	S 0.9922 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1366 [0.108, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.9, 37.9] resolved	[800, ∞] right censored	S 0.9847 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3533 [0.3188, 0.3849] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[800, ∞] right censored	S 0.9823 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4987 [0.4698, 0.5309] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.66, 10.66] resolved	[800, ∞] right censored	S 0.9803 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.658 [0.6436, 0.6719] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.808, 6.808] resolved	[800, ∞] right censored	S 0.9744 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7891 [0.7747, 0.8021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.953, 4.953] resolved	[800, ∞] right censored	S 0.9749 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.842 [0.8407, 0.8432] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_180_S0_H3_candidates_common_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.101, 4.101] resolved	[800, ∞] right censored	S 0.9887 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7633 [0.7395, 0.787] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_181_S0_H3_candidates_common_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8682, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_182_S0_H3_fingerprints_common_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3856, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9124, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_183_S0_H3_fingerprints_common_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01546; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008224, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_184_S0_H3_fingerprints_common_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01145, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8082, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_185_S0_H3_fingerprints_common_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_186_S0_H3_fingerprints_common_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1726, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5119, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5596, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7444, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6049, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_187_S0_H3_fingerprints_common_cadence_x2_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_188_S0_H3_pairs_common_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_189_S0_H3_pairs_common_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_190_S0_H3_pairs_common_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_191_S0_H3_pairs_common_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_192_S0_H3_pairs_common_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_193_S0_H3_pairs_common_cadence_x2_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_194_S0_H3_pairs_common_cadence_x2_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_195_S0_H3_pairs_common_cadence_x2_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_196_S0_H3_pairs_common_cadence_x2_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009369 [0, 0.002186] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01712 [0.007027, 0.0304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9892 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.235 [0.1245, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9798 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6387 [0.5516, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.9, 505.9] resolved	[8000, ∞] right censored	S 0.9422 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9548 [0.9448, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.3, 128.3] resolved	[8000, ∞] right censored	S 0.9207 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.9844, 0.9881] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[83.27, 83.27] resolved	[8000, ∞] right censored	S 0.9319 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9606 [0.9582, 0.9631] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_197_S0_H3_candidates_common_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[475.4, 475.4] resolved	[8000, ∞] right censored	S 0.2373 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07594 [0.06543, 0.08484] S low 0.8565 [0.8397, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.6, 563.6] resolved	[8000, ∞] right censored	S 0.131 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06283 [0.05564, 0.07001] S low 0.9004 [0.8879, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[733.2, 733.2] resolved	[8000, ∞] right censored	S 0.05288 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.0266 [0.02082, 0.03206] S low 0.9579 [0.9503, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1251, 1251] resolved	[8000, ∞] right censored	S 0.01525 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004997 [0.001874, 0.00812] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_198_S0_H3_candidates_common_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004164 [0, 0.001145] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.979 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1969 [0.03435, 0.3848] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3903, 3903] resolved	[8000, ∞] right censored	S 0.9796 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4837 [0.3914, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[8000, ∞] right censored	S 0.985 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7161 [0.6474, 0.7783] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.9, 986.9] resolved	[8000, ∞] right censored	S 0.9749 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.857 [0.832, 0.8798] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.5, 388.5] resolved	[8000, ∞] right censored	S 0.9627 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.919 [0.9007, 0.9373] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104, 104] resolved	[8000, ∞] right censored	S 0.9616 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9335 [0.9332, 0.9338] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_199_S0_H3_candidates_common_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[51.19, 51.19] resolved	[8000, ∞] right censored	S 0.9708 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.852 [0.8463, 0.8576] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0007287 [5.205e-05, 0.001666] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.5, 533.5] resolved	[800, ∞] right censored	S 0.9981 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02363 [0.006558, 0.04497] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[800, ∞] right censored	S 0.9922 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1366 [0.108, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.9, 37.9] resolved	[800, ∞] right censored	S 0.9847 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3533 [0.3188, 0.3849] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[800, ∞] right censored	S 0.9823 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4987 [0.4698, 0.5309] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.66, 10.66] resolved	[800, ∞] right censored	S 0.9803 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.658 [0.6436, 0.6719] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.808, 6.808] resolved	[800, ∞] right censored	S 0.9744 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7891 [0.7747, 0.8021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.953, 4.953] resolved	[800, ∞] right censored	S 0.9749 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.842 [0.8407, 0.8432] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_200_S0_H3_candidates_common_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.101, 4.101] resolved	[800, ∞] right censored	S 0.9887 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7633 [0.7395, 0.787] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_201_S0_H3_candidates_common_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8682, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_202_S0_H3_fingerprints_common_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3856, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9124, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_203_S0_H3_fingerprints_common_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01546; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008224, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_204_S0_H3_fingerprints_common_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01145, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8082, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_205_S0_H3_fingerprints_common_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_206_S0_H3_fingerprints_common_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1726, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5119, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5596, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7444, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6049, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_207_S0_H3_fingerprints_common_cadence_x2_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_208_S0_H3_pairs_common_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_209_S0_H3_pairs_common_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_210_S0_H3_pairs_common_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_211_S0_H3_pairs_common_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_212_S0_H3_pairs_common_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_213_S0_H3_pairs_common_cadence_x2_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_214_S0_H3_pairs_common_cadence_x2_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_215_S0_H3_pairs_common_cadence_x2_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_216_S0_H3_pairs_common_cadence_x2_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009363 [0, 0.002185] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01737 [0.007176, 0.03059] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.989 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2348 [0.1243, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9804 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6395 [0.5528, 0.7213] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.7, 504.7] resolved	[8000, ∞] right censored	S 0.9445 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.954 [0.9441, 0.964] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.6, 128.6] resolved	[8000, ∞] right censored	S 0.9195 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.985, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[84, 84] resolved	[8000, ∞] right censored	S 0.9341 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9638 [0.96, 0.9675] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_217_S0_H3_candidates_common_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[478, 478] resolved	[8000, ∞] right censored	S 0.2386 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07563 [0.06471, 0.0849] S low 0.8562 [0.8387, 0.8753] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.3, 563.3] resolved	[8000, ∞] right censored	S 0.13 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06149 [0.05379, 0.06898] S low 0.9012 [0.8878, 0.915] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[736.8, 736.8] resolved	[8000, ∞] right censored	S 0.05301 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02643 [0.0205, 0.03215] S low 0.9581 [0.9501, 0.9672] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1264, 1264] resolved	[8000, ∞] right censored	S 0.01493 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.005618 [0.002497, 0.008739] S low 0.9919 [0.9863, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_218_S0_H3_candidates_common_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004161 [0, 0.001248] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9788 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1964 [0.03391, 0.3844] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3907, 3907] resolved	[8000, ∞] right censored	S 0.9785 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4842 [0.3916, 0.5754] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2013, 2013] resolved	[8000, ∞] right censored	S 0.9844 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7165 [0.6475, 0.7785] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[985.7, 985.7] resolved	[8000, ∞] right censored	S 0.975 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8579 [0.8328, 0.8807] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[390.9, 390.9] resolved	[8000, ∞] right censored	S 0.9621 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9184 [0.9004, 0.9365] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.4, 104.4] resolved	[8000, ∞] right censored	S 0.9597 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9332 [0.9313, 0.9351] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_219_S0_H3_candidates_common_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[52.68, 52.68] resolved	[8000, ∞] right censored	S 0.9712 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8508 [0.8452, 0.8564] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0006242 [0, 0.001457] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.5, 533.5] resolved	[800, ∞] right censored	S 0.998 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02362 [0.006656, 0.04484] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[800, ∞] right censored	S 0.9919 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1363 [0.1071, 0.1705] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.89, 37.89] resolved	[800, ∞] right censored	S 0.9848 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3548 [0.3199, 0.3867] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.15, 18.15] resolved	[800, ∞] right censored	S 0.983 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4988 [0.4698, 0.5304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.72, 10.72] resolved	[800, ∞] right censored	S 0.9789 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6572 [0.6426, 0.6712] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.891, 6.891] resolved	[800, ∞] right censored	S 0.9746 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7887 [0.7737, 0.8026] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.058, 5.058] resolved	[800, ∞] right censored	S 0.9724 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.8452 [0.8427, 0.8477] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_220_S0_H3_candidates_common_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.068, 4.068] resolved	[800, ∞] right censored	S 0.9862 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7665 [0.7403, 0.7928] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_221_S0_H3_candidates_common_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8685, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_222_S0_H3_fingerprints_common_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3855, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9143, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_223_S0_H3_fingerprints_common_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.0155; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008219, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_224_S0_H3_fingerprints_common_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01144, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8553, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8081, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_225_S0_H3_fingerprints_common_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_226_S0_H3_fingerprints_common_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1728, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5112, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5605, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.743, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6073, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_227_S0_H3_fingerprints_common_cadence_x4_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_228_S0_H3_pairs_common_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_229_S0_H3_pairs_common_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_230_S0_H3_pairs_common_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_231_S0_H3_pairs_common_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_232_S0_H3_pairs_common_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_233_S0_H3_pairs_common_cadence_x4_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_234_S0_H3_pairs_common_cadence_x4_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_235_S0_H3_pairs_common_cadence_x4_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_236_S0_H3_pairs_common_cadence_x4_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0009363 [0, 0.002185] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01737 [0.007176, 0.03059] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.989 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2348 [0.1243, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[8000, ∞] right censored	S 0.9804 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6395 [0.5528, 0.7213] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.7, 504.7] resolved	[8000, ∞] right censored	S 0.9445 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.954 [0.9441, 0.964] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.6, 128.6] resolved	[8000, ∞] right censored	S 0.9195 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9863 [0.985, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[84, 84] resolved	[8000, ∞] right censored	S 0.9341 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9638 [0.96, 0.9675] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_237_S0_H3_candidates_common_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[478, 478] resolved	[8000, ∞] right censored	S 0.2386 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07563 [0.06471, 0.0849] S low 0.8562 [0.8387, 0.8753] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.3, 563.3] resolved	[8000, ∞] right censored	S 0.13 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06149 [0.05379, 0.06898] S low 0.9012 [0.8878, 0.915] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[736.8, 736.8] resolved	[8000, ∞] right censored	S 0.05301 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02643 [0.0205, 0.03215] S low 0.9581 [0.9501, 0.9672] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1264, 1264] resolved	[8000, ∞] right censored	S 0.01493 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.005618 [0.002497, 0.008739] S low 0.9919 [0.9863, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_238_S0_H3_candidates_common_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004161 [0, 0.001248] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9788 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1964 [0.03391, 0.3844] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3907, 3907] resolved	[8000, ∞] right censored	S 0.9785 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4842 [0.3916, 0.5754] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2013, 2013] resolved	[8000, ∞] right censored	S 0.9844 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.7165 [0.6475, 0.7785] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[985.7, 985.7] resolved	[8000, ∞] right censored	S 0.975 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8579 [0.8328, 0.8807] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[390.9, 390.9] resolved	[8000, ∞] right censored	S 0.9621 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9184 [0.9004, 0.9365] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.4, 104.4] resolved	[8000, ∞] right censored	S 0.9597 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9332 [0.9313, 0.9351] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_239_S0_H3_candidates_common_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[52.68, 52.68] resolved	[8000, ∞] right censored	S 0.9712 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8508 [0.8452, 0.8564] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0006242 [0, 0.001457] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[533.5, 533.5] resolved	[800, ∞] right censored	S 0.998 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02362 [0.006656, 0.04484] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[800, ∞] right censored	S 0.9919 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1363 [0.1071, 0.1705] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.89, 37.89] resolved	[800, ∞] right censored	S 0.9848 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3548 [0.3199, 0.3867] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.15, 18.15] resolved	[800, ∞] right censored	S 0.983 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4988 [0.4698, 0.5304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.72, 10.72] resolved	[800, ∞] right censored	S 0.9789 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6572 [0.6426, 0.6712] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.891, 6.891] resolved	[800, ∞] right censored	S 0.9746 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7887 [0.7737, 0.8026] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.058, 5.058] resolved	[800, ∞] right censored	S 0.9724 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.8452 [0.8427, 0.8477] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_240_S0_H3_candidates_common_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.068, 4.068] resolved	[800, ∞] right censored	S 0.9862 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7665 [0.7403, 0.7928] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_241_S0_H3_candidates_common_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8685, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_242_S0_H3_fingerprints_common_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3855, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9143, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_243_S0_H3_fingerprints_common_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.0155; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008219, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_244_S0_H3_fingerprints_common_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01144, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8553, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8081, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_245_S0_H3_fingerprints_common_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_246_S0_H3_fingerprints_common_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1728, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5112, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5605, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.743, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6073, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_247_S0_H3_fingerprints_common_cadence_x4_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_248_S0_H3_pairs_common_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_249_S0_H3_pairs_common_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_250_S0_H3_pairs_common_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_251_S0_H3_pairs_common_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_252_S0_H3_pairs_common_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_253_S0_H3_pairs_common_cadence_x4_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_254_S0_H3_pairs_common_cadence_x4_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_255_S0_H3_pairs_common_cadence_x4_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_256_S0_H3_pairs_common_cadence_x4_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.001247 [0, 0.002702] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01808 [0.008105, 0.03096] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9888 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2344 [0.1236, 0.3753] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2702, 2702] resolved	[8000, ∞] right censored	S 0.98 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6394 [0.5521, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.9, 504.9] resolved	[8000, ∞] right censored	S 0.9444 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.953 [0.9426, 0.9634] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[130.7, 130.7] resolved	[8000, ∞] right censored	S 0.9131 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.99 [0.985, 0.995] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[82.42, 82.42] resolved	[8000, ∞] right censored	S 0.9286 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.9663 [0.9626, 0.9701] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_257_S0_H3_candidates_common_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[479.3, 479.3] resolved	[8000, ∞] right censored	S 0.2363 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07461 [0.06234, 0.08541] S low 0.8585 [0.841, 0.8786] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[556.9, 556.9] resolved	[8000, ∞] right censored	S 0.1307 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06234 [0.05486, 0.07004] S low 0.9009 [0.8872, 0.9156] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[767.4, 767.4] resolved	[8000, ∞] right censored	S 0.05192 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02702 [0.02078, 0.03304] S low 0.9572 [0.9493, 0.9663] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1269, 1269] resolved	[8000, ∞] right censored	S 0.01491 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004988 [0.002494, 0.007481] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_258_S0_H3_candidates_common_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004156 [0, 0.001247] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9797 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1966 [0.0349, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3906, 3906] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4823 [0.3896, 0.5738] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2015, 2015] resolved	[8000, ∞] right censored	S 0.9841 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.718 [0.6498, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[981.7, 981.7] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8603 [0.8356, 0.8832] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[392.1, 392.1] resolved	[8000, ∞] right censored	S 0.9591 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9175 [0.899, 0.936] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.9, 104.9] resolved	[8000, ∞] right censored	S 0.9648 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9302 [0.9252, 0.9352] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_259_S0_H3_candidates_common_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[0, 80] unresolved below two cadences	[8000, ∞] right censored	S 0.9724 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.8354 [0.8204, 0.8504] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0008313 [0, 0.00187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[534.3, 534.3] resolved	[800, ∞] right censored	S 0.9978 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02328 [0.006234, 0.04489] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146.2, 146.2] resolved	[800, ∞] right censored	S 0.9913 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1357 [0.1062, 0.1704] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.83, 37.83] resolved	[800, ∞] right censored	S 0.984 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3554 [0.3215, 0.3874] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.25, 18.25] resolved	[800, ∞] right censored	S 0.9838 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4985 [0.4701, 0.5299] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.83, 10.83] resolved	[800, ∞] right censored	S 0.9799 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6586 [0.6424, 0.6735] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9745 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.787 [0.7695, 0.802] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9599 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.8628 [0.8454, 0.8803] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_260_S0_H3_candidates_common_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9925 / W 1 H 3200 / 3200 95% bound check: ?	S multi 0.7793 [0.7456, 0.813] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_261_S0_H3_candidates_common_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.868, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_262_S0_H3_fingerprints_common_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3859, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9092, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_263_S0_H3_fingerprints_common_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01559; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008105, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_264_S0_H3_fingerprints_common_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01143, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8543, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7014, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8103, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_265_S0_H3_fingerprints_common_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_266_S0_H3_fingerprints_common_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5106, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5576, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6027, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_267_S0_H3_fingerprints_common_cadence_x8_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_268_S0_H3_pairs_common_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_269_S0_H3_pairs_common_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_270_S0_H3_pairs_common_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_271_S0_H3_pairs_common_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_272_S0_H3_pairs_common_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_273_S0_H3_pairs_common_cadence_x8_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_274_S0_H3_pairs_common_cadence_x8_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_275_S0_H3_pairs_common_cadence_x8_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_276_S0_H3_pairs_common_cadence_x8_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.001247 [0, 0.002702] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.01808 [0.008105, 0.03096] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9888 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.2344 [0.1236, 0.3753] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2702, 2702] resolved	[8000, ∞] right censored	S 0.98 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.6394 [0.5521, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.9, 504.9] resolved	[8000, ∞] right censored	S 0.9444 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.953 [0.9426, 0.9634] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[130.7, 130.7] resolved	[8000, ∞] right censored	S 0.9131 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.99 [0.985, 0.995] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[82.42, 82.42] resolved	[8000, ∞] right censored	S 0.9286 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9663 [0.9626, 0.9701] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_277_S0_H3_candidates_common_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[479.3, 479.3] resolved	[8000, ∞] right censored	S 0.2363 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.07461 [0.06234, 0.08541] S low 0.8585 [0.841, 0.8786] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[556.9, 556.9] resolved	[8000, ∞] right censored	S 0.1307 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.06234 [0.05486, 0.07004] S low 0.9009 [0.8872, 0.9156] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[767.4, 767.4] resolved	[8000, ∞] right censored	S 0.05192 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.02702 [0.02078, 0.03304] S low 0.9572 [0.9493, 0.9663] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1269, 1269] resolved	[8000, ∞] right censored	S 0.01491 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0.004988 [0.002494, 0.007481] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_278_S0_H3_candidates_common_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 32000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.0004156 [0, 0.001247] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9797 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.1966 [0.0349, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3906, 3906] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: N	S multi 0.4823 [0.3896, 0.5738] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2015, 2015] resolved	[8000, ∞] right censored	S 0.9841 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.718 [0.6498, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[981.7, 981.7] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8603 [0.8356, 0.8832] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[392.1, 392.1] resolved	[8000, ∞] right censored	S 0.9591 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9175 [0.899, 0.936] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.9, 104.9] resolved	[8000, ∞] right censored	S 0.9648 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.9302 [0.9252, 0.9352] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_279_S0_H3_candidates_common_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[0, 80] unresolved below two cadences	[8000, ∞] right censored	S 0.9724 / W 1 H 32000 / 32000 95% bound check: Y	S multi 0.8354 [0.8204, 0.8504] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.0008313 [0, 0.00187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[534.3, 534.3] resolved	[800, ∞] right censored	S 0.9978 / W 1 H 3200 / 3200 95% bound check: N	S multi 0.02328 [0.006234, 0.04489] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146.2, 146.2] resolved	[800, ∞] right censored	S 0.9913 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.1357 [0.1062, 0.1704] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.83, 37.83] resolved	[800, ∞] right censored	S 0.984 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.3554 [0.3215, 0.3874] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.25, 18.25] resolved	[800, ∞] right censored	S 0.9838 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.4985 [0.4701, 0.5299] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.83, 10.83] resolved	[800, ∞] right censored	S 0.9799 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.6586 [0.6424, 0.6735] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9745 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.787 [0.7695, 0.802] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9599 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.8628 [0.8454, 0.8803] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_280_S0_H3_candidates_common_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · common · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[0, 8] unresolved below two cadences	[800, ∞] right censored	S 0.9925 / W 1 H 3200 / 3200 95% bound check: Y	S multi 0.7793 [0.7456, 0.813] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_281_S0_H3_candidates_common_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.868, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_282_S0_H3_fingerprints_common_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3859, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9092, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1.008 [1.007, 1.01] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.03 [1.029, 1.031] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.121] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.345 [3.282, 3.41] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_283_S0_H3_fingerprints_common_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01559; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7685 [0.7559, 0.7812] N 12/12; valid 1	2.616 [2.548, 2.681] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008105, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07067 [0.05517, 0.08696] N 2/2; valid 1	0.4228 [0.3957, 0.4538] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_284_S0_H3_fingerprints_common_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01143, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8543, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7014, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8103, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_285_S0_H3_fingerprints_common_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 2/2; valid 1	1 [1, 1] N 2/2; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_286_S0_H3_fingerprints_common_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · common · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5106, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5576, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6027, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_287_S0_H3_fingerprints_common_cadence_x8_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_288_S0_H3_pairs_common_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_289_S0_H3_pairs_common_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07067 [0.05517, 0.08696] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4228 [0.3957, 0.4538] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_290_S0_H3_pairs_common_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_291_S0_H3_pairs_common_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_292_S0_H3_pairs_common_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_293_S0_H3_pairs_common_cadence_x8_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_294_S0_H3_pairs_common_cadence_x8_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_295_S0_H3_pairs_common_cadence_x8_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · common · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07067 [0.05517, 0.08696] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4228 [0.3957, 0.4538] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_296_S0_H3_pairs_common_cadence_x8_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [0.0002929, 0.00222] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0171 [0.007002, 0.03033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9893 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2347 [0.124, 0.3757] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9795 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6389 [0.5518, 0.7217] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.5, 505.5] resolved	[3.2e+04, ∞] right censored	S 0.9425 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9545 [0.9444, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[127.9, 127.9] resolved	[3.2e+04, ∞] right censored	S 0.9218 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.9853, 0.9872] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[83.26, 83.26] resolved	[3.2e+04, ∞] right censored	S 0.9328 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9613 [0.9603, 0.9622] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_297_S0_H3_candidates_longest_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9574 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.003893 [0, 0.01091] S low 0.03678 [0, 0.1046] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[474.5, 474.5] resolved	[3.2e+04, ∞] right censored	S 0.2376 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07688 [0.06599, 0.08612] S low 0.856 [0.8388, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[564.4, 564.4] resolved	[3.2e+04, ∞] right censored	S 0.1314 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0624 [0.05545, 0.06938] S low 0.9004 [0.8883, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[734.3, 734.3] resolved	[3.2e+04, ∞] right censored	S 0.05307 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02684 [0.02106, 0.03265] S low 0.9579 [0.9502, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1257, 1257] resolved	[3.2e+04, ∞] right censored	S 0.01543 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004842 [0.001874, 0.00781] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_298_S0_H3_candidates_longest_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001367 [0, 0.0004101] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0003905 [0, 0.001119] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9789 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1968 [0.0342, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3901, 3901] resolved	[3.2e+04, ∞] right censored	S 0.9798 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4833 [0.3906, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[3.2e+04, ∞] right censored	S 0.985 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7162 [0.6475, 0.7784] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.5, 986.5] resolved	[3.2e+04, ∞] right censored	S 0.9754 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8569 [0.8322, 0.8799] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.1, 388.1] resolved	[3.2e+04, ∞] right censored	S 0.9626 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9196 [0.901, 0.9378] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[103.8, 103.8] resolved	[3.2e+04, ∞] right censored	S 0.964 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9311 [0.9288, 0.9335] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_299_S0_H3_candidates_longest_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[51.37, 51.37] resolved	[3.2e+04, ∞] right censored	S 0.9693 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8552 [0.8485, 0.8619] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.000781 [7.81e-05, 0.001744] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.4, 533.4] resolved	[3200, ∞] right censored	S 0.9981 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02353 [0.006378, 0.04493] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146, 146] resolved	[3200, ∞] right censored	S 0.9921 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1367 [0.1076, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.92, 37.92] resolved	[3200, ∞] right censored	S 0.9845 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3529 [0.3182, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[3200, ∞] right censored	S 0.9828 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4992 [0.4707, 0.531] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.65, 10.65] resolved	[3200, ∞] right censored	S 0.9792 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6586 [0.6433, 0.6732] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.794, 6.794] resolved	[3200, ∞] right censored	S 0.9745 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7892 [0.7759, 0.8008] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.964, 4.964] resolved	[3200, ∞] right censored	S 0.9746 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.8399 [0.8372, 0.8425] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_300_S0_H3_candidates_longest_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.121, 4.121] resolved	[3200, ∞] right censored	S 0.9903 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.768 [0.7476, 0.7885] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_301_S0_H3_candidates_longest_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8679, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_302_S0_H3_fingerprints_longest_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3857, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9133, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_303_S0_H3_fingerprints_longest_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01544; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008175, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_304_S0_H3_fingerprints_longest_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01148, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8567, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7005, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8086, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_305_S0_H3_fingerprints_longest_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_306_S0_H3_fingerprints_longest_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5114, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5599, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.744, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6043, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_307_S0_H3_fingerprints_longest_main_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_308_S0_H3_pairs_longest_main_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_309_S0_H3_pairs_longest_main_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_310_S0_H3_pairs_longest_main_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_311_S0_H3_pairs_longest_main_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_312_S0_H3_pairs_longest_main_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_313_S0_H3_pairs_longest_main_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_314_S0_H3_pairs_longest_main_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_315_S0_H3_pairs_longest_main_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_316_S0_H3_pairs_longest_main_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [0.0002929, 0.00222] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0171 [0.007002, 0.03033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9893 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2347 [0.124, 0.3757] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9795 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6389 [0.5518, 0.7217] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.5, 505.5] resolved	[3.2e+04, ∞] right censored	S 0.9425 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9545 [0.9444, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[127.9, 127.9] resolved	[3.2e+04, ∞] right censored	S 0.9218 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.9853, 0.9872] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[83.26, 83.26] resolved	[3.2e+04, ∞] right censored	S 0.9328 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9613 [0.9603, 0.9622] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_317_S0_H3_candidates_longest_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9574 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.003893 [0, 0.01091] S low 0.03678 [0, 0.1046] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[474.5, 474.5] resolved	[3.2e+04, ∞] right censored	S 0.2376 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07688 [0.06599, 0.08612] S low 0.856 [0.8388, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[564.4, 564.4] resolved	[3.2e+04, ∞] right censored	S 0.1314 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0624 [0.05545, 0.06938] S low 0.9004 [0.8883, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[734.3, 734.3] resolved	[3.2e+04, ∞] right censored	S 0.05307 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02684 [0.02106, 0.03265] S low 0.9579 [0.9502, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1257, 1257] resolved	[3.2e+04, ∞] right censored	S 0.01543 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004842 [0.001874, 0.00781] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_318_S0_H3_candidates_longest_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001367 [0, 0.0004101] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0003905 [0, 0.001119] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9789 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1968 [0.0342, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3901, 3901] resolved	[3.2e+04, ∞] right censored	S 0.9798 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4833 [0.3906, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[3.2e+04, ∞] right censored	S 0.985 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7162 [0.6475, 0.7784] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.5, 986.5] resolved	[3.2e+04, ∞] right censored	S 0.9754 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8569 [0.8322, 0.8799] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.1, 388.1] resolved	[3.2e+04, ∞] right censored	S 0.9626 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9196 [0.901, 0.9378] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[103.8, 103.8] resolved	[3.2e+04, ∞] right censored	S 0.964 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9311 [0.9288, 0.9335] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_319_S0_H3_candidates_longest_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[51.37, 51.37] resolved	[3.2e+04, ∞] right censored	S 0.9693 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8552 [0.8485, 0.8619] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.000781 [7.81e-05, 0.001744] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.4, 533.4] resolved	[3200, ∞] right censored	S 0.9981 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02353 [0.006378, 0.04493] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146, 146] resolved	[3200, ∞] right censored	S 0.9921 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1367 [0.1076, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.92, 37.92] resolved	[3200, ∞] right censored	S 0.9845 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3529 [0.3182, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[3200, ∞] right censored	S 0.9828 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4992 [0.4707, 0.531] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.65, 10.65] resolved	[3200, ∞] right censored	S 0.9792 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6586 [0.6433, 0.6732] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.794, 6.794] resolved	[3200, ∞] right censored	S 0.9745 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7892 [0.7759, 0.8008] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.964, 4.964] resolved	[3200, ∞] right censored	S 0.9746 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.8399 [0.8372, 0.8425] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_320_S0_H3_candidates_longest_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · main · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.121, 4.121] resolved	[3200, ∞] right censored	S 0.9903 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.768 [0.7476, 0.7885] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_321_S0_H3_candidates_longest_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8679, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_322_S0_H3_fingerprints_longest_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3857, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9133, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_323_S0_H3_fingerprints_longest_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01544; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008175, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_324_S0_H3_fingerprints_longest_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01148, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8567, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7005, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8086, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_325_S0_H3_fingerprints_longest_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_326_S0_H3_fingerprints_longest_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · main · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5114, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5599, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.744, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6043, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_327_S0_H3_fingerprints_longest_main_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_328_S0_H3_pairs_longest_main_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_329_S0_H3_pairs_longest_main_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_330_S0_H3_pairs_longest_main_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_331_S0_H3_pairs_longest_main_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_332_S0_H3_pairs_longest_main_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_333_S0_H3_pairs_longest_main_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_334_S0_H3_pairs_longest_main_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_335_S0_H3_pairs_longest_main_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · main · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_336_S0_H3_pairs_longest_main_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [2.604e-05, 0.002956] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.02441 [0.006141, 0.0519] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9964 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2057 [0.0673, 0.3891] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2704, 2704] resolved	[1.6e+04, ∞] right censored	S 0.9964 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6192 [0.5258, 0.7165] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[487, 487] resolved	[1.6e+04, ∞] right censored	S 0.9483 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9579 [0.946, 0.9681] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[132.3, 132.3] resolved	[1.6e+04, ∞] right censored	S 0.9308 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.985 [0.985, 0.985] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[77.8, 77.8] resolved	[1.6e+04, ∞] right censored	S 0.9445 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9622 [0.9594, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_337_S0_H3_candidates_longest_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[540.8, 540.8] resolved	[1.6e+04, ∞] right censored	S 0.2132 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06949 [0.05278, 0.08584] S low 0.8661 [0.844, 0.8911] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[606, 606] resolved	[1.6e+04, ∞] right censored	S 0.1347 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.05569 [0.04679, 0.06673] S low 0.9029 [0.8896, 0.9143] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[748.5, 748.5] resolved	[1.6e+04, ∞] right censored	S 0.05051 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0266 [0.01764, 0.03618] S low 0.9619 [0.9517, 0.9717] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable lag gap	[1.6e+04, ∞] right censored	S 0.0003448 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.00812 [0.003748, 0.01249] S low 0.9866 [0.9781, 0.995] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_338_S0_H3_candidates_longest_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[1.6e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0002734 [0, 0.0008202] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9932 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1835 [0.03965, 0.375] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3999, 3999] resolved	[1.6e+04, ∞] right censored	S 0.998 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.367 [0.2444, 0.4962] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[1827, 1827] resolved	[1.6e+04, ∞] right censored	S 0.9906 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7034 [0.6057, 0.7941] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[975.5, 975.5] resolved	[1.6e+04, ∞] right censored	S 0.9642 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8384 [0.7979, 0.8804] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[397.5, 397.5] resolved	[1.6e+04, ∞] right censored	S 0.9622 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9261 [0.9032, 0.9492] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[98.02, 98.02] resolved	[1.6e+04, ∞] right censored	S 0.9469 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9341 [0.9288, 0.9394] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_339_S0_H3_candidates_longest_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[49.28, 49.28] resolved	[1.6e+04, ∞] right censored	S 0.9661 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8695 [0.8682, 0.8707] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0009369 [0, 0.002655] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[347.3, 347.3] resolved	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02056 [0.002342, 0.05127] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[134.2, 134.2] resolved	[1600, ∞] right censored	S 0.9921 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1433 [0.09447, 0.193] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[34.54, 34.54] resolved	[1600, ∞] right censored	S 0.9857 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3674 [0.3309, 0.4097] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.22, 18.22] resolved	[1600, ∞] right censored	S 0.9847 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4844 [0.4473, 0.5205] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.31, 10.31] resolved	[1600, ∞] right censored	S 0.9774 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6649 [0.6391, 0.6876] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.701, 6.701] resolved	[1600, ∞] right censored	S 0.975 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7977 [0.7819, 0.814] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.789, 4.789] resolved	[1600, ∞] right censored	S 0.9742 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.8267 [0.8226, 0.8307] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_340_S0_H3_candidates_longest_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.337, 4.337] resolved	[1600, ∞] right censored	S 0.9893 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7561 [0.727, 0.7851] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_341_S0_H3_candidates_longest_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.001, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.018 [1.002, 1.042] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8815, low 0; forced mean no	1.178 [1.031, 1.372] N 12/12; valid 1	1.007 [1.003, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.651 [1.505, 1.778] N 12/12; valid 1	1.012 [1.007, 1.017] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_342_S0_H3_fingerprints_longest_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3779, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9141, low 0; forced mean no	2.309 [2.253, 2.364] N 12/12; valid 1	1.023 [1.014, 1.033] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.506 [2.388, 2.633] N 2/2; valid 1	1.112 [1.078, 1.147] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.459 [2.368, 2.554] N 2/2; valid 1	1.286 [1.259, 1.314] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.009] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.116 [1.113, 1.119] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9412 [0.9057, 0.9772] N 12/12; valid 1	1.593 [1.418, 1.838] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7942 [0.7712, 0.8197] N 12/12; valid 1	3.39 [3.312, 3.459] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_343_S0_H3_fingerprints_longest_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0.08333; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7682 [0.7542, 0.7843] N 12/12; valid 1	2.646 [2.553, 2.746] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.01228, low 0; forced mean yes	1.429 [1.342, 1.51] N 12/12; valid 1	0.4705 [0.4524, 0.4893] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.381 [1.271, 1.485] N 12/12; valid 1	0.4573 [0.4425, 0.4726] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.219 [1.139, 1.304] N 12/12; valid 1	0.4295 [0.4176, 0.4423] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.6434 [0.6143, 0.6742] N 2/2; valid 1	0.4216 [0.4195, 0.4237] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.04648 [0, 0.09419] N 12/2; valid 1	0.4256 [0.4173, 0.4334] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_344_S0_H3_fingerprints_longest_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.179 [1.032, 1.363] N 12/12; valid 1	1.012 [1.011, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.351 [1.211, 1.499] N 12/12; valid 1	1.029 [1.025, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8559, low 0; forced mean no	1.771 [1.622, 1.937] N 12/12; valid 1	1.049 [1.044, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.019 [1.911, 2.12] N 12/12; valid 1	1.073 [1.066, 1.08] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7088, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8021, low 0; forced mean no	2.151 [2.065, 2.239] N 12/12; valid 1	1.135 [1.119, 1.15] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_345_S0_H3_fingerprints_longest_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.496 [2.397, 2.6] N 2/2; valid 1	1.265 [1.254, 1.276] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.185 [2.056, 2.315] N 2/2; valid 1	1.6 [1.54, 1.663] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1.001 [1, 1.003] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.019 [1.002, 1.049] N 12/12; valid 1	1.005 [1.002, 1.007] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.128 [1.082, 1.177] N 12/12; valid 1	1.001 [0.995, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.312 [1.274, 1.352] N 12/12; valid 1	1 [0.9924, 1.009] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_346_S0_H3_fingerprints_longest_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1632, low 0; forced mean no	no eligible check single front; H 3200 multi 0.4919, low 0; forced mean no	1.418 [1.371, 1.465] N 12/12; valid 1	1.003 [0.9936, 1.013] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.569 [1.521, 1.612] N 12/12; valid 1	1.006 [0.989, 1.022] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5564, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7449, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5985, low 0; forced mean no	1.7 [1.675, 1.726] N 12/12; valid 1	1.039 [1.017, 1.061] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.738 [1.693, 1.782] N 2/2; valid 1	1.17 [1.097, 1.253] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.723 [1.693, 1.752] N 2/2; valid 1	1.271 [1.236, 1.306] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_347_S0_H3_fingerprints_longest_analysis_first_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_348_S0_H3_pairs_longest_analysis_first_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_349_S0_H3_pairs_longest_analysis_first_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_350_S0_H3_pairs_longest_analysis_first_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_351_S0_H3_pairs_longest_analysis_first_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_352_S0_H3_pairs_longest_analysis_first_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_353_S0_H3_pairs_longest_analysis_first_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_354_S0_H3_pairs_longest_analysis_first_half_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_355_S0_H3_pairs_longest_analysis_first_half_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_356_S0_H3_pairs_longest_analysis_first_half_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [2.604e-05, 0.002956] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.02441 [0.006141, 0.0519] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9964 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2057 [0.0673, 0.3891] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2704, 2704] resolved	[1.6e+04, ∞] right censored	S 0.9964 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6192 [0.5258, 0.7165] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[487, 487] resolved	[1.6e+04, ∞] right censored	S 0.9483 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9579 [0.946, 0.9681] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[132.3, 132.3] resolved	[1.6e+04, ∞] right censored	S 0.9308 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.985 [0.985, 0.985] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[77.8, 77.8] resolved	[1.6e+04, ∞] right censored	S 0.9445 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9622 [0.9594, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_357_S0_H3_candidates_longest_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[540.8, 540.8] resolved	[1.6e+04, ∞] right censored	S 0.2132 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06949 [0.05278, 0.08584] S low 0.8661 [0.844, 0.8911] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[606, 606] resolved	[1.6e+04, ∞] right censored	S 0.1347 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.05569 [0.04679, 0.06673] S low 0.9029 [0.8896, 0.9143] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[748.5, 748.5] resolved	[1.6e+04, ∞] right censored	S 0.05051 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0266 [0.01764, 0.03618] S low 0.9619 [0.9517, 0.9717] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable lag gap	[1.6e+04, ∞] right censored	S 0.0003448 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.00812 [0.003748, 0.01249] S low 0.9866 [0.9781, 0.995] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_358_S0_H3_candidates_longest_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[1.6e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0002734 [0, 0.0008202] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9932 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1835 [0.03965, 0.375] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3999, 3999] resolved	[1.6e+04, ∞] right censored	S 0.998 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.367 [0.2444, 0.4962] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[1827, 1827] resolved	[1.6e+04, ∞] right censored	S 0.9906 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7034 [0.6057, 0.7941] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[975.5, 975.5] resolved	[1.6e+04, ∞] right censored	S 0.9642 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8384 [0.7979, 0.8804] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[397.5, 397.5] resolved	[1.6e+04, ∞] right censored	S 0.9622 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9261 [0.9032, 0.9492] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[98.02, 98.02] resolved	[1.6e+04, ∞] right censored	S 0.9469 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9341 [0.9288, 0.9394] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_359_S0_H3_candidates_longest_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[49.28, 49.28] resolved	[1.6e+04, ∞] right censored	S 0.9661 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8695 [0.8682, 0.8707] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0009369 [0, 0.002655] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[347.3, 347.3] resolved	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02056 [0.002342, 0.05127] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[134.2, 134.2] resolved	[1600, ∞] right censored	S 0.9921 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1433 [0.09447, 0.193] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[34.54, 34.54] resolved	[1600, ∞] right censored	S 0.9857 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3674 [0.3309, 0.4097] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.22, 18.22] resolved	[1600, ∞] right censored	S 0.9847 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4844 [0.4473, 0.5205] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.31, 10.31] resolved	[1600, ∞] right censored	S 0.9774 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6649 [0.6391, 0.6876] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.701, 6.701] resolved	[1600, ∞] right censored	S 0.975 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7977 [0.7819, 0.814] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.789, 4.789] resolved	[1600, ∞] right censored	S 0.9742 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.8267 [0.8226, 0.8307] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_360_S0_H3_candidates_longest_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis first half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.337, 4.337] resolved	[1600, ∞] right censored	S 0.9893 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7561 [0.727, 0.7851] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_361_S0_H3_candidates_longest_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.001, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.018 [1.002, 1.042] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8815, low 0; forced mean no	1.178 [1.031, 1.372] N 12/12; valid 1	1.007 [1.003, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.651 [1.505, 1.778] N 12/12; valid 1	1.012 [1.007, 1.017] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_362_S0_H3_fingerprints_longest_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3779, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9141, low 0; forced mean no	2.309 [2.253, 2.364] N 12/12; valid 1	1.023 [1.014, 1.033] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.506 [2.388, 2.633] N 2/2; valid 1	1.112 [1.078, 1.147] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.459 [2.368, 2.554] N 2/2; valid 1	1.286 [1.259, 1.314] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.009] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.116 [1.113, 1.119] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9412 [0.9057, 0.9772] N 12/12; valid 1	1.593 [1.418, 1.838] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7942 [0.7712, 0.8197] N 12/12; valid 1	3.39 [3.312, 3.459] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_363_S0_H3_fingerprints_longest_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0.08333; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.7682 [0.7542, 0.7843] N 12/12; valid 1	2.646 [2.553, 2.746] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.01228, low 0; forced mean yes	1.429 [1.342, 1.51] N 12/12; valid 1	0.4705 [0.4524, 0.4893] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.381 [1.271, 1.485] N 12/12; valid 1	0.4573 [0.4425, 0.4726] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.219 [1.139, 1.304] N 12/12; valid 1	0.4295 [0.4176, 0.4423] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.6434 [0.6143, 0.6742] N 2/2; valid 1	0.4216 [0.4195, 0.4237] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.04648 [0, 0.09419] N 12/2; valid 1	0.4256 [0.4173, 0.4334] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_364_S0_H3_fingerprints_longest_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.179 [1.032, 1.363] N 12/12; valid 1	1.012 [1.011, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.351 [1.211, 1.499] N 12/12; valid 1	1.029 [1.025, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8559, low 0; forced mean no	1.771 [1.622, 1.937] N 12/12; valid 1	1.049 [1.044, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.019 [1.911, 2.12] N 12/12; valid 1	1.073 [1.066, 1.08] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7088, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8021, low 0; forced mean no	2.151 [2.065, 2.239] N 12/12; valid 1	1.135 [1.119, 1.15] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_365_S0_H3_fingerprints_longest_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.496 [2.397, 2.6] N 2/2; valid 1	1.265 [1.254, 1.276] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.185 [2.056, 2.315] N 2/2; valid 1	1.6 [1.54, 1.663] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1.001 [1, 1.003] N 12/12; valid 1	1.002 [1.001, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.019 [1.002, 1.049] N 12/12; valid 1	1.005 [1.002, 1.007] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.128 [1.082, 1.177] N 12/12; valid 1	1.001 [0.995, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.312 [1.274, 1.352] N 12/12; valid 1	1 [0.9924, 1.009] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_366_S0_H3_fingerprints_longest_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis first half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1632, low 0; forced mean no	no eligible check single front; H 3200 multi 0.4919, low 0; forced mean no	1.418 [1.371, 1.465] N 12/12; valid 1	1.003 [0.9936, 1.013] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.569 [1.521, 1.612] N 12/12; valid 1	1.006 [0.989, 1.022] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5564, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7449, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5985, low 0; forced mean no	1.7 [1.675, 1.726] N 12/12; valid 1	1.039 [1.017, 1.061] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.738 [1.693, 1.782] N 2/2; valid 1	1.17 [1.097, 1.253] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.723 [1.693, 1.752] N 2/2; valid 1	1.271 [1.236, 1.306] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_367_S0_H3_fingerprints_longest_analysis_first_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.651 [1.505, 1.778] N 12/12; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.012 [1.007, 1.017] N 12/12; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_368_S0_H3_pairs_longest_analysis_first_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.253, 2.364] N 12/12; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.023 [1.014, 1.033] N 12/12; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.506 [2.388, 2.633] N 2/2; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.112 [1.078, 1.147] N 2/2; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.429 [1.342, 1.51] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4705 [0.4524, 0.4893] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_369_S0_H3_pairs_longest_analysis_first_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.381 [1.271, 1.485] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4573 [0.4425, 0.4726] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 1.219 [1.139, 1.304] N 12/12; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4295 [0.4176, 0.4423] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.6434 [0.6143, 0.6742] N 2/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4216 [0.4195, 0.4237] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.368, 2.554] N 2/2; valid 1 B 0.04648 [0, 0.09419] N 12/2; valid 1	A 1.286 [1.259, 1.314] N 2/2; valid 1 B 0.4256 [0.4173, 0.4334] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_370_S0_H3_pairs_longest_analysis_first_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.429 [1.342, 1.51] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4705 [0.4524, 0.4893] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_371_S0_H3_pairs_longest_analysis_first_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.381 [1.271, 1.485] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4573 [0.4425, 0.4726] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_372_S0_H3_pairs_longest_analysis_first_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.219 [1.139, 1.304] N 12/12; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4295 [0.4176, 0.4423] N 12/12; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_373_S0_H3_pairs_longest_analysis_first_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6434 [0.6143, 0.6742] N 2/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4216 [0.4195, 0.4237] N 2/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_374_S0_H3_pairs_longest_analysis_first_half_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.771 [1.622, 1.937] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.049 [1.044, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.019 [1.911, 2.12] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.073 [1.066, 1.08] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.151 [2.065, 2.239] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.135 [1.119, 1.15] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.496 [2.397, 2.6] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.265 [1.254, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 2.185 [2.056, 2.315] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.6 [1.54, 1.663] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.569 [1.521, 1.612] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.006 [0.989, 1.022] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.7 [1.675, 1.726] N 12/12; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.039 [1.017, 1.061] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.738 [1.693, 1.782] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.17 [1.097, 1.253] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_375_S0_H3_pairs_longest_analysis_first_half_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis first half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.04648 [0, 0.09419] N 12/2; valid 1 B 1.723 [1.693, 1.752] N 2/2; valid 1	A 0.4256 [0.4173, 0.4334] N 12/2; valid 1 B 1.271 [1.236, 1.306] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_376_S0_H3_pairs_longest_analysis_first_half_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [7.811e-05, 0.002591] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.009786 [0.003748, 0.01713] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9823 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2636 [0.1409, 0.4316] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2591, 2591] resolved	[1.6e+04, ∞] right censored	S 0.9641 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6587 [0.5212, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[519.6, 519.6] resolved	[1.6e+04, ∞] right censored	S 0.9356 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9512 [0.9327, 0.9693] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[120.8, 120.8] resolved	[1.6e+04, ∞] right censored	S 0.9141 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9875 [0.9856, 0.9894] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[89.93, 89.93] resolved	[1.6e+04, ∞] right censored	S 0.921 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9603 [0.9557, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_377_S0_H3_candidates_longest_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9282 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.007785 [0, 0.02182] S low 0.07356 [0, 0.2091] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[445.8, 445.8] resolved	[1.6e+04, ∞] right censored	S 0.261 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.08432 [0.07131, 0.09593] S low 0.8457 [0.8243, 0.8687] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[513, 513] resolved	[1.6e+04, ∞] right censored	S 0.1159 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06907 [0.05767, 0.08078] S low 0.898 [0.8794, 0.9162] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[778.2, 778.2] resolved	[1.6e+04, ∞] right censored	S 0.05398 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02712 [0.01973, 0.03409] S low 0.9539 [0.9411, 0.9668] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1169, 1169] resolved	[1.6e+04, ∞] right censored	S 0.02487 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.001562 [0, 0.003123] S low 0.9984 [0.9969, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_378_S0_H3_candidates_longest_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[1.6e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0007808 [0, 0.002238] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9646 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2101 [0.01353, 0.4176] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	criterion met	inconclusive	[3164, 3164] resolved	[1.6e+04, ∞] right censored	S 0.9536 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.5997 [0.4581, 0.7432] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2436, 2436] resolved	[1.6e+04, ∞] right censored	S 0.9774 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.729 [0.6299, 0.8188] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[932.3, 932.3] resolved	[1.6e+04, ∞] right censored	S 0.9844 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8754 [0.8284, 0.9147] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[382.3, 382.3] resolved	[1.6e+04, ∞] right censored	S 0.9625 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9132 [0.8915, 0.9359] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[110.7, 110.7] resolved	[1.6e+04, ∞] right censored	S 0.9823 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9279 [0.9269, 0.9288] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_379_S0_H3_candidates_longest_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[53.4, 53.4] resolved	[1.6e+04, ∞] right censored	S 0.972 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8407 [0.8289, 0.8526] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0006246 [0, 0.00177] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 0.995 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02649 [0.00406, 0.05945] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[199, 199] resolved	[1600, ∞] right censored	S 0.9903 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.13 [0.08468, 0.182] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[42.25, 42.25] resolved	[1600, ∞] right censored	S 0.9823 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3384 [0.2924, 0.388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.06, 18.06] resolved	[1600, ∞] right censored	S 0.9805 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.5139 [0.4664, 0.562] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.07, 11.07] resolved	[1600, ∞] right censored	S 0.9809 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6522 [0.6354, 0.669] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.896, 6.896] resolved	[1600, ∞] right censored	S 0.9739 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7807 [0.759, 0.8005] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.19, 5.19] resolved	[1600, ∞] right censored	S 0.9749 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.8532 [0.8438, 0.8626] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_380_S0_H3_candidates_longest_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[3.881, 3.881] resolved	[1600, ∞] right censored	S 0.9912 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7801 [0.7683, 0.792] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_381_S0_H3_candidates_longest_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.003 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.004 [1, 1.009] N 12/12; valid 1	1.006 [1.005, 1.008] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8543, low 0; forced mean no	1.246 [1.112, 1.416] N 12/12; valid 1	1.01 [1.007, 1.014] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.685 [1.497, 1.846] N 12/12; valid 1	1.007 [1.001, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_382_S0_H3_fingerprints_longest_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3933, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9126, low 0; forced mean no	2.31 [2.228, 2.389] N 12/12; valid 1	1.025 [1.017, 1.035] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.459 [2.364, 2.555] N 2/2; valid 1	1.137 [1.132, 1.143] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.556 [2.481, 2.633] N 2/2; valid 1	1.216 [1.208, 1.224] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.114 [1.111, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9592, 0.9901] N 12/12; valid 1	1.39 [1.33, 1.457] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7884 [0.7767, 0.8] N 12/12; valid 1	3.296 [3.212, 3.387] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_383_S0_H3_fingerprints_longest_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.03087; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1978; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8323 [0.7592, 0.954] N 12/12; valid 1	2.465 [2.105, 2.697] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.00406, low 0; forced mean yes	1.478 [1.369, 1.585] N 12/12; valid 1	0.4817 [0.4616, 0.5027] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.316 [1.238, 1.397] N 12/12; valid 1	0.4644 [0.4506, 0.4779] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.193 [1.074, 1.325] N 12/12; valid 1	0.4449 [0.4296, 0.4626] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.6643 [0.5959, 0.7388] N 2/2; valid 1	0.4279 [0.4122, 0.4438] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.0961 [0.07751, 0.1151] N 12/2; valid 1	0.44 [0.4345, 0.4466] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_384_S0_H3_fingerprints_longest_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.208 [1.002, 1.431] N 12/12; valid 1	1.014 [1.013, 1.016] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.669 [1.476, 1.851] N 12/12; valid 1	1.028 [1.024, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.02295, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.77 [1.621, 1.919] N 12/12; valid 1	1.049 [1.042, 1.057] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.004 [1.889, 2.119] N 12/12; valid 1	1.079 [1.071, 1.088] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.6923, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8151, low 0; forced mean no	2.07 [1.996, 2.152] N 12/12; valid 1	1.134 [1.12, 1.148] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_385_S0_H3_fingerprints_longest_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.218 [2.101, 2.339] N 2/2; valid 1	1.347 [1.313, 1.381] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.245 [2.098, 2.405] N 2/2; valid 1	1.787 [1.703, 1.878] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.023 [1.002, 1.056] N 12/12; valid 1	1.009 [1.005, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.113 [1.068, 1.163] N 12/12; valid 1	1.001 [0.9955, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.284 [1.242, 1.334] N 12/12; valid 1	0.9997 [0.9923, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_386_S0_H3_fingerprints_longest_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1823, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5307, low 0; forced mean no	1.449 [1.399, 1.503] N 12/12; valid 1	0.9915 [0.9815, 1] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.564 [1.538, 1.588] N 12/12; valid 1	1.019 [0.9997, 1.038] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5634, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7431, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.61, low 0; forced mean no	1.702 [1.667, 1.739] N 12/12; valid 1	1.027 [1, 1.053] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.787 [1.716, 1.86] N 2/2; valid 1	1.173 [1.09, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.796 [1.762, 1.83] N 2/2; valid 1	1.23 [1.172, 1.291] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_387_S0_H3_fingerprints_longest_analysis_second_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_388_S0_H3_pairs_longest_analysis_second_half_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_389_S0_H3_pairs_longest_analysis_second_half_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_390_S0_H3_pairs_longest_analysis_second_half_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_391_S0_H3_pairs_longest_analysis_second_half_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_392_S0_H3_pairs_longest_analysis_second_half_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_393_S0_H3_pairs_longest_analysis_second_half_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_394_S0_H3_pairs_longest_analysis_second_half_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_395_S0_H3_pairs_longest_analysis_second_half_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_396_S0_H3_pairs_longest_analysis_second_half_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_397_S0_H3_pairs_longest_analysis_second_half_primary_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001146 [7.811e-05, 0.002591] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.009786 [0.003748, 0.01713] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9823 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2636 [0.1409, 0.4316] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2591, 2591] resolved	[1.6e+04, ∞] right censored	S 0.9641 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6587 [0.5212, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[519.6, 519.6] resolved	[1.6e+04, ∞] right censored	S 0.9356 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9512 [0.9327, 0.9693] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[120.8, 120.8] resolved	[1.6e+04, ∞] right censored	S 0.9141 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9875 [0.9856, 0.9894] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[89.93, 89.93] resolved	[1.6e+04, ∞] right censored	S 0.921 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9603 [0.9557, 0.965] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_398_S0_H3_candidates_longest_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9282 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.007785 [0, 0.02182] S low 0.07356 [0, 0.2091] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[445.8, 445.8] resolved	[1.6e+04, ∞] right censored	S 0.261 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.08432 [0.07131, 0.09593] S low 0.8457 [0.8243, 0.8687] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[513, 513] resolved	[1.6e+04, ∞] right censored	S 0.1159 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06907 [0.05767, 0.08078] S low 0.898 [0.8794, 0.9162] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[778.2, 778.2] resolved	[1.6e+04, ∞] right censored	S 0.05398 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02712 [0.01973, 0.03409] S low 0.9539 [0.9411, 0.9668] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1169, 1169] resolved	[1.6e+04, ∞] right censored	S 0.02487 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.001562 [0, 0.003123] S low 0.9984 [0.9969, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_399_S0_H3_candidates_longest_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[1.6e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[1.6e+04, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0007808 [0, 0.002238] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[4000, ∞] right censored	[1.6e+04, ∞] right censored	S 0.9646 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2101 [0.01353, 0.4176] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	criterion met	inconclusive	[3164, 3164] resolved	[1.6e+04, ∞] right censored	S 0.9536 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.5997 [0.4581, 0.7432] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2436, 2436] resolved	[1.6e+04, ∞] right censored	S 0.9774 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.729 [0.6299, 0.8188] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[932.3, 932.3] resolved	[1.6e+04, ∞] right censored	S 0.9844 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8754 [0.8284, 0.9147] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[382.3, 382.3] resolved	[1.6e+04, ∞] right censored	S 0.9625 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9132 [0.8915, 0.9359] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[110.7, 110.7] resolved	[1.6e+04, ∞] right censored	S 0.9823 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9279 [0.9269, 0.9288] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_400_S0_H3_candidates_longest_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[53.4, 53.4] resolved	[1.6e+04, ∞] right censored	S 0.972 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8407 [0.8289, 0.8526] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[1600, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0006246 [0, 0.00177] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[400, ∞] right censored	[1600, ∞] right censored	S 0.995 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02649 [0.00406, 0.05945] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[199, 199] resolved	[1600, ∞] right censored	S 0.9903 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.13 [0.08468, 0.182] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[42.25, 42.25] resolved	[1600, ∞] right censored	S 0.9823 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3384 [0.2924, 0.388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.06, 18.06] resolved	[1600, ∞] right censored	S 0.9805 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.5139 [0.4664, 0.562] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.07, 11.07] resolved	[1600, ∞] right censored	S 0.9809 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6522 [0.6354, 0.669] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.896, 6.896] resolved	[1600, ∞] right censored	S 0.9739 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7807 [0.759, 0.8005] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.19, 5.19] resolved	[1600, ∞] right censored	S 0.9749 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.8532 [0.8438, 0.8626] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_401_S0_H3_candidates_longest_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · analysis second half · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[3.881, 3.881] resolved	[1600, ∞] right censored	S 0.9912 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7801 [0.7683, 0.792] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_402_S0_H3_candidates_longest_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.003 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.004 [1, 1.009] N 12/12; valid 1	1.006 [1.005, 1.008] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8543, low 0; forced mean no	1.246 [1.112, 1.416] N 12/12; valid 1	1.01 [1.007, 1.014] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.685 [1.497, 1.846] N 12/12; valid 1	1.007 [1.001, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_403_S0_H3_fingerprints_longest_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3933, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9126, low 0; forced mean no	2.31 [2.228, 2.389] N 12/12; valid 1	1.025 [1.017, 1.035] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.459 [2.364, 2.555] N 2/2; valid 1	1.137 [1.132, 1.143] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.556 [2.481, 2.633] N 2/2; valid 1	1.216 [1.208, 1.224] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.114 [1.111, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9592, 0.9901] N 12/12; valid 1	1.39 [1.33, 1.457] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7884 [0.7767, 0.8] N 12/12; valid 1	3.296 [3.212, 3.387] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_404_S0_H3_fingerprints_longest_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.03087; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1978; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8323 [0.7592, 0.954] N 12/12; valid 1	2.465 [2.105, 2.697] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.00406, low 0; forced mean yes	1.478 [1.369, 1.585] N 12/12; valid 1	0.4817 [0.4616, 0.5027] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.316 [1.238, 1.397] N 12/12; valid 1	0.4644 [0.4506, 0.4779] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.193 [1.074, 1.325] N 12/12; valid 1	0.4449 [0.4296, 0.4626] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.6643 [0.5959, 0.7388] N 2/2; valid 1	0.4279 [0.4122, 0.4438] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.0961 [0.07751, 0.1151] N 12/2; valid 1	0.44 [0.4345, 0.4466] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_405_S0_H3_fingerprints_longest_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.009] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.208 [1.002, 1.431] N 12/12; valid 1	1.014 [1.013, 1.016] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.669 [1.476, 1.851] N 12/12; valid 1	1.028 [1.024, 1.033] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.02295, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.77 [1.621, 1.919] N 12/12; valid 1	1.049 [1.042, 1.057] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.004 [1.889, 2.119] N 12/12; valid 1	1.079 [1.071, 1.088] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.6923, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8151, low 0; forced mean no	2.07 [1.996, 2.152] N 12/12; valid 1	1.134 [1.12, 1.148] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_406_S0_H3_fingerprints_longest_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.218 [2.101, 2.339] N 2/2; valid 1	1.347 [1.313, 1.381] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.245 [2.098, 2.405] N 2/2; valid 1	1.787 [1.703, 1.878] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.003, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.023 [1.002, 1.056] N 12/12; valid 1	1.009 [1.005, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.113 [1.068, 1.163] N 12/12; valid 1	1.001 [0.9955, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.284 [1.242, 1.334] N 12/12; valid 1	0.9997 [0.9923, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_407_S0_H3_fingerprints_longest_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · analysis second half · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1823, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5307, low 0; forced mean no	1.449 [1.399, 1.503] N 12/12; valid 1	0.9915 [0.9815, 1] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.564 [1.538, 1.588] N 12/12; valid 1	1.019 [0.9997, 1.038] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5634, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7431, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.61, low 0; forced mean no	1.702 [1.667, 1.739] N 12/12; valid 1	1.027 [1, 1.053] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.787 [1.716, 1.86] N 2/2; valid 1	1.173 [1.09, 1.268] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.796 [1.762, 1.83] N 2/2; valid 1	1.23 [1.172, 1.291] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_408_S0_H3_fingerprints_longest_analysis_second_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.685 [1.497, 1.846] N 12/12; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.007 [1.001, 1.013] N 12/12; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_409_S0_H3_pairs_longest_analysis_second_half_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.31 [2.228, 2.389] N 12/12; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.025 [1.017, 1.035] N 12/12; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.459 [2.364, 2.555] N 2/2; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.137 [1.132, 1.143] N 2/2; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.478 [1.369, 1.585] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4817 [0.4616, 0.5027] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_410_S0_H3_pairs_longest_analysis_second_half_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.316 [1.238, 1.397] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4644 [0.4506, 0.4779] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 1.193 [1.074, 1.325] N 12/12; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4449 [0.4296, 0.4626] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.6643 [0.5959, 0.7388] N 2/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.4279 [0.4122, 0.4438] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.556 [2.481, 2.633] N 2/2; valid 1 B 0.0961 [0.07751, 0.1151] N 12/2; valid 1	A 1.216 [1.208, 1.224] N 2/2; valid 1 B 0.44 [0.4345, 0.4466] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_411_S0_H3_pairs_longest_analysis_second_half_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.478 [1.369, 1.585] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4817 [0.4616, 0.5027] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_412_S0_H3_pairs_longest_analysis_second_half_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_413_S0_H3_pairs_longest_analysis_second_half_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.316 [1.238, 1.397] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4644 [0.4506, 0.4779] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_414_S0_H3_pairs_longest_analysis_second_half_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.193 [1.074, 1.325] N 12/12; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4449 [0.4296, 0.4626] N 12/12; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_415_S0_H3_pairs_longest_analysis_second_half_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.6643 [0.5959, 0.7388] N 2/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.4279 [0.4122, 0.4438] N 2/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_416_S0_H3_pairs_longest_analysis_second_half_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.669 [1.476, 1.851] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.028 [1.024, 1.033] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.77 [1.621, 1.919] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.049 [1.042, 1.057] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.004 [1.889, 2.119] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.079 [1.071, 1.088] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.07 [1.996, 2.152] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.134 [1.12, 1.148] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.218 [2.101, 2.339] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.347 [1.313, 1.381] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 2.245 [2.098, 2.405] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.787 [1.703, 1.878] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.449 [1.399, 1.503] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 0.9915 [0.9815, 1] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.564 [1.538, 1.588] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.019 [0.9997, 1.038] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_417_S0_H3_pairs_longest_analysis_second_half_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · analysis second half · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.702 [1.667, 1.739] N 12/12; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.027 [1, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.787 [1.716, 1.86] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.173 [1.09, 1.268] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.0961 [0.07751, 0.1151] N 12/2; valid 1 B 1.796 [1.762, 1.83] N 2/2; valid 1	A 0.44 [0.4345, 0.4466] N 12/2; valid 1 B 1.23 [1.172, 1.291] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_418_S0_H3_pairs_longest_analysis_second_half_inclusive_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0004686 [0, 0.00125] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.009155 [0.001144, 0.01987] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9757 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2917 [0.1244, 0.5021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9902 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6283 [0.4561, 0.7684] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[523.6, 523.6] resolved	[8000, ∞] right censored	S 0.9389 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9597 [0.9421, 0.9774] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[113.6, 113.6] resolved	[8000, ∞] right censored	S 0.9101 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.985 [0.9825, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[90.81, 90.81] resolved	[8000, ∞] right censored	S 0.9227 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9638 [0.9563, 0.9713] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_419_S0_H3_candidates_longest_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9147 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.008148 [0, 0.02031] S low 0.09125 [0, 0.2506] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[450, 450] resolved	[8000, ∞] right censored	S 0.2955 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0748 [0.0594, 0.09124] S low 0.8537 [0.8408, 0.8707] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[428, 428] resolved	[8000, ∞] right censored	S 0.128 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0671 [0.05452, 0.07886] S low 0.9087 [0.892, 0.926] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[978, 978] resolved	[8000, ∞] right censored	S 0.05124 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.03069 [0.01987, 0.04203] S low 0.9511 [0.9319, 0.9681] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_420_S0_H3_candidates_longest_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9919 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.162 [0.01007, 0.3558] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9537 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.5145 [0.3285, 0.706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9877 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6821 [0.5204, 0.8242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[1347, 1347] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.8719 [0.8317, 0.9134] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[367.8, 367.8] resolved	[8000, ∞] right censored	S 0.9576 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9181 [0.8925, 0.944] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[97.62, 97.62] resolved	[8000, ∞] right censored	S 0.9923 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9295 [0.9201, 0.9388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_421_S0_H3_candidates_longest_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[58.16, 58.16] resolved	[8000, ∞] right censored	S 0.9759 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8102 [0.7928, 0.8277] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[200, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.00104 [0, 0.003121] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[200, ∞] right censored	[800, ∞] right censored	S 0.99 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.04359 [0.00437, 0.1033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[179.1, 179.1] resolved	[800, ∞] right censored	S 0.9928 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.1256 [0.0723, 0.1927] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[48.74, 48.74] resolved	[800, ∞] right censored	S 0.9833 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3343 [0.27, 0.4022] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.09, 18.09] resolved	[800, ∞] right censored	S 0.9776 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.533 [0.4707, 0.6013] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.29, 11.29] resolved	[800, ∞] right censored	S 0.9828 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6471 [0.6185, 0.6756] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.988, 6.988] resolved	[800, ∞] right censored	S 0.9724 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7762 [0.7507, 0.7998] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.513, 5.513] resolved	[800, ∞] right censored	S 0.9747 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.829 [0.7903, 0.8677] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_422_S0_H3_candidates_longest_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[3.991, 3.991] resolved	[800, ∞] right censored	S 0.9924 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7709 [0.7191, 0.8227] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_423_S0_H3_candidates_longest_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.005 [1, 1.012] N 12/12; valid 1	1.008 [1.005, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8688, low 0; forced mean no	1.29 [1.103, 1.521] N 12/12; valid 1	1.011 [1.006, 1.015] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.641 [1.428, 1.853] N 12/12; valid 1	1.008 [0.9981, 1.018] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_424_S0_H3_fingerprints_longest_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3682, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9189, low 0; forced mean no	2.312 [2.243, 2.38] N 12/12; valid 1	1.027 [1.017, 1.04] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.442 [2.348, 2.54] N 2/2; valid 1	1.143 [1.112, 1.175] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.617 [2.484, 2.75] N 2/2; valid 1	1.187 [1.123, 1.253] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.028, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.12] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9419, 1] N 12/12; valid 1	1.39 [1.313, 1.511] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7926 [0.7663, 0.8219] N 12/12; valid 1	3.221 [3.09, 3.355] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_425_S0_H3_fingerprints_longest_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.06169; forced mean no	no eligible check single front; H 32000 multi 0, low 0.2398; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8531 [0.7673, 0.976] N 12/12; valid 1	2.315 [1.928, 2.588] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.007074, low 0; forced mean yes	1.452 [1.346, 1.561] N 12/12; valid 1	0.472 [0.4473, 0.4987] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.317 [1.203, 1.428] N 12/12; valid 1	0.4675 [0.45, 0.4849] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.191 [1.04, 1.369] N 12/12; valid 1	0.4325 [0.4136, 0.4551] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.5441 [0.4932, 0.6032] N 2/2; valid 1	0.4432 [0.4413, 0.4451] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.03468 [0.02741, 0.04213] N 12/2; valid 1	0.4227 [0.4056, 0.4392] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_426_S0_H3_fingerprints_longest_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.009 [1.008, 1.01] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.161 [0.9983, 1.358] N 12/12; valid 1	1.015 [1.012, 1.018] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.614 [1.355, 1.89] N 12/12; valid 1	1.028 [1.022, 1.035] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8859, low 0; forced mean no	1.678 [1.477, 1.875] N 12/12; valid 1	1.05 [1.041, 1.059] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	1.975 [1.841, 2.108] N 12/12; valid 1	1.081 [1.069, 1.093] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7209, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8092, low 0; forced mean no	2.112 [2.008, 2.218] N 12/12; valid 1	1.132 [1.114, 1.153] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_427_S0_H3_fingerprints_longest_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.363 [2.228, 2.5] N 2/2; valid 1	1.379 [1.31, 1.45] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.164 [2.06, 2.279] N 2/2; valid 1	1.827 [1.787, 1.867] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.038 [1, 1.107] N 12/12; valid 1	1.009 [1.003, 1.014] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.107 [1.053, 1.173] N 12/12; valid 1	1.004 [0.9956, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.27 [1.203, 1.352] N 12/12; valid 1	0.9951 [0.9836, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_428_S0_H3_fingerprints_longest_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1828, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5309, low 0; forced mean no	1.473 [1.403, 1.553] N 12/12; valid 1	0.9857 [0.9721, 0.9981] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.548 [1.522, 1.573] N 12/12; valid 1	1.025 [1.001, 1.048] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5624, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6064, low 0; forced mean no	1.677 [1.64, 1.716] N 12/12; valid 1	1.023 [0.9881, 1.059] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.743 [1.686, 1.8] N 2/2; valid 1	1.153 [1.086, 1.221] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.752 [1.686, 1.82] N 2/2; valid 1	1.277 [1.102, 1.491] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_429_S0_H3_fingerprints_longest_late_quarter_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_430_S0_H3_pairs_longest_late_quarter_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_431_S0_H3_pairs_longest_late_quarter_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_432_S0_H3_pairs_longest_late_quarter_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_433_S0_H3_pairs_longest_late_quarter_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_434_S0_H3_pairs_longest_late_quarter_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_435_S0_H3_pairs_longest_late_quarter_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_436_S0_H3_pairs_longest_late_quarter_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_437_S0_H3_pairs_longest_late_quarter_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_438_S0_H3_pairs_longest_late_quarter_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_439_S0_H3_pairs_longest_late_quarter_primary_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0004686 [0, 0.00125] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.009155 [0.001144, 0.01987] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9757 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2917 [0.1244, 0.5021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9902 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6283 [0.4561, 0.7684] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[523.6, 523.6] resolved	[8000, ∞] right censored	S 0.9389 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9597 [0.9421, 0.9774] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[113.6, 113.6] resolved	[8000, ∞] right censored	S 0.9101 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.985 [0.9825, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[90.81, 90.81] resolved	[8000, ∞] right censored	S 0.9227 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9638 [0.9563, 0.9713] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_440_S0_H3_candidates_longest_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 0.9147 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.008148 [0, 0.02031] S low 0.09125 [0, 0.2506] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[450, 450] resolved	[8000, ∞] right censored	S 0.2955 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0748 [0.0594, 0.09124] S low 0.8537 [0.8408, 0.8707] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[428, 428] resolved	[8000, ∞] right censored	S 0.128 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0671 [0.05452, 0.07886] S low 0.9087 [0.892, 0.926] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[978, 978] resolved	[8000, ∞] right censored	S 0.05124 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.03069 [0.01987, 0.04203] S low 0.9511 [0.9319, 0.9681] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_441_S0_H3_candidates_longest_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[8000, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[8000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9919 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.162 [0.01007, 0.3558] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	multiple fronts	inconclusive	inconclusive	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9537 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.5145 [0.3285, 0.706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	inconclusive	criterion met	[2000, ∞] right censored	[8000, ∞] right censored	S 0.9877 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.6821 [0.5204, 0.8242] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[1347, 1347] resolved	[8000, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.8719 [0.8317, 0.9134] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[367.8, 367.8] resolved	[8000, ∞] right censored	S 0.9576 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9181 [0.8925, 0.944] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[97.62, 97.62] resolved	[8000, ∞] right censored	S 0.9923 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9295 [0.9201, 0.9388] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_442_S0_H3_candidates_longest_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[58.16, 58.16] resolved	[8000, ∞] right censored	S 0.9759 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8102 [0.7928, 0.8277] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[800, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[200, ∞] right censored	[800, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.00104 [0, 0.003121] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	inconclusive	inconclusive	[200, ∞] right censored	[800, ∞] right censored	S 0.99 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.04359 [0.00437, 0.1033] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[179.1, 179.1] resolved	[800, ∞] right censored	S 0.9928 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.1256 [0.0723, 0.1927] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[48.74, 48.74] resolved	[800, ∞] right censored	S 0.9833 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3343 [0.27, 0.4022] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	multiple fronts	criterion met	criterion met	[18.09, 18.09] resolved	[800, ∞] right censored	S 0.9776 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.533 [0.4707, 0.6013] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[11.29, 11.29] resolved	[800, ∞] right censored	S 0.9828 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6471 [0.6185, 0.6756] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.988, 6.988] resolved	[800, ∞] right censored	S 0.9724 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7762 [0.7507, 0.7998] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.513, 5.513] resolved	[800, ∞] right censored	S 0.9747 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.829 [0.7903, 0.8677] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_443_S0_H3_candidates_longest_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · late quarter · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[3.991, 3.991] resolved	[800, ∞] right censored	S 0.9924 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7709 [0.7191, 0.8227] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_444_S0_H3_candidates_longest_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.005 [1, 1.012] N 12/12; valid 1	1.008 [1.005, 1.011] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8688, low 0; forced mean no	1.29 [1.103, 1.521] N 12/12; valid 1	1.011 [1.006, 1.015] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.641 [1.428, 1.853] N 12/12; valid 1	1.008 [0.9981, 1.018] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_445_S0_H3_fingerprints_longest_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3682, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9189, low 0; forced mean no	2.312 [2.243, 2.38] N 12/12; valid 1	1.027 [1.017, 1.04] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.442 [2.348, 2.54] N 2/2; valid 1	1.143 [1.112, 1.175] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.617 [2.484, 2.75] N 2/2; valid 1	1.187 [1.123, 1.253] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.028, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.11, 1.12] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9756 [0.9419, 1] N 12/12; valid 1	1.39 [1.313, 1.511] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7926 [0.7663, 0.8219] N 12/12; valid 1	3.221 [3.09, 3.355] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_446_S0_H3_fingerprints_longest_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.06169; forced mean no	no eligible check single front; H 32000 multi 0, low 0.2398; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8531 [0.7673, 0.976] N 12/12; valid 1	2.315 [1.928, 2.588] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.007074, low 0; forced mean yes	1.452 [1.346, 1.561] N 12/12; valid 1	0.472 [0.4473, 0.4987] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.317 [1.203, 1.428] N 12/12; valid 1	0.4675 [0.45, 0.4849] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.191 [1.04, 1.369] N 12/12; valid 1	0.4325 [0.4136, 0.4551] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.5441 [0.4932, 0.6032] N 2/2; valid 1	0.4432 [0.4413, 0.4451] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.03468 [0.02741, 0.04213] N 12/2; valid 1	0.4227 [0.4056, 0.4392] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_447_S0_H3_fingerprints_longest_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.004 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.009 [1.008, 1.01] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.161 [0.9983, 1.358] N 12/12; valid 1	1.015 [1.012, 1.018] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.614 [1.355, 1.89] N 12/12; valid 1	1.028 [1.022, 1.035] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8859, low 0; forced mean no	1.678 [1.477, 1.875] N 12/12; valid 1	1.05 [1.041, 1.059] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	1.975 [1.841, 2.108] N 12/12; valid 1	1.081 [1.069, 1.093] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7209, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8092, low 0; forced mean no	2.112 [2.008, 2.218] N 12/12; valid 1	1.132 [1.114, 1.153] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_448_S0_H3_fingerprints_longest_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.363 [2.228, 2.5] N 2/2; valid 1	1.379 [1.31, 1.45] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.164 [2.06, 2.279] N 2/2; valid 1	1.827 [1.787, 1.867] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.003 [1, 1.006] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.038 [1, 1.107] N 12/12; valid 1	1.009 [1.003, 1.014] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.107 [1.053, 1.173] N 12/12; valid 1	1.004 [0.9956, 1.012] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.27 [1.203, 1.352] N 12/12; valid 1	0.9951 [0.9836, 1.007] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_449_S0_H3_fingerprints_longest_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · late quarter · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	single front; H 3200 qualified yes; same no multi 0, low 0; forced mean no	single front; H 3200 qualified yes; same no multi 0.1828, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.5309, low 0; forced mean no	1.473 [1.403, 1.553] N 12/12; valid 1	0.9857 [0.9721, 0.9981] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.548 [1.522, 1.573] N 12/12; valid 1	1.025 [1.001, 1.048] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5624, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6064, low 0; forced mean no	1.677 [1.64, 1.716] N 12/12; valid 1	1.023 [0.9881, 1.059] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.743 [1.686, 1.8] N 2/2; valid 1	1.153 [1.086, 1.221] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.752 [1.686, 1.82] N 2/2; valid 1	1.277 [1.102, 1.491] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_450_S0_H3_fingerprints_longest_late_quarter_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.641 [1.428, 1.853] N 12/12; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.008 [0.9981, 1.018] N 12/12; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_451_S0_H3_pairs_longest_late_quarter_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.312 [2.243, 2.38] N 12/12; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.027 [1.017, 1.04] N 12/12; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.442 [2.348, 2.54] N 2/2; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.143 [1.112, 1.175] N 2/2; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.452 [1.346, 1.561] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.472 [0.4473, 0.4987] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_452_S0_H3_pairs_longest_late_quarter_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.317 [1.203, 1.428] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4675 [0.45, 0.4849] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 1.191 [1.04, 1.369] N 12/12; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4325 [0.4136, 0.4551] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.5441 [0.4932, 0.6032] N 2/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4432 [0.4413, 0.4451] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.617 [2.484, 2.75] N 2/2; valid 1 B 0.03468 [0.02741, 0.04213] N 12/2; valid 1	A 1.187 [1.123, 1.253] N 2/2; valid 1 B 0.4227 [0.4056, 0.4392] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_453_S0_H3_pairs_longest_late_quarter_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.452 [1.346, 1.561] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.472 [0.4473, 0.4987] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_454_S0_H3_pairs_longest_late_quarter_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_455_S0_H3_pairs_longest_late_quarter_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.317 [1.203, 1.428] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4675 [0.45, 0.4849] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	inconclusive	count inconclusive contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_456_S0_H3_pairs_longest_late_quarter_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.191 [1.04, 1.369] N 12/12; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4325 [0.4136, 0.4551] N 12/12; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_457_S0_H3_pairs_longest_late_quarter_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.5441 [0.4932, 0.6032] N 2/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4432 [0.4413, 0.4451] N 2/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_458_S0_H3_pairs_longest_late_quarter_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.02 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.614 [1.355, 1.89] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.028 [1.022, 1.035] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.678 [1.477, 1.875] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.05 [1.041, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.975 [1.841, 2.108] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.081 [1.069, 1.093] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.112 [2.008, 2.218] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.132 [1.114, 1.153] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.363 [2.228, 2.5] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.379 [1.31, 1.45] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 2.164 [2.06, 2.279] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.827 [1.787, 1.867] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.473 [1.403, 1.553] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 0.9857 [0.9721, 0.9981] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.548 [1.522, 1.573] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.025 [1.001, 1.048] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_459_S0_H3_pairs_longest_late_quarter_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · late quarter · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.677 [1.64, 1.716] N 12/12; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.023 [0.9881, 1.059] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.743 [1.686, 1.8] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.153 [1.086, 1.221] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.03468 [0.02741, 0.04213] N 12/2; valid 1 B 1.752 [1.686, 1.82] N 2/2; valid 1	A 0.4227 [0.4056, 0.4392] N 12/2; valid 1 B 1.277 [1.102, 1.491] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_460_S0_H3_pairs_longest_late_quarter_inclusive_010.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001159 [0.0002991, 0.002239] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01712 [0.007027, 0.0304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9892 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.235 [0.1245, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9798 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6387 [0.5516, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.9, 505.9] resolved	[3.2e+04, ∞] right censored	S 0.9422 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9548 [0.9448, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.3, 128.3] resolved	[3.2e+04, ∞] right censored	S 0.9207 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.9844, 0.9881] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[83.27, 83.27] resolved	[3.2e+04, ∞] right censored	S 0.9319 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9606 [0.9582, 0.9631] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_461_S0_H3_candidates_longest_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9573 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.00401 [0, 0.01127] S low 0.03671 [0, 0.1044] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[475.4, 475.4] resolved	[3.2e+04, ∞] right censored	S 0.2373 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07594 [0.06543, 0.08484] S low 0.8565 [0.8397, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.6, 563.6] resolved	[3.2e+04, ∞] right censored	S 0.131 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06283 [0.05564, 0.07001] S low 0.9004 [0.8879, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[733.2, 733.2] resolved	[3.2e+04, ∞] right censored	S 0.05288 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0266 [0.02082, 0.03206] S low 0.9579 [0.9503, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1251, 1251] resolved	[3.2e+04, ∞] right censored	S 0.01525 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004997 [0.001874, 0.00812] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_462_S0_H3_candidates_longest_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001302 [0, 0.0003906] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004164 [0, 0.001145] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.979 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1969 [0.03435, 0.3848] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3903, 3903] resolved	[3.2e+04, ∞] right censored	S 0.9796 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4837 [0.3914, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[3.2e+04, ∞] right censored	S 0.985 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7161 [0.6474, 0.7783] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.9, 986.9] resolved	[3.2e+04, ∞] right censored	S 0.9749 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.857 [0.832, 0.8798] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.5, 388.5] resolved	[3.2e+04, ∞] right censored	S 0.9627 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.919 [0.9007, 0.9373] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104, 104] resolved	[3.2e+04, ∞] right censored	S 0.9616 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9335 [0.9332, 0.9338] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_463_S0_H3_candidates_longest_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[51.19, 51.19] resolved	[3.2e+04, ∞] right censored	S 0.9708 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.852 [0.8463, 0.8576] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0007287 [5.205e-05, 0.001666] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.5, 533.5] resolved	[3200, ∞] right censored	S 0.9981 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02363 [0.006558, 0.04497] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[3200, ∞] right censored	S 0.9922 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1366 [0.108, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.9, 37.9] resolved	[3200, ∞] right censored	S 0.9847 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3533 [0.3188, 0.3849] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[3200, ∞] right censored	S 0.9823 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4987 [0.4698, 0.5309] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.66, 10.66] resolved	[3200, ∞] right censored	S 0.9803 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.658 [0.6436, 0.6719] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.808, 6.808] resolved	[3200, ∞] right censored	S 0.9744 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7891 [0.7747, 0.8021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[4.953, 4.953] resolved	[3200, ∞] right censored	S 0.9749 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.842 [0.8407, 0.8432] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_464_S0_H3_candidates_longest_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.101, 4.101] resolved	[3200, ∞] right censored	S 0.9887 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7633 [0.7395, 0.787] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_465_S0_H3_candidates_longest_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8682, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_466_S0_H3_fingerprints_longest_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3856, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9124, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_467_S0_H3_fingerprints_longest_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01546; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008224, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_468_S0_H3_fingerprints_longest_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01145, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8082, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_469_S0_H3_fingerprints_longest_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_470_S0_H3_fingerprints_longest_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1726, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5119, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5596, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7444, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6049, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_471_S0_H3_fingerprints_longest_cadence_x2_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_472_S0_H3_pairs_longest_cadence_x2_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_473_S0_H3_pairs_longest_cadence_x2_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_474_S0_H3_pairs_longest_cadence_x2_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_475_S0_H3_pairs_longest_cadence_x2_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_476_S0_H3_pairs_longest_cadence_x2_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_477_S0_H3_pairs_longest_cadence_x2_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_478_S0_H3_pairs_longest_cadence_x2_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_479_S0_H3_pairs_longest_cadence_x2_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_480_S0_H3_pairs_longest_cadence_x2_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001159 [0.0002991, 0.002239] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01712 [0.007027, 0.0304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9892 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.235 [0.1245, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9798 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6387 [0.5516, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[505.9, 505.9] resolved	[3.2e+04, ∞] right censored	S 0.9422 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9548 [0.9448, 0.9647] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.3, 128.3] resolved	[3.2e+04, ∞] right censored	S 0.9207 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.9844, 0.9881] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[83.27, 83.27] resolved	[3.2e+04, ∞] right censored	S 0.9319 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9606 [0.9582, 0.9631] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_481_S0_H3_candidates_longest_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9573 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.00401 [0, 0.01127] S low 0.03671 [0, 0.1044] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[475.4, 475.4] resolved	[3.2e+04, ∞] right censored	S 0.2373 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07594 [0.06543, 0.08484] S low 0.8565 [0.8397, 0.8749] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.6, 563.6] resolved	[3.2e+04, ∞] right censored	S 0.131 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06283 [0.05564, 0.07001] S low 0.9004 [0.8879, 0.9136] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[733.2, 733.2] resolved	[3.2e+04, ∞] right censored	S 0.05288 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.0266 [0.02082, 0.03206] S low 0.9579 [0.9503, 0.9669] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1251, 1251] resolved	[3.2e+04, ∞] right censored	S 0.01525 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004997 [0.001874, 0.00812] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_482_S0_H3_candidates_longest_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001302 [0, 0.0003906] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004164 [0, 0.001145] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.979 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1969 [0.03435, 0.3848] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3903, 3903] resolved	[3.2e+04, ∞] right censored	S 0.9796 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4837 [0.3914, 0.5751] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2012, 2012] resolved	[3.2e+04, ∞] right censored	S 0.985 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7161 [0.6474, 0.7783] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[986.9, 986.9] resolved	[3.2e+04, ∞] right censored	S 0.9749 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.857 [0.832, 0.8798] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[388.5, 388.5] resolved	[3.2e+04, ∞] right censored	S 0.9627 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.919 [0.9007, 0.9373] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104, 104] resolved	[3.2e+04, ∞] right censored	S 0.9616 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9335 [0.9332, 0.9338] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_483_S0_H3_candidates_longest_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[51.19, 51.19] resolved	[3.2e+04, ∞] right censored	S 0.9708 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.852 [0.8463, 0.8576] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0007287 [5.205e-05, 0.001666] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.5, 533.5] resolved	[3200, ∞] right censored	S 0.9981 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02363 [0.006558, 0.04497] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[3200, ∞] right censored	S 0.9922 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1366 [0.108, 0.1706] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.9, 37.9] resolved	[3200, ∞] right censored	S 0.9847 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3533 [0.3188, 0.3849] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.18, 18.18] resolved	[3200, ∞] right censored	S 0.9823 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4987 [0.4698, 0.5309] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.66, 10.66] resolved	[3200, ∞] right censored	S 0.9803 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.658 [0.6436, 0.6719] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.808, 6.808] resolved	[3200, ∞] right censored	S 0.9744 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7891 [0.7747, 0.8021] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[4.953, 4.953] resolved	[3200, ∞] right censored	S 0.9749 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.842 [0.8407, 0.8432] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_484_S0_H3_candidates_longest_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x2 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.101, 4.101] resolved	[3200, ∞] right censored	S 0.9887 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7633 [0.7395, 0.787] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_485_S0_H3_candidates_longest_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8682, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_486_S0_H3_fingerprints_longest_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3856, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9124, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_487_S0_H3_fingerprints_longest_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01546; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1406; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008224, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_488_S0_H3_fingerprints_longest_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01145, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8576, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8082, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_489_S0_H3_fingerprints_longest_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_490_S0_H3_fingerprints_longest_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x2 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1726, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5119, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5596, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7444, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6049, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_491_S0_H3_fingerprints_longest_cadence_x2_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_492_S0_H3_pairs_longest_cadence_x2_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_493_S0_H3_pairs_longest_cadence_x2_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_494_S0_H3_pairs_longest_cadence_x2_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_495_S0_H3_pairs_longest_cadence_x2_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_496_S0_H3_pairs_longest_cadence_x2_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_497_S0_H3_pairs_longest_cadence_x2_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_498_S0_H3_pairs_longest_cadence_x2_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_499_S0_H3_pairs_longest_cadence_x2_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x2 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_500_S0_H3_pairs_longest_cadence_x2_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001172 [0.0002603, 0.002292] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01737 [0.007176, 0.03059] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.989 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2348 [0.1243, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9804 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6395 [0.5528, 0.7213] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.7, 504.7] resolved	[3.2e+04, ∞] right censored	S 0.9445 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.954 [0.9441, 0.964] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.6, 128.6] resolved	[3.2e+04, ∞] right censored	S 0.9195 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.985, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[84, 84] resolved	[3.2e+04, ∞] right censored	S 0.9341 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9638 [0.96, 0.9675] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_501_S0_H3_candidates_longest_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9575 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.004035 [0, 0.01135] S low 0.03676 [0, 0.1045] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[478, 478] resolved	[3.2e+04, ∞] right censored	S 0.2386 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07563 [0.06471, 0.0849] S low 0.8562 [0.8387, 0.8753] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.3, 563.3] resolved	[3.2e+04, ∞] right censored	S 0.13 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06149 [0.05379, 0.06898] S low 0.9012 [0.8878, 0.915] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[736.8, 736.8] resolved	[3.2e+04, ∞] right censored	S 0.05301 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02643 [0.0205, 0.03215] S low 0.9581 [0.9501, 0.9672] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1264, 1264] resolved	[3.2e+04, ∞] right censored	S 0.01493 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.005618 [0.002497, 0.008739] S low 0.9919 [0.9863, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_502_S0_H3_candidates_longest_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001302 [0, 0.0003905] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004161 [0, 0.001248] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9788 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1964 [0.03391, 0.3844] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3907, 3907] resolved	[3.2e+04, ∞] right censored	S 0.9785 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4842 [0.3916, 0.5754] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2013, 2013] resolved	[3.2e+04, ∞] right censored	S 0.9844 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7165 [0.6475, 0.7785] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[985.7, 985.7] resolved	[3.2e+04, ∞] right censored	S 0.975 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8579 [0.8328, 0.8807] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[390.9, 390.9] resolved	[3.2e+04, ∞] right censored	S 0.9621 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9184 [0.9004, 0.9365] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.4, 104.4] resolved	[3.2e+04, ∞] right censored	S 0.9597 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9332 [0.9313, 0.9351] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_503_S0_H3_candidates_longest_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[52.68, 52.68] resolved	[3.2e+04, ∞] right censored	S 0.9712 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8508 [0.8452, 0.8564] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0006242 [0, 0.001457] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.5, 533.5] resolved	[3200, ∞] right censored	S 0.998 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02362 [0.006656, 0.04484] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[3200, ∞] right censored	S 0.9919 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1363 [0.1071, 0.1705] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.89, 37.89] resolved	[3200, ∞] right censored	S 0.9848 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3548 [0.3199, 0.3867] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.15, 18.15] resolved	[3200, ∞] right censored	S 0.983 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4988 [0.4698, 0.5304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.72, 10.72] resolved	[3200, ∞] right censored	S 0.9789 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6572 [0.6426, 0.6712] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.891, 6.891] resolved	[3200, ∞] right censored	S 0.9746 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7887 [0.7737, 0.8026] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[5.058, 5.058] resolved	[3200, ∞] right censored	S 0.9724 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.8452 [0.8427, 0.8477] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_504_S0_H3_candidates_longest_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[4.068, 4.068] resolved	[3200, ∞] right censored	S 0.9862 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7665 [0.7403, 0.7928] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_505_S0_H3_candidates_longest_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8685, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_506_S0_H3_fingerprints_longest_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3855, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9143, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_507_S0_H3_fingerprints_longest_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.0155; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008219, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_508_S0_H3_fingerprints_longest_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01144, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8553, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8081, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_509_S0_H3_fingerprints_longest_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_510_S0_H3_fingerprints_longest_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1728, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5112, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5605, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.743, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6073, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_511_S0_H3_fingerprints_longest_cadence_x4_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_512_S0_H3_pairs_longest_cadence_x4_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_513_S0_H3_pairs_longest_cadence_x4_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_514_S0_H3_pairs_longest_cadence_x4_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_515_S0_H3_pairs_longest_cadence_x4_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_516_S0_H3_pairs_longest_cadence_x4_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_517_S0_H3_pairs_longest_cadence_x4_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_518_S0_H3_pairs_longest_cadence_x4_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_519_S0_H3_pairs_longest_cadence_x4_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_520_S0_H3_pairs_longest_cadence_x4_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001172 [0.0002603, 0.002292] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01737 [0.007176, 0.03059] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.989 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2348 [0.1243, 0.3759] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2696, 2696] resolved	[3.2e+04, ∞] right censored	S 0.9804 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6395 [0.5528, 0.7213] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.7, 504.7] resolved	[3.2e+04, ∞] right censored	S 0.9445 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.954 [0.9441, 0.964] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[128.6, 128.6] resolved	[3.2e+04, ∞] right censored	S 0.9195 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9863 [0.985, 0.9875] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[84, 84] resolved	[3.2e+04, ∞] right censored	S 0.9341 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9638 [0.96, 0.9675] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_521_S0_H3_candidates_longest_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9575 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.004035 [0, 0.01135] S low 0.03676 [0, 0.1045] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[478, 478] resolved	[3.2e+04, ∞] right censored	S 0.2386 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07563 [0.06471, 0.0849] S low 0.8562 [0.8387, 0.8753] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[563.3, 563.3] resolved	[3.2e+04, ∞] right censored	S 0.13 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06149 [0.05379, 0.06898] S low 0.9012 [0.8878, 0.915] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[736.8, 736.8] resolved	[3.2e+04, ∞] right censored	S 0.05301 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02643 [0.0205, 0.03215] S low 0.9581 [0.9501, 0.9672] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1264, 1264] resolved	[3.2e+04, ∞] right censored	S 0.01493 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.005618 [0.002497, 0.008739] S low 0.9919 [0.9863, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_522_S0_H3_candidates_longest_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001302 [0, 0.0003905] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004161 [0, 0.001248] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9788 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1964 [0.03391, 0.3844] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3907, 3907] resolved	[3.2e+04, ∞] right censored	S 0.9785 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4842 [0.3916, 0.5754] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2013, 2013] resolved	[3.2e+04, ∞] right censored	S 0.9844 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.7165 [0.6475, 0.7785] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[985.7, 985.7] resolved	[3.2e+04, ∞] right censored	S 0.975 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8579 [0.8328, 0.8807] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[390.9, 390.9] resolved	[3.2e+04, ∞] right censored	S 0.9621 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9184 [0.9004, 0.9365] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.4, 104.4] resolved	[3.2e+04, ∞] right censored	S 0.9597 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9332 [0.9313, 0.9351] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_523_S0_H3_candidates_longest_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[52.68, 52.68] resolved	[3.2e+04, ∞] right censored	S 0.9712 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8508 [0.8452, 0.8564] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0006242 [0, 0.001457] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[533.5, 533.5] resolved	[3200, ∞] right censored	S 0.998 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02362 [0.006656, 0.04484] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[145.8, 145.8] resolved	[3200, ∞] right censored	S 0.9919 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1363 [0.1071, 0.1705] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.89, 37.89] resolved	[3200, ∞] right censored	S 0.9848 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3548 [0.3199, 0.3867] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.15, 18.15] resolved	[3200, ∞] right censored	S 0.983 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4988 [0.4698, 0.5304] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.72, 10.72] resolved	[3200, ∞] right censored	S 0.9789 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6572 [0.6426, 0.6712] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[6.891, 6.891] resolved	[3200, ∞] right censored	S 0.9746 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7887 [0.7737, 0.8026] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[5.058, 5.058] resolved	[3200, ∞] right censored	S 0.9724 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.8452 [0.8427, 0.8477] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_524_S0_H3_candidates_longest_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x4 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[4.068, 4.068] resolved	[3200, ∞] right censored	S 0.9862 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7665 [0.7403, 0.7928] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_525_S0_H3_candidates_longest_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.8685, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_526_S0_H3_fingerprints_longest_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3855, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9143, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_527_S0_H3_fingerprints_longest_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.0155; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008219, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_528_S0_H3_fingerprints_longest_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01144, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8553, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7007, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8081, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_529_S0_H3_fingerprints_longest_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_530_S0_H3_fingerprints_longest_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x4 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1728, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5112, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5605, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.743, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6073, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_531_S0_H3_fingerprints_longest_cadence_x4_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_532_S0_H3_pairs_longest_cadence_x4_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_533_S0_H3_pairs_longest_cadence_x4_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_534_S0_H3_pairs_longest_cadence_x4_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_535_S0_H3_pairs_longest_cadence_x4_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_536_S0_H3_pairs_longest_cadence_x4_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_537_S0_H3_pairs_longest_cadence_x4_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_538_S0_H3_pairs_longest_cadence_x4_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_539_S0_H3_pairs_longest_cadence_x4_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x4 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_540_S0_H3_pairs_longest_cadence_x4_inclusive_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001093 [0.0002082, 0.002186] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01808 [0.008105, 0.03096] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9888 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2344 [0.1236, 0.3753] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2702, 2702] resolved	[3.2e+04, ∞] right censored	S 0.98 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6394 [0.5521, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.9, 504.9] resolved	[3.2e+04, ∞] right censored	S 0.9444 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.953 [0.9426, 0.9634] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[130.7, 130.7] resolved	[3.2e+04, ∞] right censored	S 0.9131 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.99 [0.985, 0.995] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	inconclusive	inconclusive	[82.42, 82.42] resolved	[3.2e+04, ∞] right censored	S 0.9286 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.9663 [0.9626, 0.9701] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_541_S0_H3_candidates_longest_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9577 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.00406 [0, 0.0114] S low 0.0368 [0, 0.1046] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	inconclusive	[479.3, 479.3] resolved	[3.2e+04, ∞] right censored	S 0.2363 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07461 [0.06234, 0.08541] S low 0.8585 [0.841, 0.8786] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[556.9, 556.9] resolved	[3.2e+04, ∞] right censored	S 0.1307 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06234 [0.05486, 0.07004] S low 0.9009 [0.8872, 0.9156] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[767.4, 767.4] resolved	[3.2e+04, ∞] right censored	S 0.05192 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02702 [0.02078, 0.03304] S low 0.9572 [0.9493, 0.9663] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1269, 1269] resolved	[3.2e+04, ∞] right censored	S 0.01491 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004988 [0.002494, 0.007481] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_542_S0_H3_candidates_longest_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001562 [0, 0.0004685] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004156 [0, 0.001247] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9797 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1966 [0.0349, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3906, 3906] resolved	[3.2e+04, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4823 [0.3896, 0.5738] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2015, 2015] resolved	[3.2e+04, ∞] right censored	S 0.9841 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.718 [0.6498, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[981.7, 981.7] resolved	[3.2e+04, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8603 [0.8356, 0.8832] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[392.1, 392.1] resolved	[3.2e+04, ∞] right censored	S 0.9591 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9175 [0.899, 0.936] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.9, 104.9] resolved	[3.2e+04, ∞] right censored	S 0.9648 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9302 [0.9252, 0.9352] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_543_S0_H3_candidates_longest_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	inconclusive	inconclusive	[0, 80] unresolved below two cadences	[3.2e+04, ∞] right censored	S 0.9724 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.8354 [0.8204, 0.8504] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0008313 [0, 0.00187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[534.3, 534.3] resolved	[3200, ∞] right censored	S 0.9978 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02328 [0.006234, 0.04489] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146.2, 146.2] resolved	[3200, ∞] right censored	S 0.9913 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1357 [0.1062, 0.1704] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.83, 37.83] resolved	[3200, ∞] right censored	S 0.984 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3554 [0.3215, 0.3874] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.25, 18.25] resolved	[3200, ∞] right censored	S 0.9838 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4985 [0.4701, 0.5299] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.83, 10.83] resolved	[3200, ∞] right censored	S 0.9799 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6586 [0.6424, 0.6735] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9745 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.787 [0.7695, 0.802] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	inconclusive	inconclusive	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9599 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.8628 [0.8454, 0.8803] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_544_S0_H3_candidates_longest_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · operationally unforced conditions

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	inconclusive	inconclusive	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9925 / W 1 H 3200 / 12800 95% bound check: ?	S multi 0.7793 [0.7456, 0.813] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_545_S0_H3_candidates_longest_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.868, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_546_S0_H3_fingerprints_longest_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3859, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9092, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_547_S0_H3_fingerprints_longest_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01559; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.03	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check low contrast; H 32000 multi 0, low 1; forced mean yes	no eligible check single front; H 32000 multi 0.008105, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count unavailable contrast unavailable
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count unavailable contrast unavailable
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count unavailable contrast unavailable
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_548_S0_H3_fingerprints_longest_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01143, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8543, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7014, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8103, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_549_S0_H3_fingerprints_longest_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_550_S0_H3_fingerprints_longest_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · operationally unforced conditions

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5106, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5576, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6027, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count unavailable contrast unavailable
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count unavailable contrast unavailable

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_551_S0_H3_fingerprints_longest_cadence_x8_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_552_S0_H3_pairs_longest_cadence_x8_primary_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_553_S0_H3_pairs_longest_cadence_x8_primary_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_554_S0_H3_pairs_longest_cadence_x8_primary_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_555_S0_H3_pairs_longest_cadence_x8_primary_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_556_S0_H3_pairs_longest_cadence_x8_primary_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_557_S0_H3_pairs_longest_cadence_x8_primary_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_558_S0_H3_pairs_longest_cadence_x8_primary_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	inconclusive	count inconclusive contrast inconclusive
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_559_S0_H3_pairs_longest_cadence_x8_primary_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · operationally unforced conditions

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / inconclusive E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	inconclusive	count inconclusive contrast inconclusive

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_560_S0_H3_pairs_longest_cadence_x8_primary_009.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Wave-pinning f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.001093 [0.0002082, 0.002186] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.02	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.01808 [0.008105, 0.03096] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.03	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9888 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.2344 [0.1236, 0.3753] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.05	multiple fronts	criterion met	inconclusive	[2702, 2702] resolved	[3.2e+04, ∞] right censored	S 0.98 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.6394 [0.5521, 0.7211] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.1	multiple fronts	criterion met	criterion met	[504.9, 504.9] resolved	[3.2e+04, ∞] right censored	S 0.9444 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.953 [0.9426, 0.9634] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=0.3	multiple fronts	criterion met	inconclusive	[130.7, 130.7] resolved	[3.2e+04, ∞] right censored	S 0.9131 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.99 [0.985, 0.995] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Wave-pinning f=1	multiple fronts	criterion met	inconclusive	[82.42, 82.42] resolved	[3.2e+04, ∞] right censored	S 0.9286 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9663 [0.9626, 0.9701] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_561_S0_H3_candidates_longest_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.01	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.015	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.02	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9577 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.00406 [0, 0.0114] S low 0.0368 [0, 0.1046] W multi 0; low 0	S N/N/N W N/N/N
Goryachev f=0.03	low contrast	inconclusive	criterion met	[479.3, 479.3] resolved	[3.2e+04, ∞] right censored	S 0.2363 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.07461 [0.06234, 0.08541] S low 0.8585 [0.841, 0.8786] W multi 0; low 0	S Y/Y/Y W N/N/N
Goryachev f=0.05	low contrast	inconclusive	inconclusive	[556.9, 556.9] resolved	[3.2e+04, ∞] right censored	S 0.1307 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.06234 [0.05486, 0.07004] S low 0.9009 [0.8872, 0.9156] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.1	low contrast	inconclusive	inconclusive	[767.4, 767.4] resolved	[3.2e+04, ∞] right censored	S 0.05192 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.02702 [0.02078, 0.03304] S low 0.9572 [0.9493, 0.9663] W multi 0; low 0	S Y/N/Y W N/N/N
Goryachev f=0.3	low contrast	inconclusive	inconclusive	[1269, 1269] resolved	[3.2e+04, ∞] right censored	S 0.01491 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0.004988 [0.002494, 0.007481] S low 0.9925 [0.9875, 0.9975] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_562_S0_H3_candidates_longest_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Goryachev f=1	low contrast	inconclusive	inconclusive	[0, ∞] unavailable no contiguous lag curve	[3.2e+04, ∞] right censored	S 0 / W 1 H 32000 / 128000 95% bound check: ?	S multi 0 [0, 0] S low 1 [1, 1] W multi 0; low 0	S Y/N/Y W N/N/N
Holmes 3 f=0.0003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.001	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.003	single front	inconclusive	inconclusive	[3.2e+04, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 128000 / 128000 95% bound check: N	S multi 0.0001562 [0, 0.0004685] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.005	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 1 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.0004156 [0, 0.001247] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.01	single front	inconclusive	inconclusive	[8000, ∞] right censored	[3.2e+04, ∞] right censored	S 0.9797 / W 1 H 32000 / 128000 95% bound check: N	S multi 0.1966 [0.0349, 0.3847] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.02	single front	criterion met	inconclusive	[3906, 3906] resolved	[3.2e+04, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.4823 [0.3896, 0.5738] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.03	multiple fronts	criterion met	criterion met	[2015, 2015] resolved	[3.2e+04, ∞] right censored	S 0.9841 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.718 [0.6498, 0.7791] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.05	multiple fronts	criterion met	inconclusive	[981.7, 981.7] resolved	[3.2e+04, ∞] right censored	S 0.9783 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8603 [0.8356, 0.8832] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.1	multiple fronts	criterion met	criterion met	[392.1, 392.1] resolved	[3.2e+04, ∞] right censored	S 0.9591 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9175 [0.899, 0.936] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Holmes 3 f=0.3	multiple fronts	criterion met	inconclusive	[104.9, 104.9] resolved	[3.2e+04, ∞] right censored	S 0.9648 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.9302 [0.9252, 0.9352] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_563_S0_H3_candidates_longest_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Holmes 3 f=1	multiple fronts	criterion met	inconclusive	[0, 80] unresolved below two cadences	[3.2e+04, ∞] right censored	S 0.9724 / W 1 H 32000 / 128000 95% bound check: Y	S multi 0.8354 [0.8204, 0.8504] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N
Spring f=0.0003	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.001	single front	inconclusive	inconclusive	[3200, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 12800 / 12800 95% bound check: N	S multi 0 [0, 0] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.003	single front	inconclusive	inconclusive	[800, ∞] right censored	[3200, ∞] right censored	S 1 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.0008313 [0, 0.00187] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.005	single front	criterion met	inconclusive	[534.3, 534.3] resolved	[3200, ∞] right censored	S 0.9978 / W 1 H 3200 / 12800 95% bound check: N	S multi 0.02328 [0.006234, 0.04489] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.01	single front	criterion met	inconclusive	[146.2, 146.2] resolved	[3200, ∞] right censored	S 0.9913 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.1357 [0.1062, 0.1704] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.02	single front	criterion met	inconclusive	[37.83, 37.83] resolved	[3200, ∞] right censored	S 0.984 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.3554 [0.3215, 0.3874] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.03	single front	criterion met	inconclusive	[18.25, 18.25] resolved	[3200, ∞] right censored	S 0.9838 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.4985 [0.4701, 0.5299] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.05	multiple fronts	criterion met	inconclusive	[10.83, 10.83] resolved	[3200, ∞] right censored	S 0.9799 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.6586 [0.6424, 0.6735] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.1	multiple fronts	criterion met	criterion met	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9745 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.787 [0.7695, 0.802] S low 0 [0, 0] W multi 0; low 0	S N/N/N W N/N/N
Spring f=0.3	multiple fronts	criterion met	inconclusive	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9599 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.8628 [0.8454, 0.8803] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_564_S0_H3_candidates_longest_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 candidate components · longest · cadence x8 · all conditions including operationally forced

Model / input	Native mode	Strong-input component	E1 component	Strong τ bounds (native units)	Weak τ bounds (native units)	Coverage S/W Native H S/W	Native largest-size fractions S=strong; W=weak	Forcing M/E1/C
Spring f=1	multiple fronts	criterion met	inconclusive	[0, 8] unresolved below two cadences	[3200, ∞] right censored	S 0.9925 / W 1 H 3200 / 12800 95% bound check: Y	S multi 0.7793 [0.7456, 0.813] S low 0 [0, 0] W multi 0; low 0	S Y/N/Y W N/N/N

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_565_S0_H3_candidates_longest_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.01	not declared	not declared	not declared	1.001 [1, 1.002] N 12/12; valid 1	1.002 [1.002, 1.003] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.02	not declared	not declared	not declared	1.011 [1.002, 1.023] N 12/12; valid 1	1.005 [1.004, 1.007] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.03	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	no eligible check multiple fronts; H 32000 multi 0.868, low 0; forced mean no	1.212 [1.09, 1.34] N 12/12; valid 1	1.009 [1.007, 1.01] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.05	not declared	not declared	not declared	1.668 [1.558, 1.771] N 12/12; valid 1	1.009 [1.006, 1.013] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_566_S0_H3_fingerprints_longest_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Wave-pinning f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.3859, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.9092, low 0; forced mean no	2.309 [2.248, 2.367] N 12/12; valid 1	1.024 [1.017, 1.031] N 12/12; valid 1	count yes contrast yes
Wave-pinning f=0.3	not declared	not declared	not declared	2.482 [2.376, 2.593] N 2/2; valid 1	1.124 [1.11, 1.139] N 2/2; valid 1	count yes contrast yes
Wave-pinning f=1	not declared	not declared	not declared	2.507 [2.432, 2.583] N 2/2; valid 1	1.25 [1.233, 1.268] N 2/2; valid 1	count yes contrast yes
Goryachev f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.008, 1.008] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.029 [1.029, 1.03] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.115 [1.113, 1.117] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.005	not declared	not declared	not declared	0.9581 [0.936, 0.9784] N 12/12; valid 1	1.484 [1.393, 1.597] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.01	not declared	not declared	not declared	0.7913 [0.7772, 0.8073] N 12/12; valid 1	3.342 [3.279, 3.408] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_567_S0_H3_fingerprints_longest_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Goryachev f=0.015	not declared	not declared	not declared	unavailable	unavailable	count unavailable contrast unavailable
Goryachev f=0.02	no eligible check single front; H 32000 multi 0, low 0.01559; forced mean no	no eligible check single front; H 32000 multi 0, low 0.1407; forced mean no	no eligible check single front; H 32000 multi 0, low 0; forced mean no	0.8003 [0.7627, 0.8608] N 12/12; valid 1	2.556 [2.371, 2.688] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.03	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	low contrast; H 32000 qualified yes; same yes multi 0, low 1; forced mean yes	single front; H 32000 qualified yes; same no multi 0.008105, low 0; forced mean yes	1.454 [1.386, 1.527] N 12/12; valid 1	0.4761 [0.4656, 0.4867] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.05	not declared	not declared	not declared	1.348 [1.277, 1.415] N 12/12; valid 1	0.4608 [0.45, 0.4716] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.1	not declared	not declared	not declared	1.206 [1.133, 1.282] N 12/12; valid 1	0.4371 [0.4272, 0.4482] N 12/12; valid 1	count yes contrast yes
Goryachev f=0.3	not declared	not declared	not declared	0.654 [0.6049, 0.7068] N 2/2; valid 1	0.4247 [0.4159, 0.4336] N 2/2; valid 1	count yes contrast yes
Goryachev f=1	not declared	not declared	not declared	0.07134 [0.05671, 0.08629] N 12/2; valid 1	0.4328 [0.4281, 0.4376] N 12/2; valid 1	count yes contrast yes
Holmes 3 f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_568_S0_H3_fingerprints_longest_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.001 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.005 [1.004, 1.005] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.005	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.008 [1.007, 1.008] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.01	not declared	not declared	not declared	1.194 [1.022, 1.375] N 12/12; valid 1	1.013 [1.012, 1.014] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.02	not declared	not declared	not declared	1.509 [1.389, 1.617] N 12/12; valid 1	1.029 [1.026, 1.031] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.03	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	single front; H 32000 qualified yes; same no multi 0.01143, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8543, low 0; forced mean no	1.771 [1.657, 1.891] N 12/12; valid 1	1.049 [1.045, 1.053] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.05	not declared	not declared	not declared	2.012 [1.942, 2.085] N 12/12; valid 1	1.076 [1.07, 1.083] N 12/12; valid 1	count yes contrast yes
Holmes 3 f=0.1	single front; H 32000 qualified yes; same no multi 0, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.7014, low 0; forced mean no	multiple fronts; H 32000 qualified yes; same yes multi 0.8103, low 0; forced mean no	2.11 [2.05, 2.176] N 12/12; valid 1	1.134 [1.123, 1.146] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_569_S0_H3_fingerprints_longest_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Holmes 3 f=0.3	not declared	not declared	not declared	2.354 [2.32, 2.389] N 2/2; valid 1	1.305 [1.285, 1.325] N 2/2; valid 1	count yes contrast yes
Holmes 3 f=1	not declared	not declared	not declared	2.215 [2.078, 2.359] N 2/2; valid 1	1.692 [1.633, 1.755] N 2/2; valid 1	count yes contrast yes
Spring f=0.0003	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1 [1, 1.001] N 12/12; valid 1	count yes contrast yes
Spring f=0.001	not declared	not declared	not declared	1 [1, 1] N 12/12; valid 1	1.002 [1.001, 1.002] N 12/12; valid 1	count yes contrast yes
Spring f=0.003	not declared	not declared	not declared	1 [1, 1.001] N 12/12; valid 1	1.003 [1.002, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.005	not declared	not declared	not declared	1.021 [1.004, 1.043] N 12/12; valid 1	1.007 [1.005, 1.009] N 12/12; valid 1	count yes contrast yes
Spring f=0.01	not declared	not declared	not declared	1.121 [1.094, 1.153] N 12/12; valid 1	1.001 [0.9976, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.02	not declared	not declared	not declared	1.298 [1.265, 1.33] N 12/12; valid 1	1 [0.9938, 1.006] N 12/12; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_570_S0_H3_fingerprints_longest_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · H3 preset-σ checks and optical quotients · longest · cadence x8 · all conditions including operationally forced

Model / input	Preset σ=0.25	Preset σ=0.5	Preset σ=2	Raw optical count large/small quotient CI	Raw normalized contrast large/small quotient CI	Candidate fingerprint available after qualification
Spring f=0.03	no eligible check single front; H 3200 multi 0, low 0; forced mean no	no eligible check single front; H 3200 multi 0.1727, low 0; forced mean no	no eligible check multiple fronts; H 3200 multi 0.5106, low 0; forced mean no	1.434 [1.397, 1.473] N 12/12; valid 1	0.9973 [0.9902, 1.004] N 12/12; valid 1	count yes contrast yes
Spring f=0.05	not declared	not declared	not declared	1.567 [1.538, 1.592] N 12/12; valid 1	1.012 [1, 1.024] N 12/12; valid 1	count yes contrast yes
Spring f=0.1	multiple fronts; H 3200 qualified yes; same yes multi 0.5576, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.7423, low 0; forced mean no	multiple fronts; H 3200 qualified yes; same yes multi 0.6027, low 0; forced mean no	1.701 [1.68, 1.722] N 12/12; valid 1	1.033 [1.018, 1.05] N 12/12; valid 1	count yes contrast yes
Spring f=0.3	not declared	not declared	not declared	1.762 [1.704, 1.821] N 2/2; valid 1	1.172 [1.093, 1.26] N 2/2; valid 1	count yes contrast yes
Spring f=1	not declared	not declared	not declared	1.759 [1.752, 1.766] N 2/2; valid 1	1.25 [1.225, 1.276] N 2/2; valid 1	count yes contrast yes

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Empty E1 checks outside the preset two-input grid mean no declared corroboration at that input, not missing executed native work. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_571_S0_H3_fingerprints_longest_cadence_x8_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.05 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 1.668 [1.558, 1.771] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.009 [1.006, 1.013] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / criterion met	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_572_S0_H3_pairs_longest_cadence_x8_inclusive_001.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=0.1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: criterion met / inconclusive	A 2.309 [2.248, 2.367] N 12/12; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.024 [1.017, 1.031] N 12/12; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=0.3 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.482 [2.376, 2.593] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.124 [1.11, 1.139] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.03 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / criterion met	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.454 [1.386, 1.527] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4761 [0.4656, 0.4867] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_573_S0_H3_pairs_longest_cadence_x8_inclusive_002.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Wave-pinning f=1 multiple fronts	Goryachev f=0.05 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.348 [1.277, 1.415] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4608 [0.45, 0.4716] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 1.206 [1.133, 1.282] N 12/12; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4371 [0.4272, 0.4482] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=0.3 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.654 [0.6049, 0.7068] N 2/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4247 [0.4159, 0.4336] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Wave-pinning f=1 multiple fronts	Goryachev f=1 low contrast	inconclusive	Strong A/B: criterion met / inconclusive E1 A/B: inconclusive / inconclusive	A 2.507 [2.432, 2.583] N 2/2; valid 1 B 0.07134 [0.05671, 0.08629] N 12/2; valid 1	A 1.25 [1.233, 1.268] N 2/2; valid 1 B 0.4328 [0.4281, 0.4376] N 12/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_574_S0_H3_pairs_longest_cadence_x8_inclusive_003.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.03 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / criterion met	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.03 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: criterion met / inconclusive	A 1.454 [1.386, 1.527] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4761 [0.4656, 0.4867] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_575_S0_H3_pairs_longest_cadence_x8_inclusive_004.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.05 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.05 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.348 [1.277, 1.415] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4608 [0.45, 0.4716] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_576_S0_H3_pairs_longest_cadence_x8_inclusive_005.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 1.206 [1.133, 1.282] N 12/12; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4371 [0.4272, 0.4482] N 12/12; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_577_S0_H3_pairs_longest_cadence_x8_inclusive_006.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=0.3 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=0.3 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.654 [0.6049, 0.7068] N 2/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4247 [0.4159, 0.4336] N 2/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_578_S0_H3_pairs_longest_cadence_x8_inclusive_007.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Holmes 3 f=0.03 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.771 [1.657, 1.891] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.049 [1.045, 1.053] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.012 [1.942, 2.085] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.076 [1.07, 1.083] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.11 [2.05, 2.176] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.134 [1.123, 1.146] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.354 [2.32, 2.389] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.305 [1.285, 1.325] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Holmes 3 f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 2.215 [2.078, 2.359] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.692 [1.633, 1.755] N 2/2; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.05 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.567 [1.538, 1.592] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.012 [1, 1.024] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / criterion met	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.701 [1.68, 1.722] N 12/12; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.033 [1.018, 1.05] N 12/12; valid 1	criterion met	count criterion met contrast criterion met
Goryachev f=1 low contrast	Spring f=0.3 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.762 [1.704, 1.821] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.172 [1.093, 1.26] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

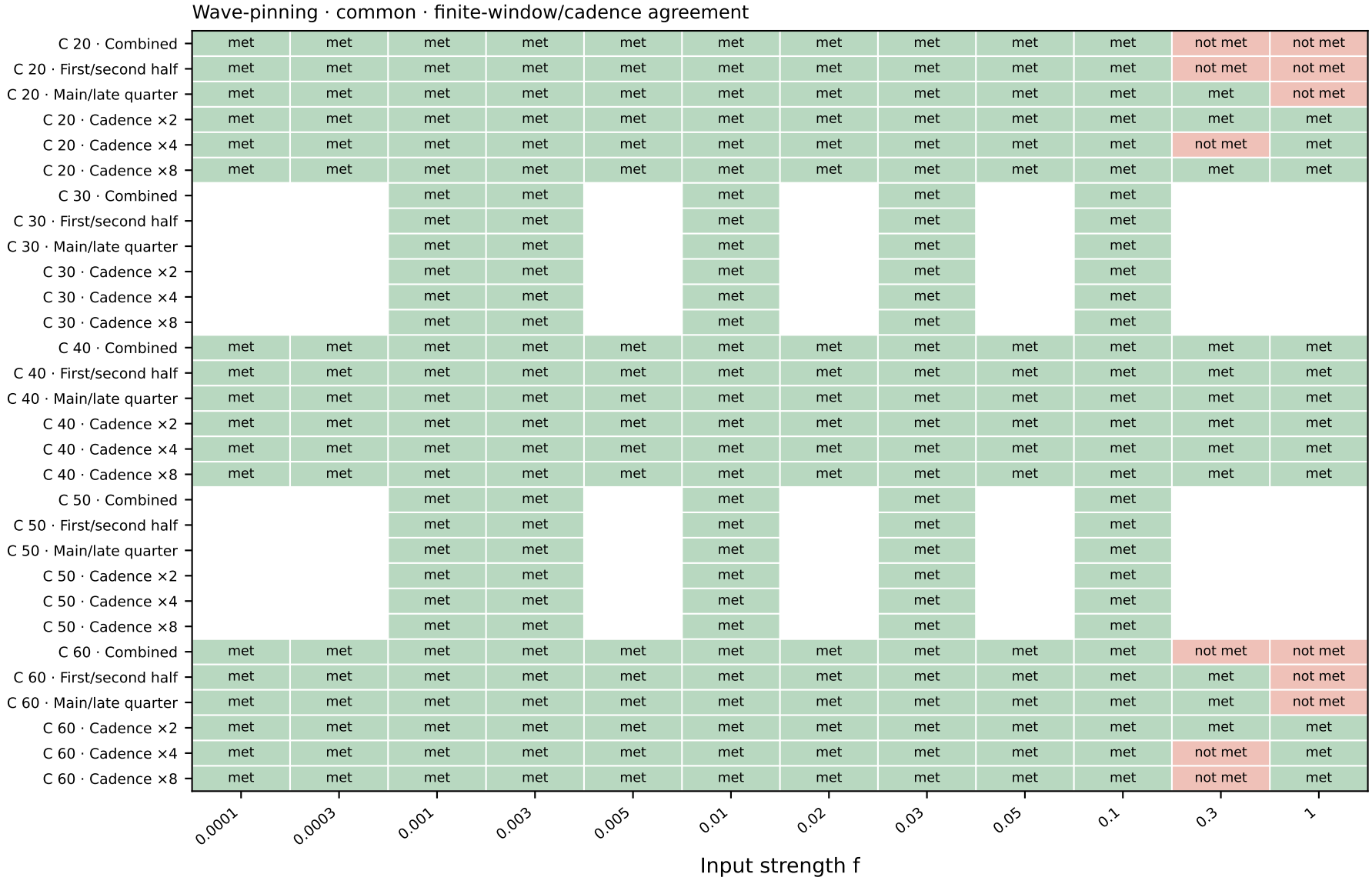
Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_579_S0_H3_pairs_longest_cadence_x8_inclusive_008.csv

S0 · Completed audited Figure 4 criteria and qualifications · Every preset H3 model pair · longest · cadence x8 · all conditions including operationally forced

First model/input (A)	Second model/input (B)	Audited pair status	Strong / E1 components	Optical count quotient A / B (each within model)	Normalized contrast quotient A / B (each within model)	Fingerprint pair separation status	Separate quotient comparison statuses
Goryachev f=1 low contrast	Spring f=1 multiple fronts	inconclusive	Strong A/B: inconclusive / criterion met E1 A/B: inconclusive / inconclusive	A 0.07134 [0.05671, 0.08629] N 12/2; valid 1 B 1.759 [1.752, 1.766] N 2/2; valid 1	A 0.4328 [0.4281, 0.4376] N 12/2; valid 1 B 1.25 [1.225, 1.276] N 2/2; valid 1	criterion met	count criterion met contrast criterion met

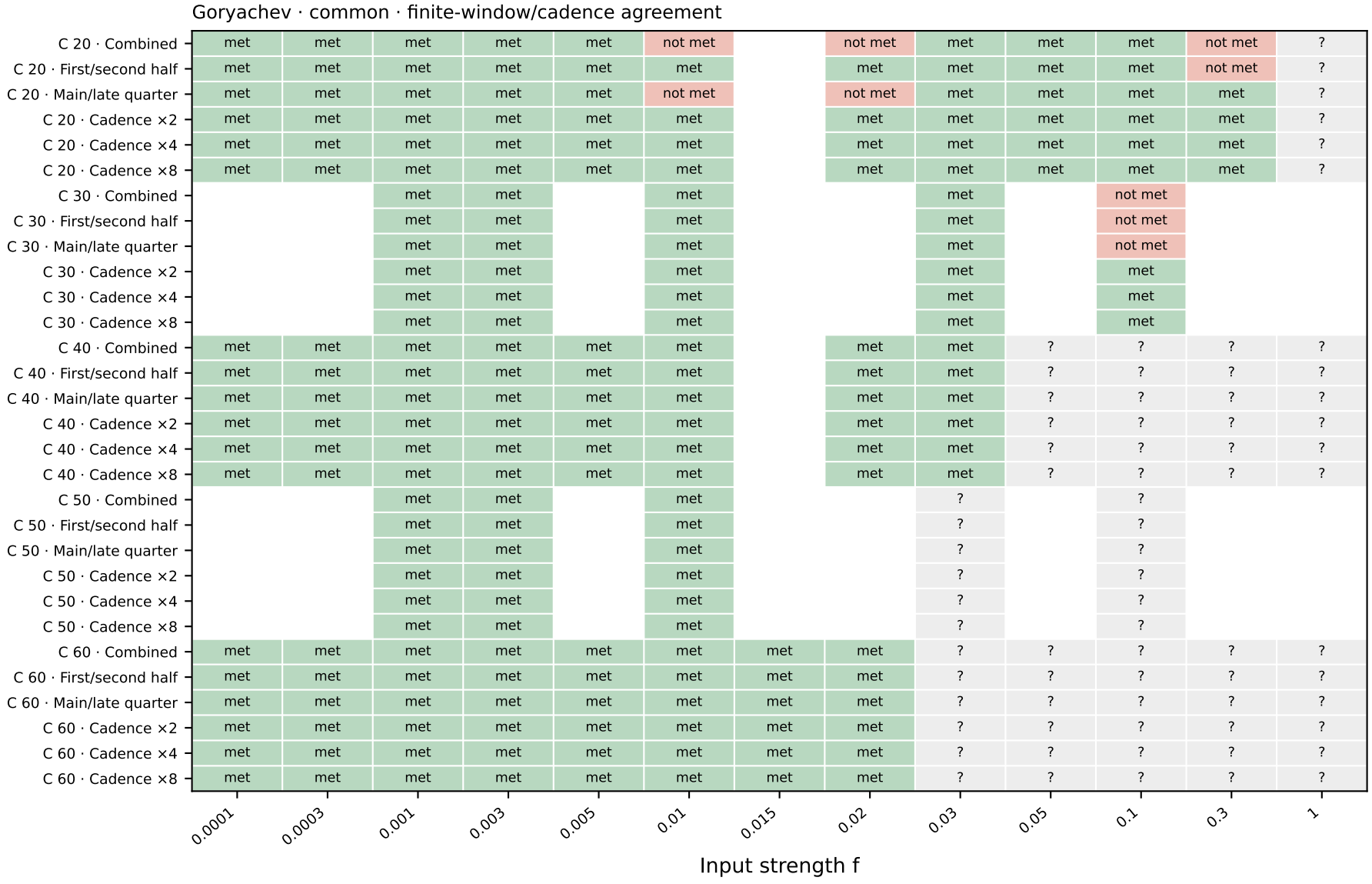
Completed audited Figure 4 criteria and qualifications. All values project existing audited truth. S/W=strong/weak. Native A4/E1 raw mode and fractions are separate from A5 optical fingerprints. Strong/weak τ intervals are observation bounds; weak input is the actual f=0.0001. The existing strong-input test uses strong upper < 0.5×weak lower, subject to source eligibility/forcing. M/E1/C flags mean matched-mean/preset-E1/combined. The 95% bound check is the existing confidence-bound qualifier. Y=yes, N=no, ?=unavailable. Raw quotient CIs do not override excluded candidate fingerprints; upper∞ follows the source unbounded flag. Whole-seed sizes/valid-bootstrap fractions remain explicit. No component or H3 gate is redefined. Pair status and every nested comparison are copied unchanged; only within-model dimensionless effects are compared across models, never raw activity/counts. Empty context pages state zero planned pairs explicitly. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_580_S0_H3_pairs_longest_cadence_x8_inclusive_009.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



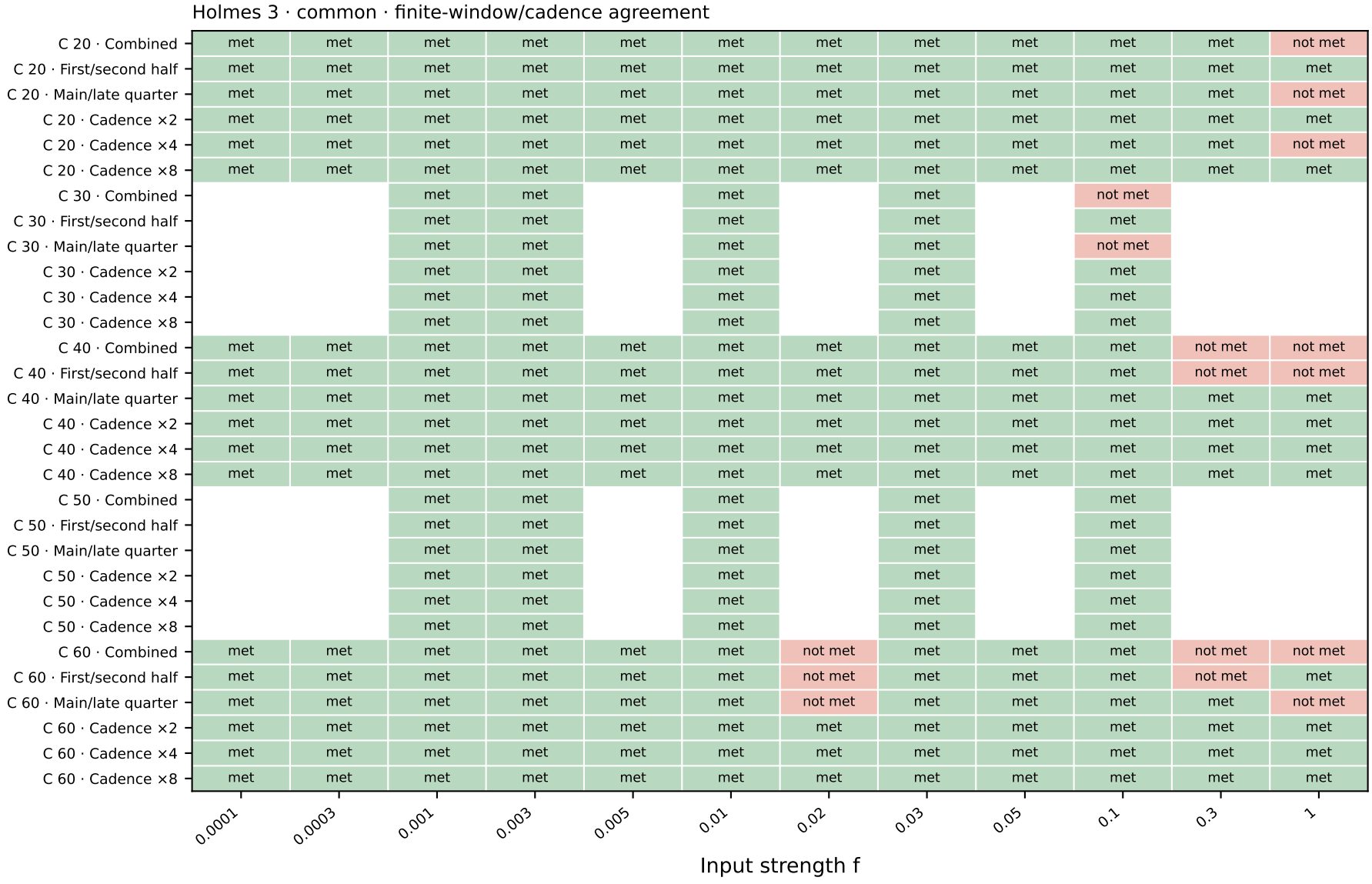
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_581_S12_windows_common_wave_pinning.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



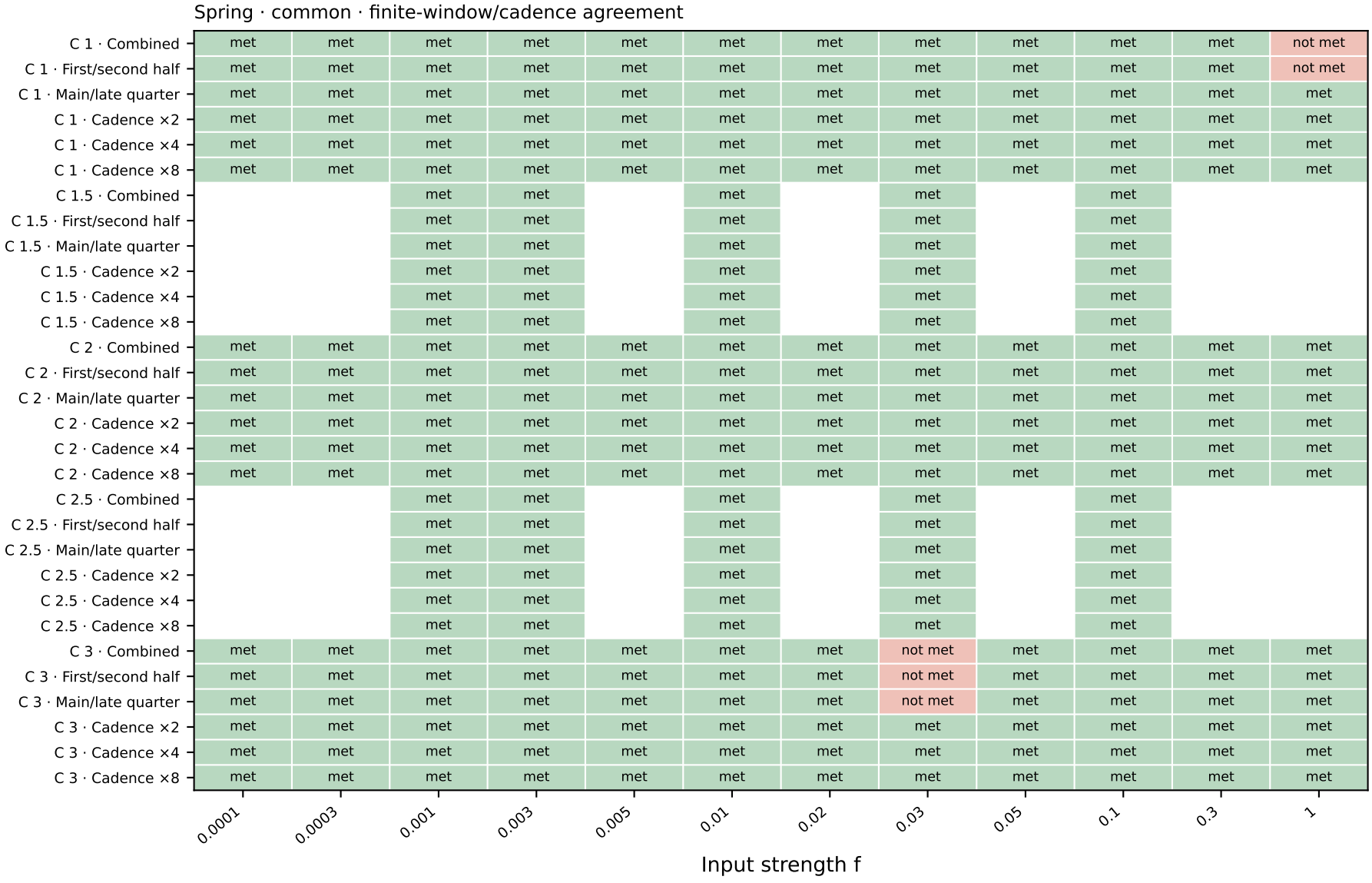
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_582_S12_windows_common_goryachev.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



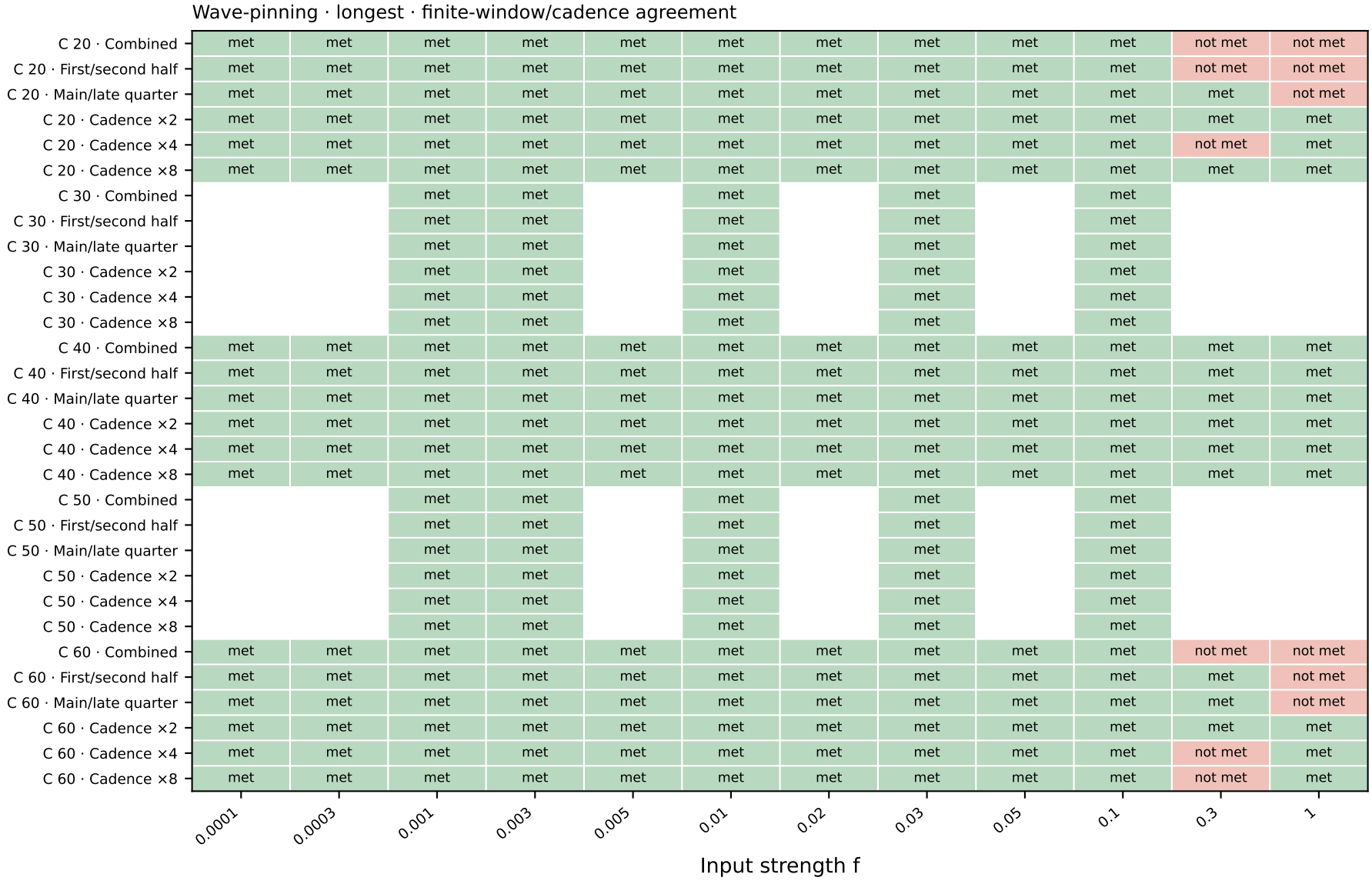
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_583_S12_windows_common_holmes_model3.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



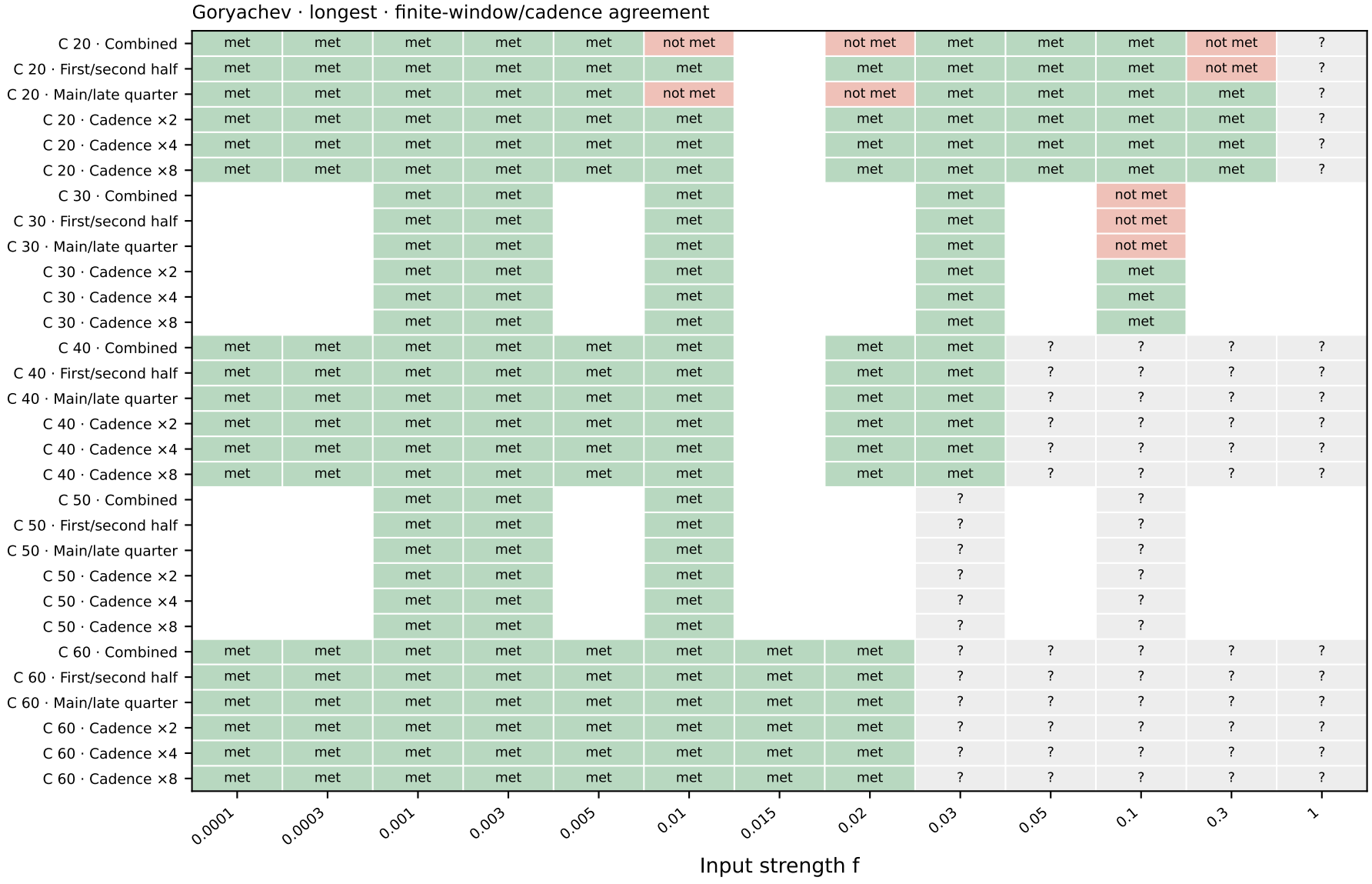
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_584_S12_windows_common_spring_ring.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



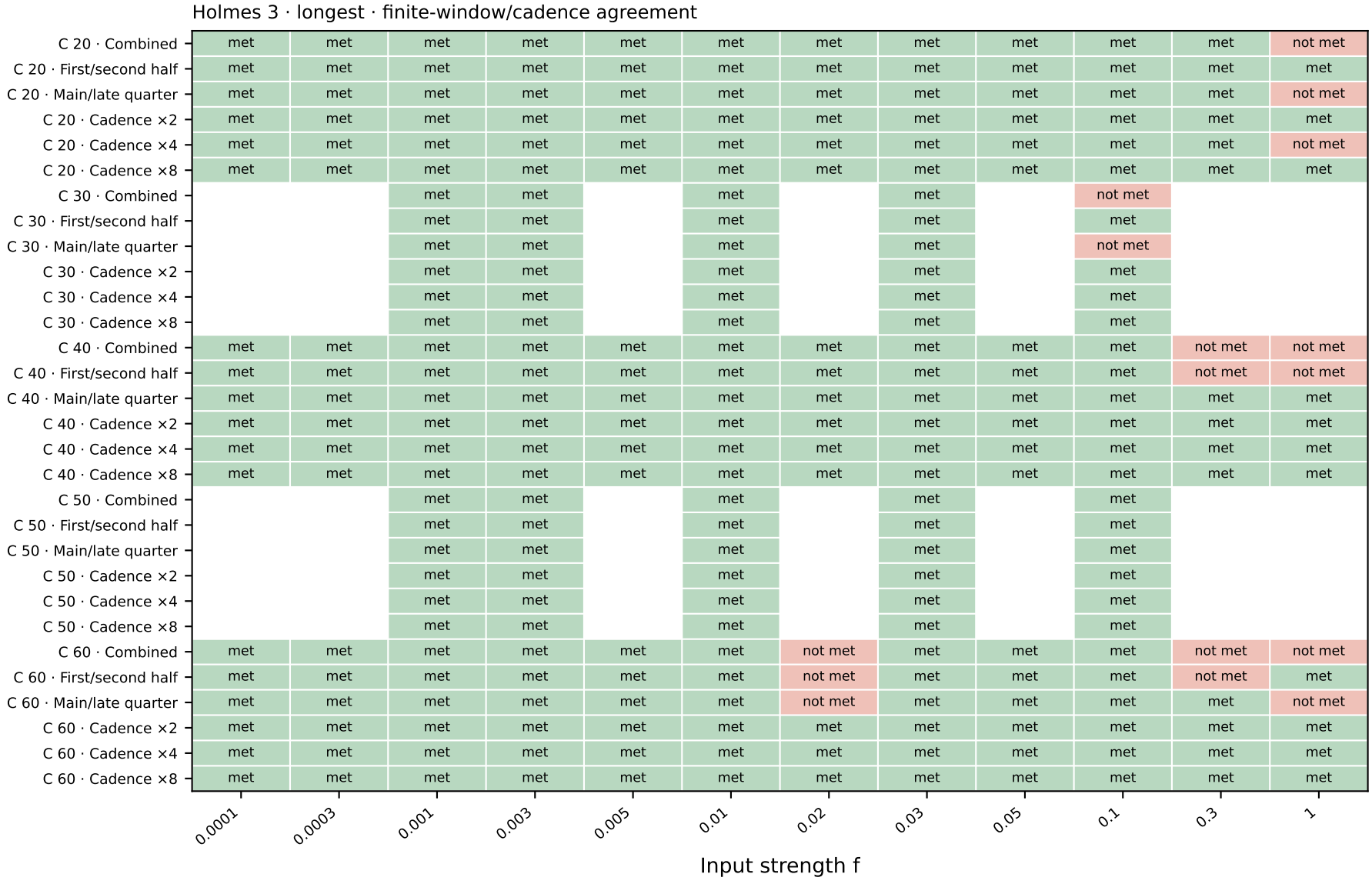
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_585_S12_windows_longest_wave_pinning.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



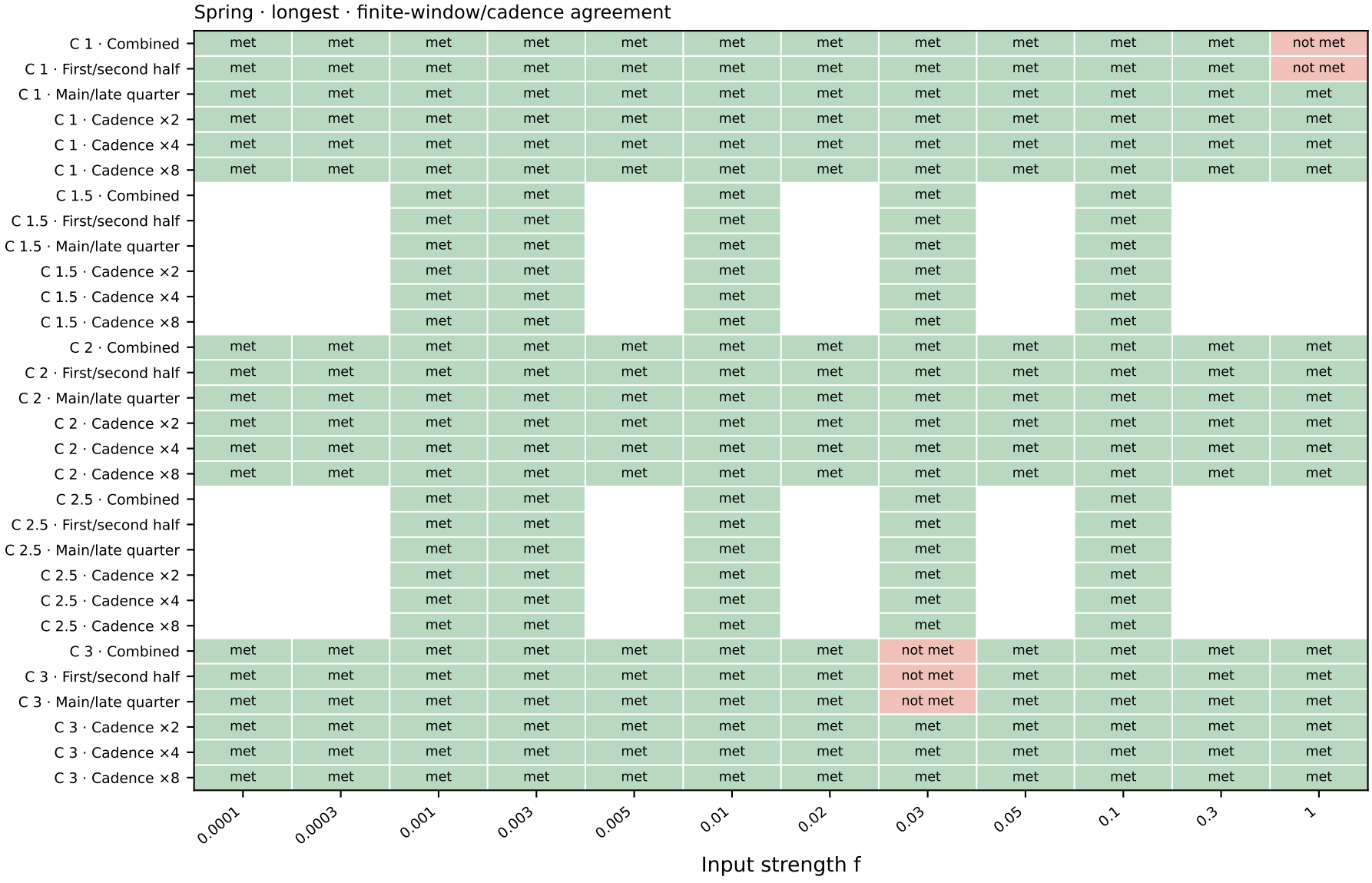
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_586_S12_windows_longest_goryachev.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



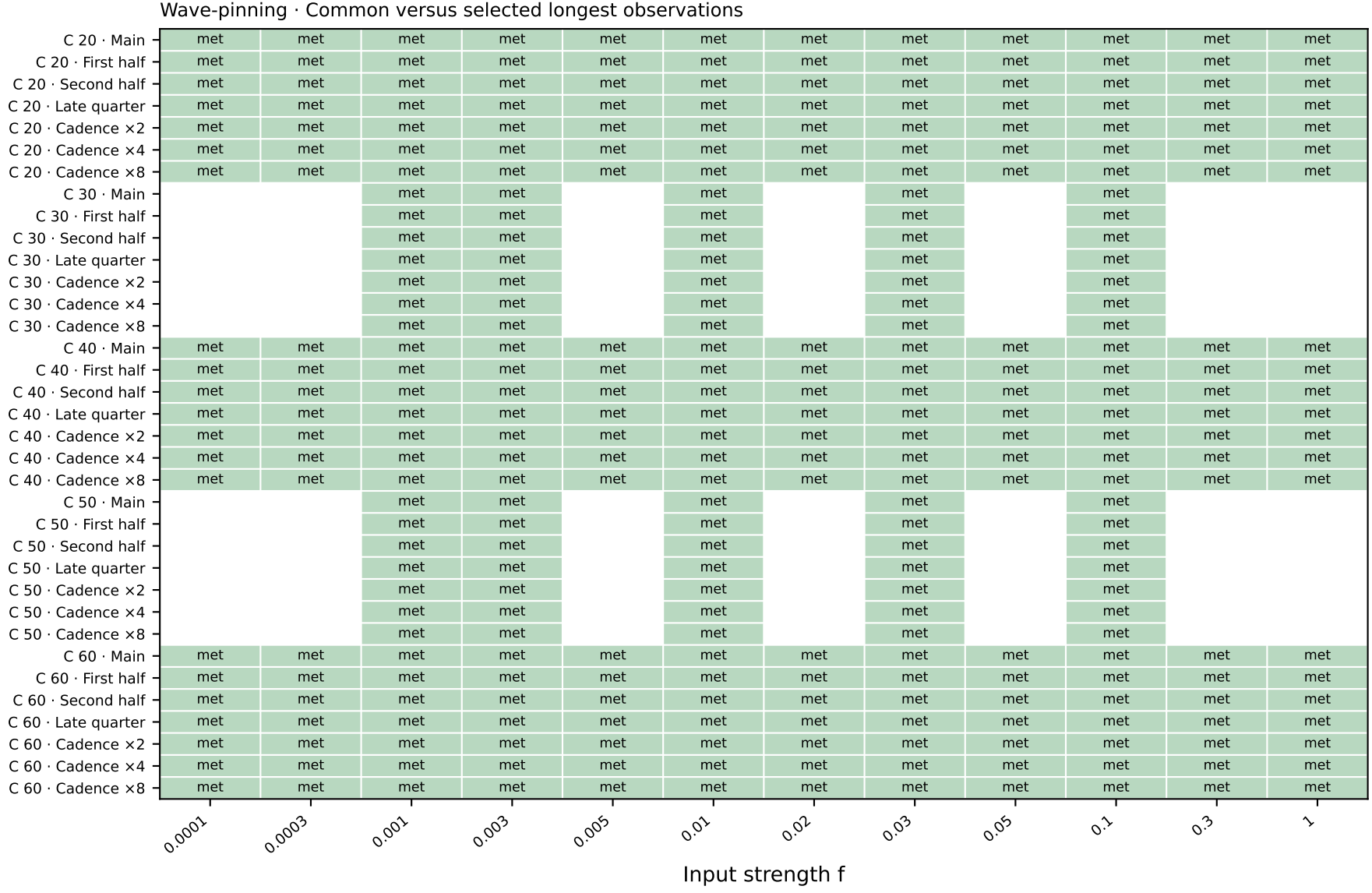
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_587_S12_windows_longest_holmes_model3.csv

S12 · Completed audited Figure 4 criteria and qualifications · A7 qualification by condition and comparison



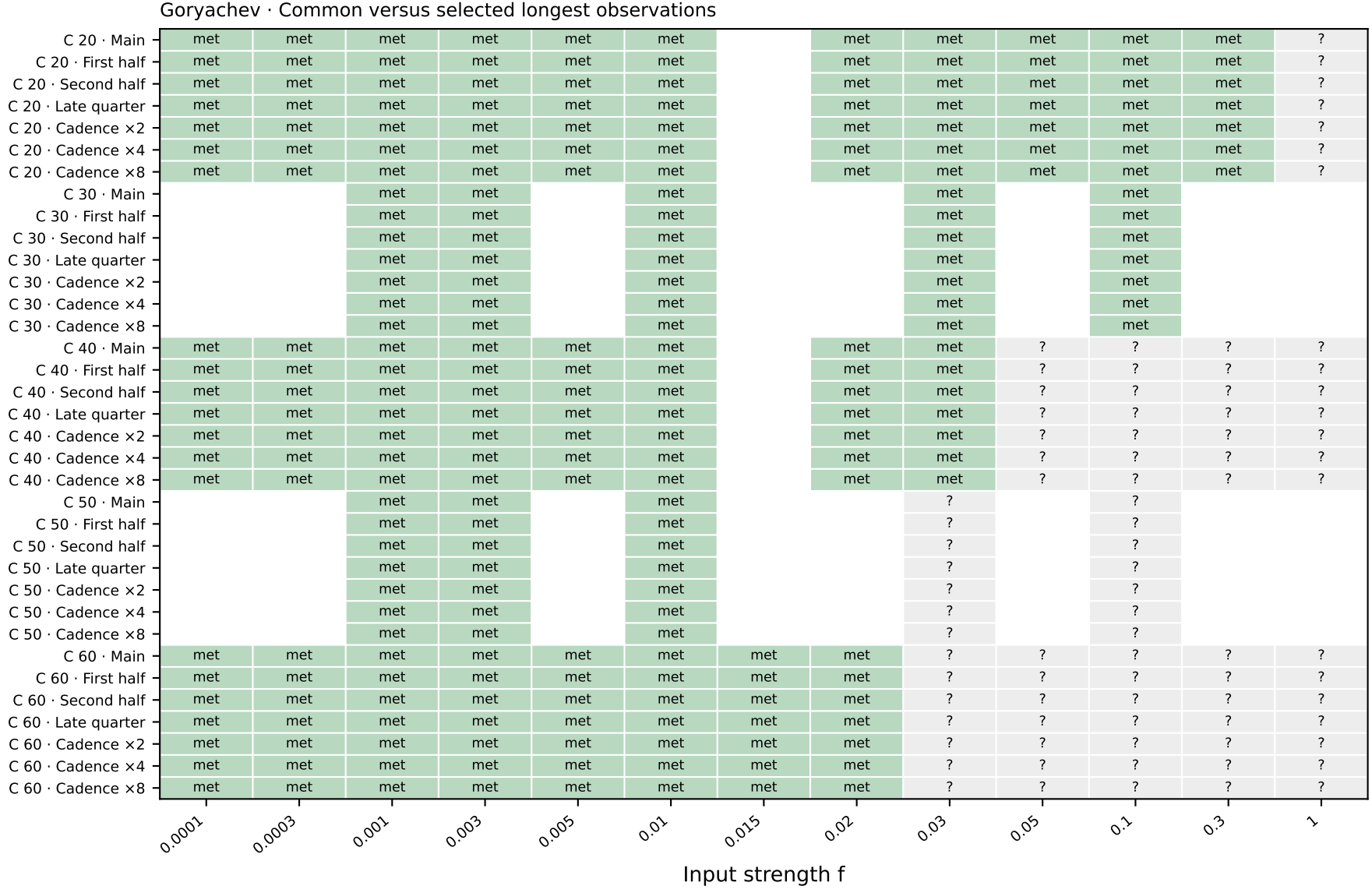
Completed audited Figure 4 criteria and qualifications. These are interval-and-native-regime agreement checks. Angular-mobility disagreement remains separately recorded in components. Blank cells are outside the declared grid. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_588_S12_windows_longest_spring_ring.csv

S12 · Completed audited Figure 4 criteria and qualifications · Common versus selected longest observations



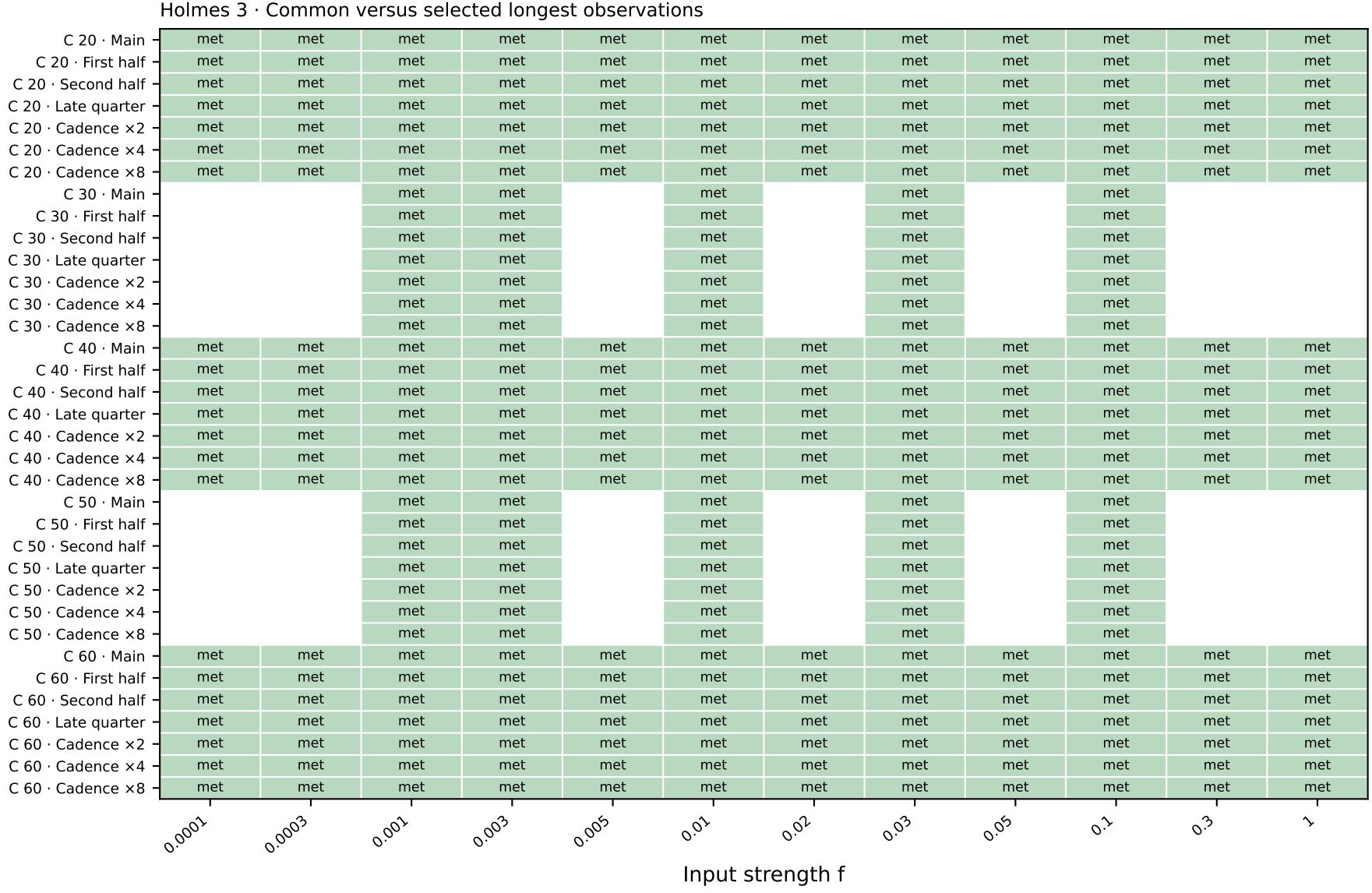
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_589_S12_horizon_qualification_wave_pinning.csv

S12 · Completed audited Figure 4 criteria and qualifications · Common versus selected longest observations



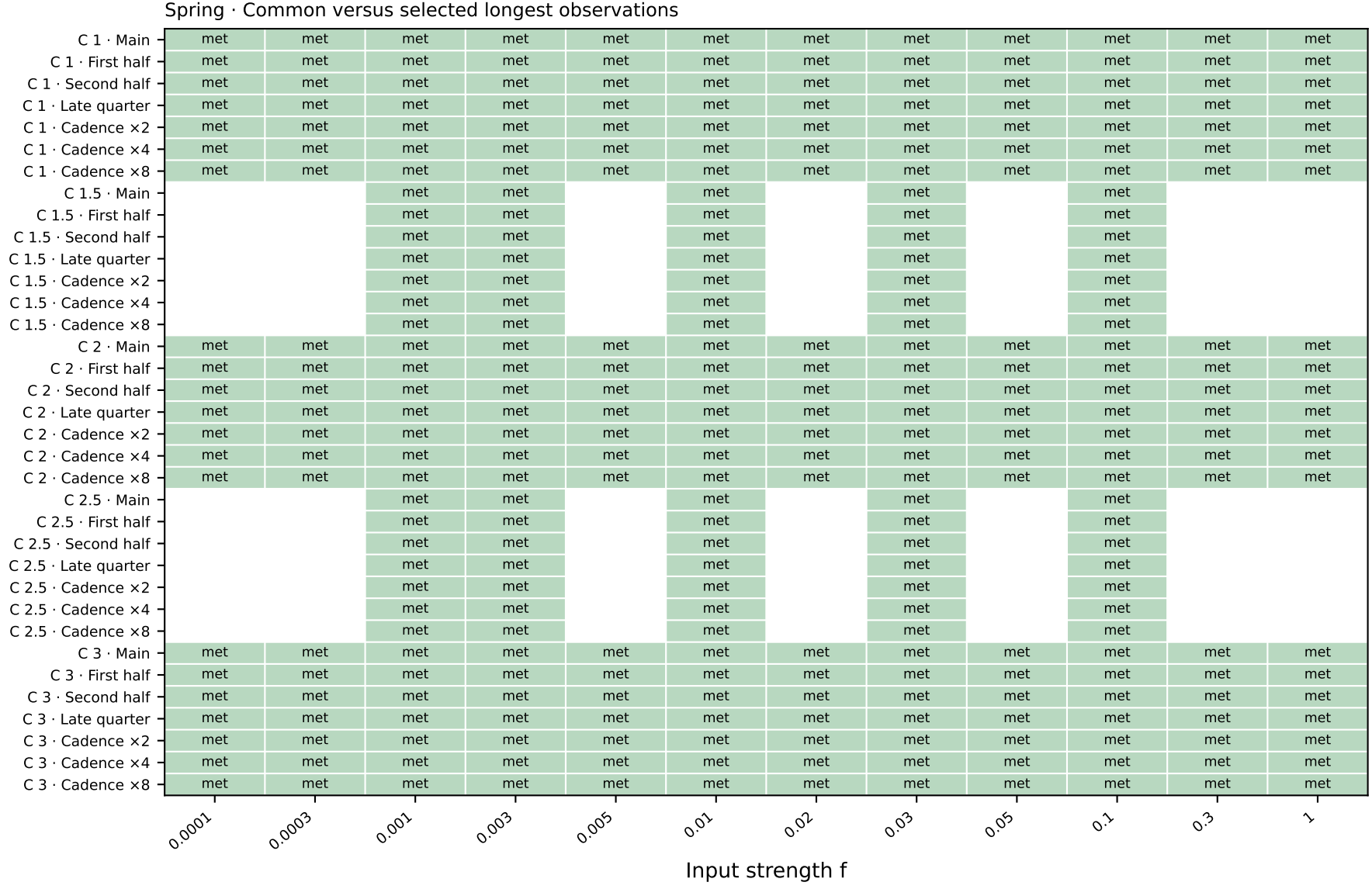
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_590_S12_horizon_qualification_goryachev.csv

S12 · Completed audited Figure 4 criteria and qualifications · Common versus selected longest observations



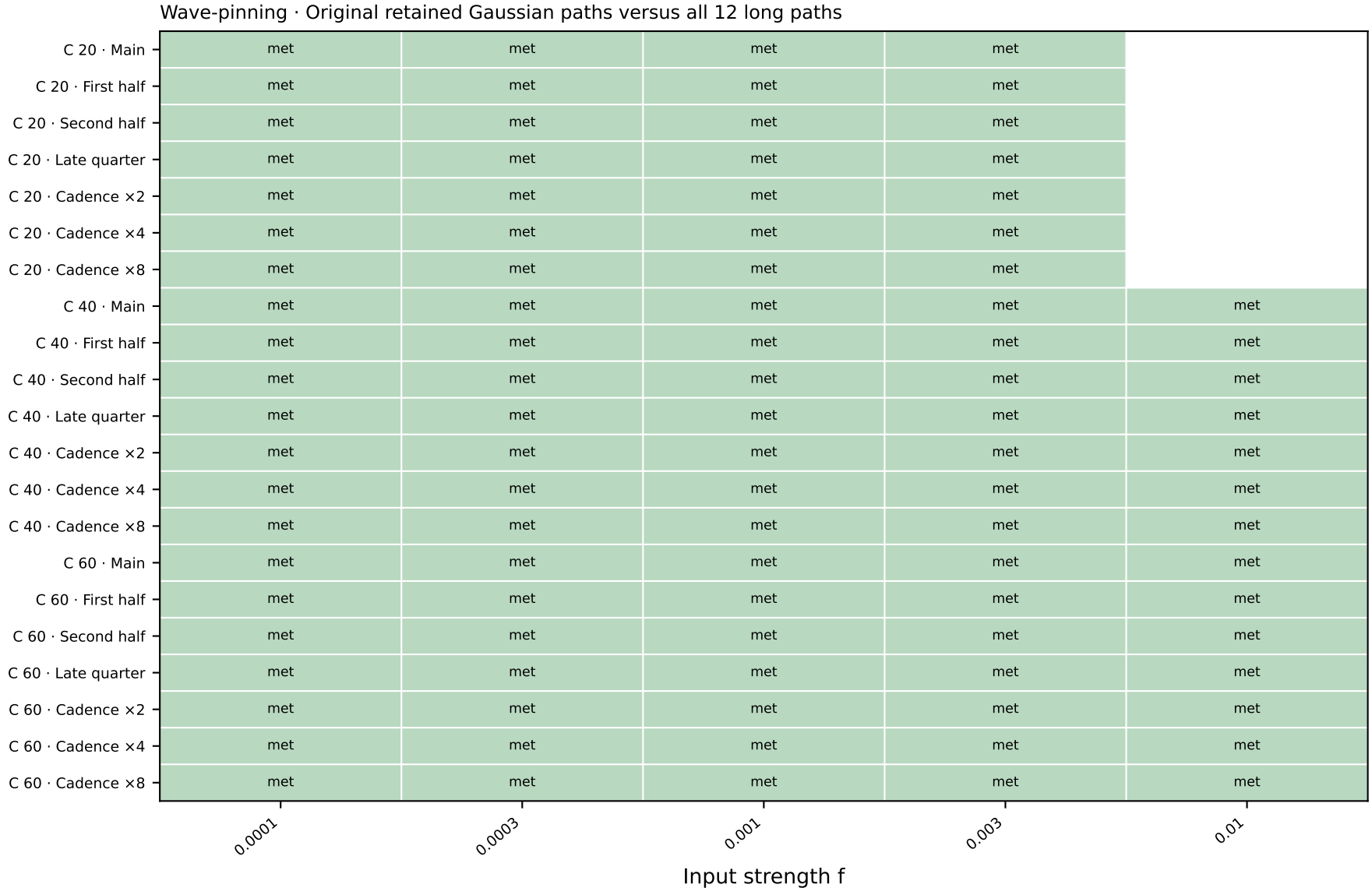
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_591_S12_horizon_qualification_holmes_model3.csv

S12 · Completed audited Figure 4 criteria and qualifications · Common versus selected longest observations



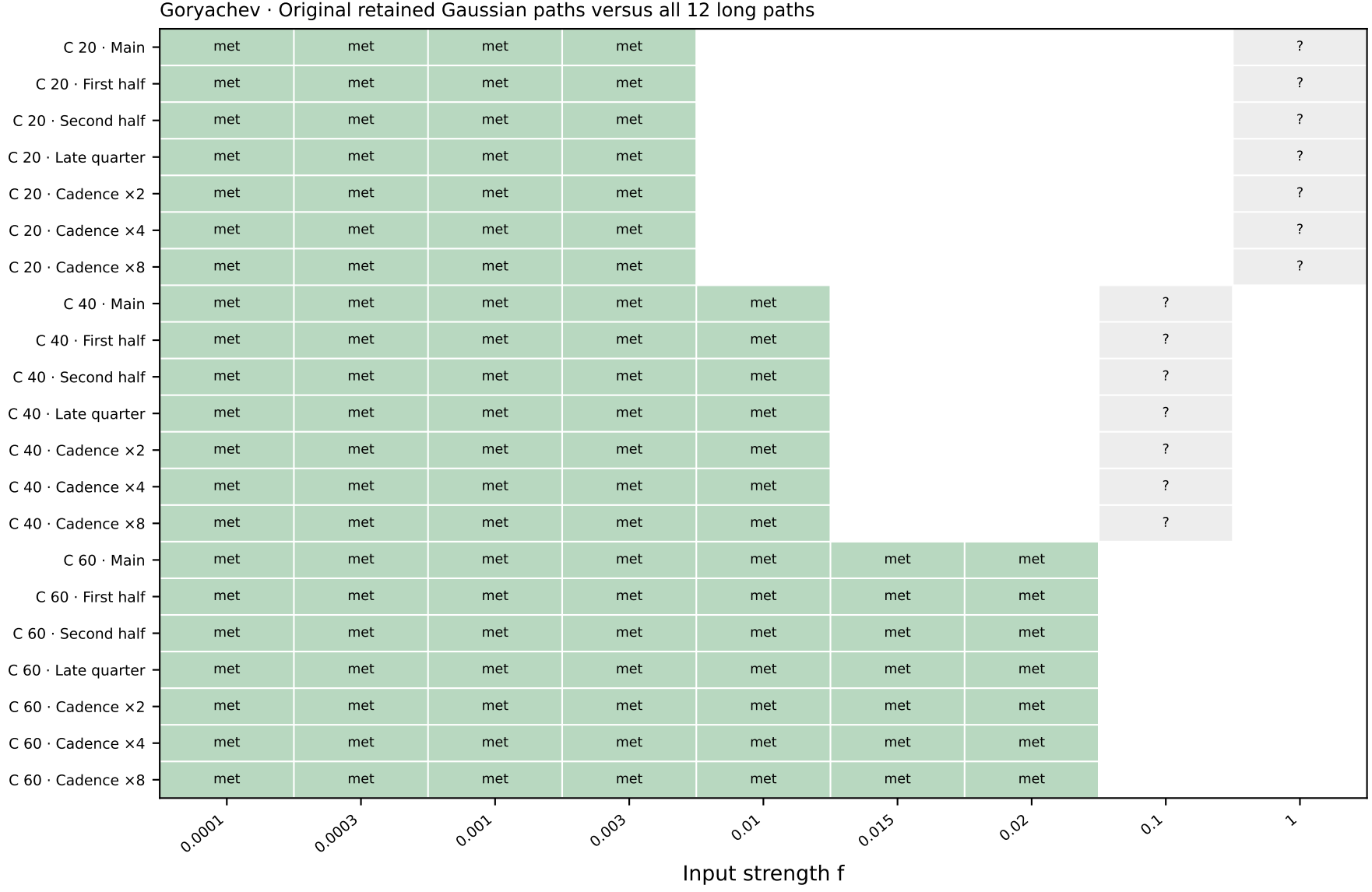
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_592_S12_horizon_qualification_spring_ring.csv

S12 · Completed audited Figure 4 criteria and qualifications · Original retained Gaussian paths versus all 12 long paths



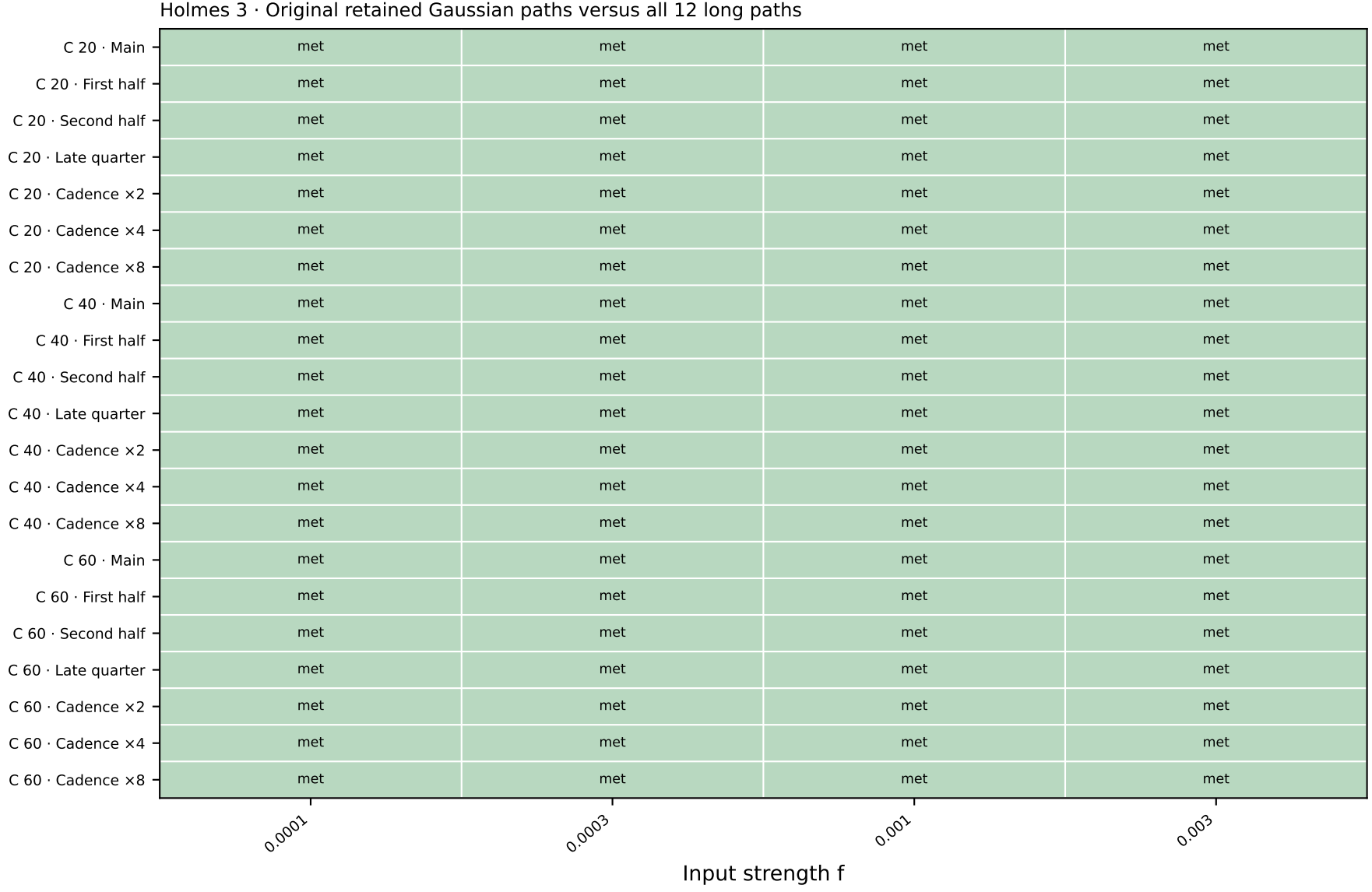
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_593_S12_original_seed_sensitivity_wave_pinning.csv

S12 · Completed audited Figure 4 criteria and qualifications · Original retained Gaussian paths versus all 12 long paths



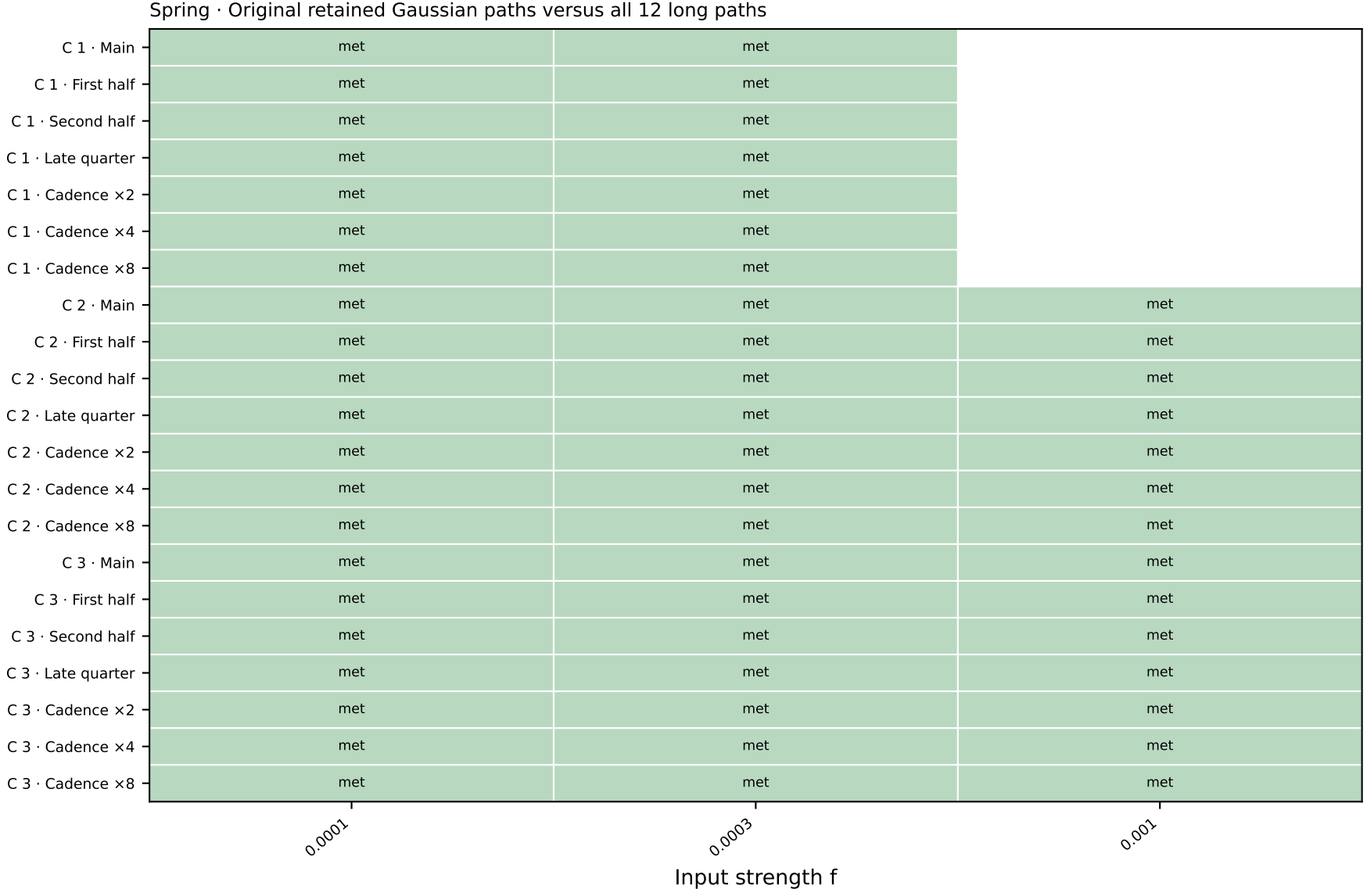
Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_594_S12_original_seed_sensitivity_goryachev.csv

S12 · Completed audited Figure 4 criteria and qualifications · Original retained Gaussian paths versus all 12 long paths



Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_595_S12_original_seed_sensitivity_holmes_model3.csv

S12 · Completed audited Figure 4 criteria and qualifications · Original retained Gaussian paths versus all 12 long paths



Completed audited Figure 4 criteria and qualifications. All seven views remain distinct. Exact original paths are a same-horizon subset, not additional independent replicates. A non-selected common observation is retained explicitly; no missing long observation is replaced. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_596_S12_original_seed_sensitivity_spring_ring.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 20, 0.0001, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.001, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.003, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.01, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_597_S12_E3_references_01.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 20, 0.03, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.03, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.1, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, main: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 0.3, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 20, 1, main: present in common arm yes; retained E3 reference.
goryachev, 20, 1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 20, 1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 20, 1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 20, 1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 20, 1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 20, 1, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0001, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_598_S12_E3_references_02.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 40, 0.0003, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.001, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.003, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.01, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.03, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_599_S12_E3_references_03.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 40, 0.1, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.1, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, main: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 0.3, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 40, 1, main: present in common arm yes; retained E3 reference.
goryachev, 40, 1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 40, 1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 40, 1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 40, 1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 40, 1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 40, 1, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.0003, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_600_S12_E3_references_04.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 60, 0.001, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.001, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.003, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.01, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.015, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.02, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_601_S12_E3_references_05.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

goryachev, 60, 0.03, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.03, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.1, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, main: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 0.3, cadence x8: present in common arm yes; retained E3 reference.
goryachev, 60, 1, main: present in common arm yes; retained E3 reference.
goryachev, 60, 1, analysis first half: present in common arm yes; retained E3 reference.
goryachev, 60, 1, analysis second half: present in common arm yes; retained E3 reference.
goryachev, 60, 1, late quarter: present in common arm yes; retained E3 reference.
goryachev, 60, 1, cadence x2: present in common arm yes; retained E3 reference.
goryachev, 60, 1, cadence x4: present in common arm yes; retained E3 reference.
goryachev, 60, 1, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0001, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_602_S12_E3_references_06.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 20, 0.0003, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.001, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.003, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.01, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.03, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_603_S12_E3_references_07.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 20, 0.1, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.1, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 0.3, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, main: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 20, 1, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.0003, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_604_S12_E3_references_08.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 40, 0.001, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.001, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.003, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.01, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.03, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.1, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_605_S12_E3_references_09.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 40, 0.3, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 0.3, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, main: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 40, 1, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.001, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_606_S12_E3_references_10.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 60, 0.003, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.003, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.01, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.03, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.1, cadence x8: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 0.3, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_607_S12_E3_references_11.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

holmes model3, 60, 1, main: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, analysis first half: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, analysis second half: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, late quarter: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, cadence x2: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, cadence x4: present in common arm yes; retained E3 reference.
holmes model3, 60, 1, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.001, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.003, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_608_S12_E3_references_12.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

spring ring, 1, 0.01, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.01, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.03, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.1, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, main: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 0.3, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 1, 1, main: present in common arm yes; retained E3 reference.
spring ring, 1, 1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 1, 1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 1, 1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 1, 1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 1, 1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 1, 1, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_609_S12_E3_references_13.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

spring ring, 2, 0.0001, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.001, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.003, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.01, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_610_S12_E3_references_14.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

spring ring, 2, 0.03, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.03, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.1, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, main: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 0.3, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 2, 1, main: present in common arm yes; retained E3 reference.
spring ring, 2, 1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 2, 1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 2, 1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 2, 1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 2, 1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 2, 1, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0001, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_611_S12_E3_references_15.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

spring ring, 3, 0.0003, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.001, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.003, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.01, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.03, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_612_S12_E3_references_16.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

spring ring, 3, 0.1, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.1, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, main: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 0.3, cadence x8: present in common arm yes; retained E3 reference.
spring ring, 3, 1, main: present in common arm yes; retained E3 reference.
spring ring, 3, 1, analysis first half: present in common arm yes; retained E3 reference.
spring ring, 3, 1, analysis second half: present in common arm yes; retained E3 reference.
spring ring, 3, 1, late quarter: present in common arm yes; retained E3 reference.
spring ring, 3, 1, cadence x2: present in common arm yes; retained E3 reference.
spring ring, 3, 1, cadence x4: present in common arm yes; retained E3 reference.
spring ring, 3, 1, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.0003, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_613_S12_E3_references_17.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

wave pinning, 20, 0.001, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.001, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.003, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.01, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.03, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.1, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_614_S12_E3_references_18.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

wave pinning, 20, 0.3, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 0.3, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, main: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 20, 1, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.001, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_615_S12_E3_references_19.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

wave pinning, 40, 0.003, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.003, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.01, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.03, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.1, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 0.3, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_616_S12_E3_references_20.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

wave pinning, 40, 1, main: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 40, 1, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0001, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.0003, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.001, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.003, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_617_S12_E3_references_21.csv

S12 · Completed audited Figure 4 criteria and qualifications · E3 retained-reference inventory

Retained E3 observations remain references; they do not replace canonical common observations.

wave pinning, 60, 0.01, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.01, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.03, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.1, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 0.3, cadence x8: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, main: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, analysis first half: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, analysis second half: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, late quarter: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, cadence x2: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, cadence x4: present in common arm yes; retained E3 reference.
wave pinning, 60, 1, cadence x8: present in common arm yes; retained E3 reference.

Completed audited Figure 4 criteria and qualifications. All retained reference identities are shown, including unmatched designed inputs. A missing designed comparison is not a missing executed run; native execution and intake remain separately audited. × at the axis floor means unidentified, not zero. Original scientific fields and display annotations remain in CSV.
CSV: figure4_S0_S12_final_decisions_20261020_618_S12_E3_references_22.csv